The

Fault Tolerant

Forehand

Succeed Under Imperfect Conditions

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Preface

What is Fault Tolerance Anyway?

"Fault tolerance" is a term from computing – it refers to a machine's ability to remain functional even in the face of a fault, a small problem. For example, imagine that one of the transistors in your processor fails. Is your entire computer going to crash? No, because your computer is *fault tolerant*; it can tolerate the fault, the transistor failure, and still function.

Elite forehands are also fault tolerant. They can tolerate small inaccuracies in preparation, timing, or execution without failing. This fault tolerance is an *essential* ingredient of a high level forehand. To gain a maximal advantage in tennis, our goal isn't *just* to produce the best possible result. Rather, it's to develop a stroke which produces that best possible result with the *highest* margin for error out of the *widest* array of different situations.

Success Under Imperfect Conditions

If a particular stroke only produces a fast, heavy, accurate shot when struck perfectly, but when struck *imperfectly*, frequently results in an outright miss, then that shot is *useless* in an actual match. The shot is *not* fault tolerant, because it can't handle a fault, a small mistake in preparation, timing, or execution, and still succeed.

On the other hand, a stroke which *succeeds* even in the face of a small preparation or execution mistake *is* fault tolerant and *is* competitively useful. This idea is critical to the rest of our discussion.

Only fault tolerant strokes grant a competitive advantage, because during stroke production, conditions will never be perfect. We'd all *like* to be in perfect position for every shot, and to practice enough that we never mistime our swings, but in practice that's impossible, *especially* if the other player is elite.

Roger Federer hitting a swinging volley – a shot for which it's very difficult to establish a consistent contact point, and thus fault tolerance is essential to success.

Instead, we're going to hit many of our shots while slightly out of position and make many *small* timing and execution mistakes throughout the match. This takes some humility to admit, but, as usual, humility is a useful virtue. To win, our stroke needs to succeed in spite of these small mistakes.

There are two different classes of skills necessary to *consistently* produce an elite forehand: first, there are all the things that *assist* in stroke production, and second, there is the stroke production itself. The first class includes things like visual tracking and footwork. The goal is to develop strong legs, fast feet, and attentive eyes such that you can avoid ever having to hit out of position, and you can avoid ever mistiming the ball. The second class concerns your stroke mechanics themselves. Here, the goal is to implement a swing technique that is *fault tolerant* enough that that you can hit out of position and mistime the ball, and still have your stroke succeed.

Tennis is Often Difficult and Awkward

When two players are competing, each is doing everything in their power to make the other's shots as *difficult and awkward* as possible. Therefore, the most advantageous competitive strokes are the strokes that *still function at a high level* out of difficult and awkward positions.

We see this concept at play all the time on the tour. Professional players *constantly* slightly mistime their strokes, and they *constantly* hit shots while slightly out of position. Everyone is moving so fast and hitting so hard that it's *humanly impossible* to position for and strike every shot perfectly.

Of course, every player *tries* to hit every ball perfectly, and they move so well that they actually do hit *a lot* of them pretty perfectly, but in the end,

Roger Federer hitting a forehand on which he didn't quite get around the ball, and it's too close to him. The shot succeeds, and he still wins the point.

even the ever graceful Roger Federer hits the occasional mistimed forehand, or gets caught by a solid, deep return and has to hit out of position.

The Modern Forehand
A Biomechanical Masterpiece

When performed correctly, the modern forehand is very fault tolerant. Proper forehand mechanics ensure that the stroke possesses a very large margin for error – when things go wrong, the swing still succeeds.

Mechanically unsound forehands are not fault tolerant. Any small inaccuracy in timing, string angle, setup, etc, will cause a miss. Most forehand inconsistency is not proximally caused by those small preparation or execution inaccuracies, but rather by a swing which lacks the fault tolerance to handle them. Once proper mechanics are mastered, the forehand is actually very difficult to miss[1].

Understanding this point is essential to evaluating your own forehand. If, as you set up for the ball, your mind feels nervous and shaky – like the smallest mistake will send the ball flying into the back fence or spiraling down into the bottom of the net, that is because you are performing the forehand *mechanically* wrong. It's not just that you need to practice more – there's something about the movement pattern that you're attempting that needs to be fundamentally changed in order to improve the stroke's success rate.

Most forehand inconsistency is not proximally caused by small inaccuracies, but rather by a swing which lacks the fault tolerance to handle them.

Small Mistakes Shouldn't Cause Misses

During a correctly executed forehand, you can non-trivially mismanage your swing and still hit the ball in. You'll notice that when professional players get a bad bounce (often near the baseline of a chewed up grass court), they are, *more often than not,* still able to hit the ball back into play. Usually, this takes the form of an awkward, jerky, arm reach in front of them

1 Of course, the instant you finally feel this "can't miss" sensation for the first time, you'll immediately start attempting ridiculous shots and missing frequently again, but this shot selection leak is a far easier problem to reign in than a mechanically inefficient forehand.

where they kind of swat the ball back over, but we'll get into the specifics of exactly what's happening there later. Here's what's important: good players consistently succeed in continuing the rally even in cases when the ball's position changes so suddenly.

This success is not merely due to natural, improvisational talent. Rather, this sudden adjustment is made possible by the forehand's exceptionally large margin for error. The quick swat or flick that they execute is the player *cashing in* on this margin. This ability to adjust is *built in* to the forehand once each body part is doing roughly what it's supposed to be doing, and that ability to adjust slowly disappears as mechanical flaws are introduced.

Even on the full stretch, Novak Djokovic maintains balance and creates the correct string angle and racket path through contact, leading to a successful cross-court passing shot. (Semi-finals of the Rogers Cup in 2015)

Strong Fundamentals Create Fault Tolerance

With weak fundamentals, any non-perfectly executed stroke is likely to fail, whereas with strong fundamentals, each individual stroke can be imperfect and still produce an adequate result. You can be a little early, or a little late. Your racket tilt can be off by four or five degrees.

You can set up a little too close to the ball, or a little too far from it. The ball will still go in. It is precisely due to this flexibility that when a professional player is faced with a bad bounce, a bounce which suddenly requires the ball to be struck 6-12 inches away from the expected contact point, they are able to quickly, often subconsciously, adjust and *still hit the ball in.* It's not going to be a great shot, but it will go in, and that's what's important.

The forehand fundamentals responsible for producing this exceptionally large margin for error are the subject of this book. Many recreational players are never told that it is a *fully attainable* goal to set up for a forehand knowing you won't miss, and the things they're told instead often cause more harm than good.

Carnage and Misconceptions from Chronically Bad Coaching

First, let's dispel this notion that "swinging faster" and "creating more topspin" creates margin for error. For 95% of tennis players, it doesn't, and in fact, it does the opposite. "Spin creates margin" is only true *after* the forehand is being performed fundamentally correctly. If a player has biomechanical issues in their swing, then swinging faster and *attempting* to generate topspin will likely *decrease* their margin for error; they'll miss more, because the added velocity and the increased verticality of the swing path will further exacerbate the core issues that are present in each of their strokes. If a player with major technical flaws wants to hit the forehand more consistently, they'll be far better off swinging flatter and slower.

Margin Comes From Fault Tolerant Mechanics

True margin for error comes from striking the ball in such a way that you can mistime or otherwise mishit it, and have it still land in. For most players, this means swinging slow and flat. Now, once a player has sufficient mastery over the modern topspin forehand, then yes, a shot with more topspin will have more margin than a shot with less, but, again, that's only true once the stroke is *already* fundamentally sound.

Further, when we discuss "adding margin for error with topspin," we're talking about something that, again, is irrelevant to 95% of tennis players. Adding topspin to a well executed aggressive shot may turn an 80% success rate into a 90% rate, and that's certainly a significant benefit, but this marginal improvement is *dwarfed* by the results when a player with a broken forehand adopts proper mechanics. After that change, they might literally increase their neutral ball's chance of going in from under 50% to upwards of 90%, an absolutely monumental difference.

I suspect the "use more topspin" or the "swing more low to high" advice is often given by coaches who don't realize how fundamental the flaws in their students' strokes really are. They're hoping to give their student the 10% bump they see in their own game from "using topspin," never stopping to realize that the student isn't missing because they're hitting too flat. They're missing because they're *striking the ball differently every time.* The solution, therefore, isn't to "use more topspin," but rather to fix the stroke such that the ball is struck *the same way* even when timing or preparation is a little off.

Incorrectly Treating the Symptom

Most recreational tennis players don't even possess a correct understanding of *how* topspin is generated during the stroke, so it's no wonder that cueing them to "try using more topspin" produces more misses, not fewer.

This issue contained in the "add topspin" cue belies a more widespread problem in coaching. Too often, a coach's instruction consists *only* of noticing a particular, specific, visually identifiable feature that is wrong with their student's stroke – their feet are in the wrong spot, their wrist should be different, the ball hit the net – and then instead of trying to figure out *why* the mistake happened, they just try to cue the student out of *that specific mistake.*

The problem here is that, in almost every case, these *observed* flaws are merely *symptoms* of an underlying systemic issue, not the cause of the failing stroke themselves. The specific issue (for example: the wrist is straight at contact) is caused by a systemic issue (the grip is not relaxed enough). So telling the student "bend your wrist" will *not* work, because the fundamental issue isn't *that* the wrist isn't bent, the issue is *why* the wrist isn't bent. All telling the student to "bend your wrist" will do is cause them to hit with a tight bent wrist instead of a tight straight wrist, a counterproductive change indeed.

As coaches, we use the clues our students' bodies give us to diagnose the underlying causes of their failing strokes.

Instead of merely informing the student of the specific, observed, incorrect result, the coach needs to search for the *fundamental gap in understanding* that's leading to that result. The problem is *caused* by something. There's some idea that's critical to the stroke, some broad movement pattern or goal, that the student either doesn't understand how to properly execute, or perhaps isn't even *attempting* to execute. *That* is what's holding them back.

The initial incorrect behavior is a *clue* as to where that gap may lie. Our role is to fill that gap in the student's understanding and then cue the student to help them correctly *implement* what they've just learned. As coaches, we use the clues our students' bodies give us to *diagnose* the underlying causes of their failing strokes. We diagnose the *cause* and then help the student with the *cause*, rather than simply relaying the clues themselves back to the student, and leaving them to blindly grope around hoping to come upon any underlying causes on their own.

This Book Teaches *Fundamentals*

Of course, the above prescription for coaching is easier said than done. In order to debug a student's stroke, the coach first must *fully understand* the stroke themselves. This is especially difficult if the coach is a very natural player on their own. The easier it is for them to effortlessly mirror a movement pattern they see, the less they'll be forced to *understand* the mechanics behind what makes that movement pattern effective.

For example, consider this question: which parts of Roger Federer's forehand stroke are essential? Some of the things he does are directly responsible for making the stroke so effective, while other aspects of it merely constitute his personal style – what feels best for *his* body, an attribute that varies widely from player to player.

If, as an athletically gifted natural athlete, you can mimic Roger's movement pattern very precisely, then there's no *need* for you to identify which of the movements you're mimicking are *actually responsible* for the various features of the stroke. What's generating the power? What's generating the spin? Why is it that you hardly ever miss? Your forehand will be great, but *why* it's great will often still remain a mystery, and therefore *teaching* a non-natural to do what you do can be quite challenging.

On the other hand, if a non-athletically gifted individual wants to mimic Roger's forehand, they can't simply watch his movement and imitate it. Their brain doesn't have that skill. Instead, they need to watch every little part of it separately, try to figure out what he's telling his body to do when, and try to figure out how those movements interact with the racket, and then the ball, to produce the desired result. It's the longer, harder road, but as you travel it, you gain a deep understanding of not only *what* the forehand stroke should look and feel like, but also *why* it should look and feel like that.

Whether you're a natural or not, whether you're a player or coach, this book will explain the forehand *from the ground up*. It will explain *how* it works, and *why* it works, and it'll provide the foundation for forehand improvement for years to come.

After finishing this book, you'll never again find yourself scouring the internet desperately trying to uncover why your forehand is so inconsistent.

We will discuss specific joint angles and limb positions *occasionally,* but mostly those are irrelevant. Instead, we'll be mostly discussing the broad body positions and movement patterns that are *universal* across all elite forehands. In each section, we'll be specific about which aspects of the stroke are essential, and which can be done effectively in multiple different ways. We'll explain where power comes from, where spin comes from, and, most importantly, where *fault tolerance* comes from.

After finishing this book, you'll never again find yourself scouring the internet desperately trying to uncover why your forehand is so inconsistent, or weak, or flat, or why it just doesn't feel right. Everyone's issues are different, but I guarantee that s*omething* in the pages that follow is your answer; you've just got to figure out what it is.

And then, of course, get out on the practice court and start training those good habits.

Part 1

Forehand Contact

Why Start With Contact?

Most instructors will tell you a whole host of things before any discussion of how to actually *strike* the ball ever takes place – "turn sideways," "move your feet," "watch the ball," "swing low to high." The problem with this is that a proper understanding of the actual *strike* you're trying to create is the starting point on any tennis shot; all of the body movement that gets you to the strike is fundamentally *subordinate* to the strike itself.

Take "move your feet," for example. It certainly is decent advice for a player at any level, but moving to a ball you have no idea how to properly strike isn't particularly useful. "Watch the ball" is similar. The student is watching it, but now how are they supposed to hit it?

More pernicious is coaching a student to "turn sideways" before teaching them *why* they're turning sideways. SO MANY recreational players hit their forehands *while still sideways,* instead of allowing their hips and chest to rotate *back* towards the target as they swing. This *destroys* the stroke's fault tolerance, as we'll discuss later.

The takeaway here is that random advice like "turn sideways" is useless without the context of *why* we're turning sideways. How does turning sideways ultimately improve our strike? The players who fail to rotate back towards the target as they swing would actually be *better* off if they'd never turned sideways at all. In fact, I once coached a 13 year old boy who naturally played his forehand just like that. He barely turned either his hips or his chest sideways at all, and he used a very abbreviated swing. As a result, his forehand was extremely fault tolerant, he could aim to both sides of the court, he practically never missed, and he routinely beat his two-year-older brother.

Random advice like "turn sideways" is useless without the context of why we're turning sideways. How does turning sideways ultimately improve our strike?

Therefore, any discussion of a fault tolerant forehand should start with a discussion of fault tolerant contact. We'll begin by answering the question: how do I strike the ball such that it flies off my racket the *same way* every time?

If you understand contact, you understand the forehand. Movement, footwork, relaxation, the kinetic chain – all of it exists to facilitate or accentuate the *same* kind of fault tolerant contact under different circumstances.

Further, this contact understanding is not purely academic. In certain circumstances, a strong understanding of contact itself is the only way to consistently succeed. The most common situation in which a strong understanding of contact makes a big difference is serve return. When you return serve, the ball is coming extremely quickly – often so quickly that you don't have time to get your body into its usual preparation position. Not only is the ball coming fast, but it's also often spinning in a novel way, making it more difficult to track, both in the air and off the bounce. The result is that it's very rare that you will be in a perfect position to strike a service return with your standard, practiced, drilled-a-thousand-times forehand motion. Returning effectively constitutes a *different* skill set than grinding forehands from the baseline. Effective returning requires an athletic understanding of how to create correct contact. Wherever the ball ends up as it gets blasted at you, you need to improvise your correct contact from there.

Important, though, is that contact understanding is a superset of baseline grinding. If you understand contact, you can *both* effectively grind at the baseline *and* effectively return serve. Further, if, on each of those typical baseline forehands, you execute the stroke by *attempting to create a certain kind of contact*, then some of that practice will translate to serve return. If, on the other hand you grind your baseline forehands *without* a deep understanding of contact itself, then a service return forehand, on which you're rushed and out of position, will feel like a completely novel, unique shot that needs to be mastered on its own.

Contact at a Glance

- The racket and the forearm always make a 90-135° angle angle.
- The racket flicks through the ball *in front* of the body.
- The racket flicks through the ball *away* from the body.
- Fling your right angle racket-arm system out, up, and through the ball.
- The arm is far freer when the body is *square* to the target.
- A slightly *closed* string angle is necessary for topspin.
- Maintain your string angle for as long as possible through the hitting zone.
- Upward friction creates topspin, not "rolling the racket."
- Relax your hand – all racket rotation is *passive.*
- The rotational kinetic chain's only purpose is to increase racket head speed through contact.

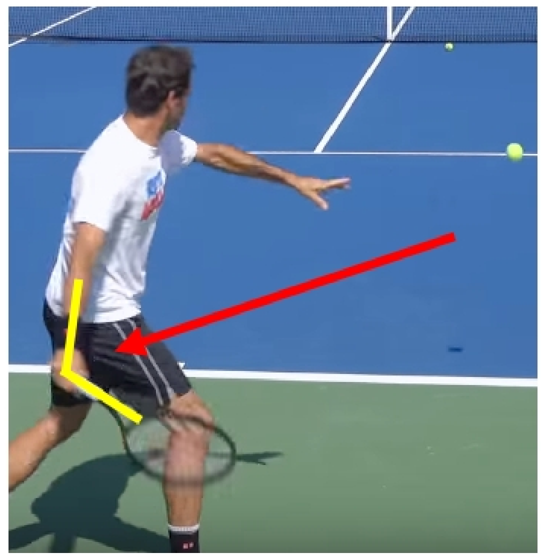
The Fundamental Theorem of Tennis

The Fundamental Theorem of Tennis is two rules that govern every shot in the game. They are as follows:

1. **The racket makes a 90-135° angle with the forearm**

2. **The racket strikes the ball in front of the body (in a pinch, in front of the forearm)**

Any shot which follows these two rules will work. I should specify that I usually verbalize this rule as "a 90 degree angle" instead of "a 90-135 degree angle," even though that's often an exaggeration. Many shots in tennis are played with a 135 degree angle, but I find that verbalizing and thinking about it as "90 degrees" produces the best results. Many students' natural intuition is to play shots with a straight, 180° racket-forearm angle, and thinking about "90 degrees" prevents this.



Roger Federer during a US Open Practice in 2019, loading his 115º forearm-racket system and preparing for a forehand swing.

Following these two rules puts the racket in a position to function as a trampoline off of which the ball can easily bounce. The forearm will facilitate velocity transfer back to the ball much better when the racket is perpendicular to it, rather than in line with it.

Control Over Contact

This alignment is also optimal for control over both the racket and the contact it produces. There are many ways to manipulate the arm-racket system while maintaining a 90° forearm-racket angle, and all of these will produce good shots. The important part isn't the method by which the 90° forearm-racket system is moved around. The important part is that the bigger movers of the body move the *entire* 90° forearm-racket system around *as a unit,* without the angle between the racket and the forearm straightening out.

Mentally cueing your racket to stay at 90° will prevent you from attempting to adjust to balls by moving *only* the racket head *without* adjusting the entire racket-arm system. Adjusting with *just* the racket head, *instead* of the entire racket-arm system elongates the angle between the arm and racket and causes you to lose fine control over the racket head's movements.

Wrist Tension

All this talk about maintaining an angle between your forearm and racket may encourage you tense up your wrist in order to do so – don't do that. It's possible to maintain this angle using very little tension. The secret is moving the racket around using the biggest muscles in the body, rather than the small muscles in the forearm and wrist.

Adjust the racket's position with your legs, your torso, or, in a pinch, your pectoral muscles (the large muscles in your chest). The chest is farthest out on the kinetic chain you should go when manipulating the racket head. The muscles farther out on the chain, the muscles of the upper arm, and *especially* the small muscles of the forearm and wrist, should *only* be used to stabilize the swing, rather than to create any force during it themselves.

"Always" Hit the Ball in Front

As a last point of interest, it's not strictly necessary to contact the ball in front of the body (though it is ideal), as long as the racket still hits the ball in front of the forearm. Think of full stretch volleys out to the side, or shots where the ball has bounced behind the player. In these cases, as long as the player is able to maintain some sort of an angle between the racket and the forearm and get the racket flicking *forward* by the time it strikes the ball, the ball will usually go in.

Novak Djokovic, on the full stretch, cocking his racket-forearm system at 90º and preparing to fling the head around his hand.

There are many ways the racket can flick around the wrist when its held loosely at roughly 90° before starting the swing. A relaxed wrist, with the racket at an angle to it, forms the foundation for improvisational tennis. A common example of where this flexibility is especially necessary is for controlling shots performed while on the dead run. You'll notice that good players on the full stretch often end up following through awkwardly in front of them, rather than around their body, because they're focusing on generating any and all *forward* flick of the racket that they can muster from such a compromised position.

The SIMPLEST Effective Forehand

At any time during a rally, if our goal is *simply* to not miss, all we need to do is take our 90° forearm-racket system, place it behind the ball, and then push the ball back over the net, contacting it in front of us. This constitutes executing **the fundamental theorem of tennis** in its simplest form.

The swing on the simplest effective forehand is almost entirely translational (back-to-front), and we ensure not to hit the ball so hard that it flies out on the other side. If you watch some very old tennis videos of guys in white shorts and wooden rackets, this is mostly what you see. You might think – okay, but we've moved past this stroke now, right? Now we use rotation, and topspin, and wrist lag! True, but not the whole story. Actually, the stroke described in the above paragraph, the simplest effective forehand, is a stroke that is still used by every player in the modern game.

On exceptionally low balls where the ground prevents a modern topspin stroke, you'll need to use this. On the dead run and the full stretch, rotation may be impossible or severely limited, and many players will elect for a version of this. When your opponent is so far off the court that all you need to do is hit the ball in to win the point, you'll use this.

In the following sections, we're going to move on to the more complex version of contact that involves utilizing topspin and violently whipping the racket through the hitting zone. Before we do, let's be clear about something: everything that follows is a way to *enhance* the contact we've already described. Fault tolerant forehand contact consists of creating the 90° forearm-racket system and translating it through the ball. No matter how much we build on top of that contact to make it more effective, that fact will not change.

Topspin Changes the Forehand

The forehand stroke changed when racket and string technology improved enough to allow players to generate a significant amount of topspin. When only gravity was acting on the ball to push it down and keep it in play, there was an inherent limit on how fast players could hit the ball over the net: hit it any harder, and it would fly out before gravity could pull it back down into the court. Topspin eliminated this limit.

When a ball is spinning in a topspin way, the air around the ball pushes it down into the court, keeping it in. This downward force is greater the faster the ball is rotating. Due to this, there is no theoretical limit on how fast a ball can be struck while still staying in play. For every ball speed, there exists an amount of topspin such that the spin will force the ball down into the court before it flies out[2].

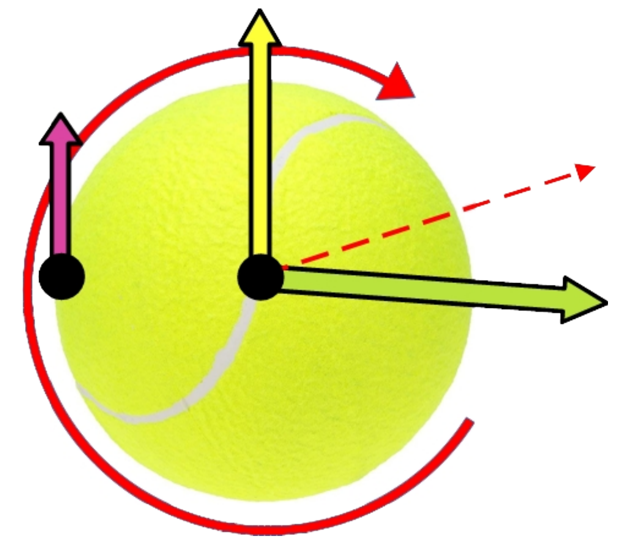
When a ball is spinning in a topspin way, the air around the ball pushes it down into the court, keeping it in.

Topspin also allows for more net clearance, a staple of fault tolerance. Though much easier to achieve than consistent contact, net clearance is essential for a high percentage forehand, because it allows a player to make small execution mistakes without actually missing. Modern players can hit fast, effective shots that pass 3-7 feet over the net, an impossibility without the power of topspin. These 3-7 feet give the player margin for error – if a player is hitting 4 feet over the net, and they make a small mistake which causes the ball to fly 3 feet lower than expected, that mistake will not cause a miss into the net, and the point will continue. This kind of fault tolerance is essential for high level play.

2 While in the air, the ball must spin such that the top of it is rotating in the direction of its velocity. When a ball rotates in this manner, air pressure pushes the ball down into the court. This downward pressure is what keeps tour level shots from flying out, even though they're struck exceptionally hard. In a vacuum (no air to push the ball down), many topspin shots would not only land out, but would actually hit the back fence long before landing.

Understanding Topspin Contact

In order to hit a topspin shot, the player's racket must exert an upward force on the back of the ball at contact, causing it to rotate. In addition to this upward force, the player must also supply the large translational force necessary to drive the ball back over the net with speed. This is accomplished by contacting the ball while the racket is moving both *upwards* and *forwards* through it, with the strings tilted slightly downward towards the court while doing so.



Green: *The normal force applied by the racket face. Note its slight downward angle, caused by the slightly closed strings.*

Yellow: *The upward translational force – friction between the ball and the strings.*

Pink: *Torque from the upward frictional force acting tangent to the ball, generating topspin.*

Red: *Approximate launch trajectory (dashed) and rotation (solid) after impact.*

The upward friction between the ball and strings at impact will both drive the ball up, giving it enough height to clear the net, and also cause the ball to rotate in a topspin manner. The translational impact – the fact that the racket is also moving *forward* through contact – will impart the ball with significant velocity towards the other side of the court.

The SIMPLEST Effective TOPSPIN Forehand

Generating topspin requires a bit more than simply adhering to the fundamental theorem of tennis, though we still won't violate it. The simplest effective topspin forehand consists of taking our right angle racket-arm system, placing it *below* the ball, and then swinging it **up** and **forward** through the ball. To create topspin, the **up** vector of the swing is necessary; if the racket face is not moving **up** when it contacts the ball, it cannot exert an upward frictional force on the back of the ball, and therefore it cannot cause the ball to rotate in a topspin way. This **up** vector is what distinguishes the simplest effective *topspin* forehand from the simplest effective forehand.

Body rotation is *not necessary* to create topspin – the rotational kinetic chain that you see utilized on TV, the technique by which a player turns away from the ball during preparation, then whips their hips and chest back around as they swing – that entire process only exists to *speed up* forehand topspin contact; it's not required to *generate* that contact in the first place.

Square to the Target at Contact

The simplest effective topspin forehand is performed with the hips square to the target, *not* with the hips turned to the side. This body orientation allows the arm far more freedom of movement than the sideways orientation does. The only reason a player turns sideways at all on the forehand is *so that* they can turn back towards their target as they swing, thereby harnessing rotation. We'll elaborate further on this "turn away, then turn back" idea in **Preparation** and **The Forward Swing,** but the important point right now is that if our goal is to perform the simplest effective topspin forehand, which doesn't harness explosive rotation, there's no need to turn sideways at all.

When the hips and chest are square to the target at contact, the racket can be freely manipulated in any direction by the strong muscles of the upper back and chest.

When the hips and chest are correctly square to the net at contact, the hand, and thereby the racket, is very free. It can move up and down easily, and it can move side to side easily. This allows for a wide range of contact points at which correct, fault tolerant contact can be made. Further, because the racket is so free in this position, your brain can use the visual information it gets while tracking the ball to subconsciously adjust your swing in real time. As we discussed earlier, we often see players on the tour suddenly and successfully adjust to bad bounces – the reason they can do so is that they're using a body orientation which *provides enough freedom* for said adjustment.

When, instead, our hips and chest are *still sideways* at contact, the arm is much more impinged. The racket has almost no freedom to move any of left, right, up, or down. Further, even if a player *technically could* awkwardly manipulate their racket from this position, the racket is no longer being controlled by the large, strong muscles of the upper back and chest, like it is when the body is square to the target. Instead, the player would need to use smaller, weaker muscles like the triceps in order to move the racket.

When the hips and chest remain sideways at contact, the arm is very impinged, and the racket loses nearly all of its freedom of adjustment.

Learn the Motion with Shadow Swings

You should experiment with the positions I've described using shadow swings. Start with the correct version, the version where your hips are square to the target. Stand with your hips and chest facing an imaginary target, and place your racket down at your side, strings down. Relax your hand, and swing the racket up and through an imaginary ball in front and away from you. Now try it if you orient your body sideways from an imaginary target. Again, swing the racket up and through an imaginary ball in front of you. Which one feels better? Which one feels freer?

Once you've convinced yourself that making contact while square to the target is optimal, you can slowly add a rotational component to your swing. Start by preparing with your body slightly turned to the side, let's say rotated away from the target by 10°, then initiate your swing by turning back towards your imaginary target, such that you're square by the time of imaginary contact. Next, turn to the side 20°, and initiate your swing by turning back those 20°. You don't need to explosively rotate, just do it lightly. Even still, you'll notice that each time you increase the amount you turn away, and then subsequently turn back, you'll gain some effortless racket head speed, even while still performing the *same exact action* with your arm and hand.

This phenomenon, the idea by which we recruit our entire body to *enhance* the very same contact we were already using – that is the foundation of the modern forehand.

Mini-Tennis and Forehand Contact Practice

I always recommend starting any practice session with a few minutes of mini-tennis – a rally type where both players stand at the service line and lightly hit balls back and forth. This is because mini-tennis allows you to re-cement your simplest effective topspin forehand in your mind *before* moving back to the baseline and adding in the extra body rotation that makes it truly explosive.

There are a few things to focus on during your few minutes of mini-tennis. First, we actually get extra vertical racked head speed for free by relaxing our hand while we swing. When you place the racket below the ball, relax your hand, and then swing **up** and **through** the ball, the racket will naturally rotate around your wrist through contact. This rotation further increases the vertical speed of the racket through contact, leading to more torque applied at the back of the ball, and thus more topspin.

Beginning your practice with mini-tennis is a great way to ensure that, on this particular practice day, you're still relaxed. It is *very* easy to randomly start gripping the racket too tightly, without even noticing that you're doing it. The issues that this increased tension creates are then also very difficult to debug. Mentally cementing your hand relaxation with a few minutes of mini-tennis helps to avoid this annoying situation.

Second, use mini-tennis to titrate your swing with body rotation. Just like in the shadow swings exercise we discussed earlier, start your mini-tennis swings using maybe 10 or 20 degrees of body rotation. As you get comfortable, increase your amount of body rotation, which will add slightly more racket head speed. You're still not explosively rotating here – you are hitting to and from the service line, after all – but you *are* performing a quick refresh of what clean, fault tolerant, topspin contact feels like, and then of how to accelerate through that contact by rotating.

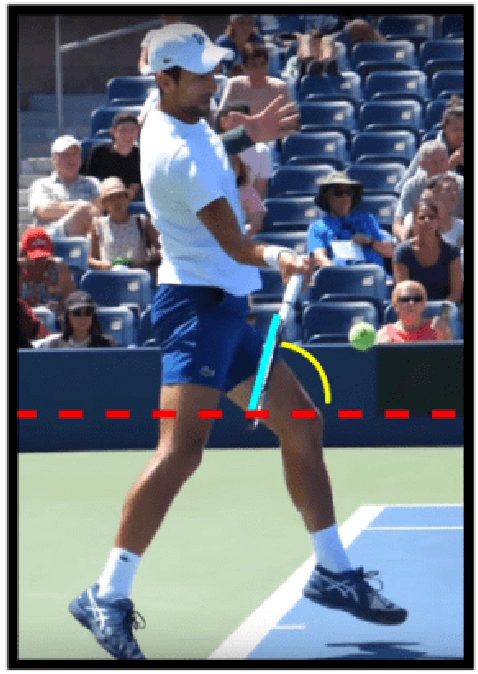
Use mini-tennis to titrate your swing with body rotation.

The Optimal Topspin Contact Point

Racket rotation around the wrist can be further accelerated by allowing the racket to flick **out** to the ball, as well as **up** and **through** it. We'll get into the specifics of how that works in **The Forward Swing**, but you should practice this during mini-tennis as well.

In order to be able to relax the wrist and have the racket flick around like we want it to, the ball needs to be struck away from the body – both in front of it and laterally away from it. By the end of your few minutes of mini-tennis, your goal is to have re-cemented a nice, relaxed, swing, during which the racket naturally flicks **out** to the ball, and then **up** and **through** it once it gets there. Once you've attained that natural, whippy feeling through contact, and you've added in maybe ¾ of your typical body rotation, you're ready to back up and hit more explosively from the baseline.

The Topspin String Angle



A graphical depiction of Novak Djokovic playing his forehand with a slightly closed string angle.

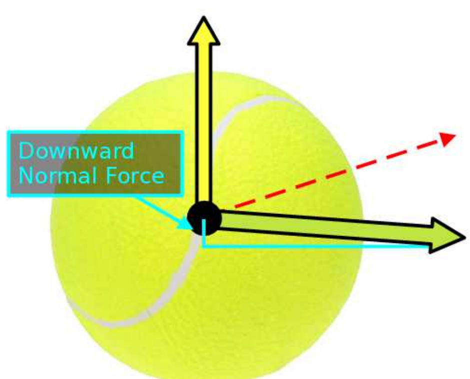
When I say "string angle," I am referring to the angle between the strings and the ground. We refer to certain string angles as "open," "neutral," or "closed" depending on orientation.

Open string angle – strings angled up towards the sky

Neutral string angle – strings parallel to the net

Closed string angle – strings angled down towards the court

During a topspin stroke, the ball should be struck with a slightly closed string angle. If, during contact, a player's strings are parallel to the net, or angled up towards the sky, then there exists a physical limit on how fast that player can swing without hitting the ball out; the frictional force pulling the ball up is just too high relative to the topspin generated, which pushes it back down.



At high swing speeds, the normal force component (green) of a topspin impact needs to act at a slightly downward angle on the ball, in order to offset the large upward frictional force (yellow) pulling the ball up.

When, instead, the strings are slightly closed, there is a very small component of the impact's normal force that actually drives the ball *down.* This slight downward normal force offsets the upward frictional force enough that a player using this string angle can swing as fast as they want, and the ball will still land in.

It's important to note that using a slightly closed string angle necessitates that the racket head also be moving **up** through contact. The two form a pair – the closed string angle and the **up** vector. If used separately, they don't work. With a level swing path and a closed string angle, the ball will go spiraling down into the net, and with an upward swing path and a neutral or open string angle, it will go flying out. Only *together* do the upward swing path and the closed string angle function to produce a high margin, high net clearance, dipping, topspin shot.

Fault Tolerant Topspin Contact

Our goal on the forehand is to create the topspin **up** and **through** contact as *consistently* as possible. We're going to be doing this really, really fast, so the timing isn't always going to be perfect. We certainty can't have flaws built into the swing that are exacerbated by swinging faster, because, again, the goal is to swing really, really fast. Instead, we want a swing technique that creates this correct topspin contact *even when the ball isn't struck perfectly*. Our swing must be fault tolerant enough to be executed extremely quickly, under a variety of different circumstances, and still succeed.

The optimal swing path on the forehand (in the back-to-front plane) is a slight *diagonal*[3]. This swing path ensures that, even if the ball is struck a little early or a little late, the forces applied during the strike will still be roughly the same. The best way to accomplish this is to prepare your hand *below* the anticipated contact height; that way, you'll be forced to fling the racket *up* through the ball as you swing. When the ball is exceptionally low, this won't be possible, and you'll be forced to prepare your hand at the level of contact, or perhaps even slightly above it. This is part of what makes low forehands more difficult – the swing is inherently less fault tolerant, because often the racket must go down, and then up during it's swing path, rather than just up[4].

We want a swing technique that creates this correct topspin contact even when the ball isn't struck perfectly.

As we've discussed, when you fling the racket **up** and **through** the ball, the string angle should be *closed* to the court. Importantly, this closed racket face angle should be *maintained*

3 Note that this is typically a much shallower diagonal than player's trying to "swing low to high" usually use.
4 If you routinely struggle with low forehands specifically, feel free to skip ahead to the ***Adjusting to Contact*** section and the ***Dominate the Slicer With Heavy Low Forehands*** section, which are both dedicated to low forehands.

throughout as much of the forward swing as possible; it doesn't change until *after* the ball is gone. This angle maintenance is one of the essential aspects facilitating the swing's fault tolerance.

Cementing Your Mental Conception of Contact

The best improvisers in tennis have trained into their brains an intuitive understanding of the exact kind of contact they're trying to create when they swing. This understanding allows them to dynamically adjust the movement patterns they attempt based on the situation. Often, there's plenty of time for a full coiling of the body, from wrist all the way down to legs, and then an explosive uncoiling leading to unparalleled racket head speed through the hitting zone. At other times, a desperate forward flick with an outstretched arm is all we can muster.

Every "forehand" is slightly different from every other "forehand." Though the *fundamental* principles dictating how the shot should be struck are very similar from shot to shot, the manner by which those fundamentals are *implemented* frequently changes. There exists an almost infinite number of distinct, different situations from which a "forehand" must be struck. If each needed to be mastered as a totally *separate* entity, it would impossible to practice all of them sufficiently. Therefore, to develop a versatile, effective forehand, a player must develop an understanding of the *fundamental goal* common to all tennis strikes: create the contact with the ball that will produce the desired ball flight. Versatile striking requires answering three questions, in order:

1. **What do I want the ball flight to look like?**

2. **What contact creates that ball flight?**

3. **How do I orient my body to produce that contact?**

Many players start with #3, deciding to hit a "forehand" (employ a particular movement pattern), without first understanding #1 and #2. The most flexible, versatile players, on the other hand, the ones who possess mastery over *all* variations of *all* kinds of forehands – those players start with ball flight and contact. Once a player has fully internalized the contact they’re trying to create, then, during actual stroke production, they instinctively work backwards from there, orienting their body to produce that contact as effectively as possible.

The Great “Rolling the Racket” Myth

Last year, I worked with a 14-year-old girl who had one of the most mismanaged forehand strokes I’ve ever seen. Her problem: “use your wrist to roll the racket over the ball.” Rarely, if ever, could she make two forehands in a row. Her wrist hurt every time she hit the ball, and she wasn’t enjoying the game.

There was a point during one of our lessons when I asked her, “Anything not making sense, any questions?”

She turned to me, scoffed, and asked, “Why do I suck?”

Multiple times during our lessons, she cried, and I don’t blame her. She was an obedient student who did exactly what her coach told her to do. Think about it from her perspective: Her previous coach clearly tells her how to strike the ball. When she hits it that way, she can’t make two shots in a row, and her wrist hurts every shot. The coach can’t be wrong, of course, so what’s the only variable left? Clearly there’s something wrong with her; the only logical conclusion is that she just inherently sucks. It’s an easy conclusion for a 14-year-old adolescent to draw.

She turned to me, scoffed, and asked, “Why do I suck?”

Further exacerbating the issue, a 14-year-old girl has no practical way of identifying when her coach has unwittingly set her up for failure. All that can really be expected of her is to blindly follow the coach’s advice, and hope everything works out. So, for her sake, let’s set the record straight on “rolling the racket over it” – the most egregious misconception and the most damaging coaching cue in all of tennis.

Origins of Confusion

If you currently think you should "turn your racket over" as you strike a forehand, I don't blame you. This misconception is *very easy* to believe, because, in fact, professional players *really do* turn their rackets over when they hit forehands. Here's the critical insight, though – this motion is part of the *follow-through,* not part of the forward swing.

*Nadal's handle-forearm angle straightens out, and his string angle closes, only **after** contact as part of the **follow-through.***

I'll say that again. This "turning over" of the racket is part of the follow-through. It happens as a natural consequence of forces generated before and during the contact phase, but, again, it is *not* part of the swing itself. During the swing, we maintain the angle of the strings as consistently as possible to increase margin for error.

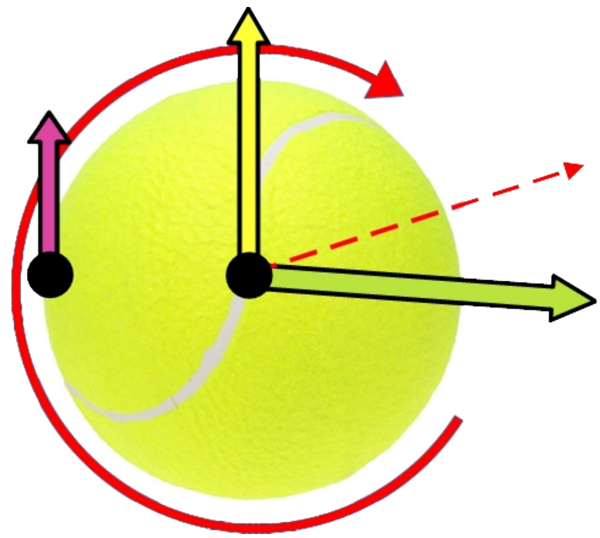
If we allow the racket-face angle to change dramatically through contact, say, "rolling over" such that the strings are pointing towards the sky at the beginning, and towards the ground after contact, then our margin for error evaporates; if we hit the ball slightly too early, our overly open strings will send the ball flying out, and if we hit the ball a little too late, our closed strings will smother it down into the net. If we don't time it perfectly, we'll miss. By instead keeping the angle consistent through the contact zone, we allow ourselves to consistently strike the ball with the proper up-and-forward diagonal contact, even in cases when the swing is slightly mistimed. It is by *maintaining* our string angle that we make our stroke fault tolerant.

"Rolling Over" Does Not Create Topspin

The idea that "rolling the racket over" through contact creates topspin is simply a physics mistake. Just because the ball is "rolling" doesn't mean your racket also needs to roll to create that motion. Topspin is generated by a strong, upward frictional force between the strings and the back of the ball at contact.

Recall our free body diagram from the **Understanding Topspin Contact** section. *This* is how topspin is generated – by transient forces that act on the ball for a few milliseconds during contact – *not* by somehow mirroring, with your racket, the ball motion that you want to cause.

This is true of all spin, by the way, not just topspin. You don't slice a serve by "carving around the ball." You don't hit a backspin shot by "scooping under the ball". That's not how physics works. Spin is created by *friction*, by a transient force which lasts a few milliseconds and applies a torque to the ball, causing it to rotate.



Green: *The normal force applied by the racket face. Note its slight downward angle, caused by the slightly closed strings.*

Yellow: *The upward translational force – friction between the ball and the strings.*

Pink: *Torque from the upward frictional force acting tangent to the ball, generating topspin.*

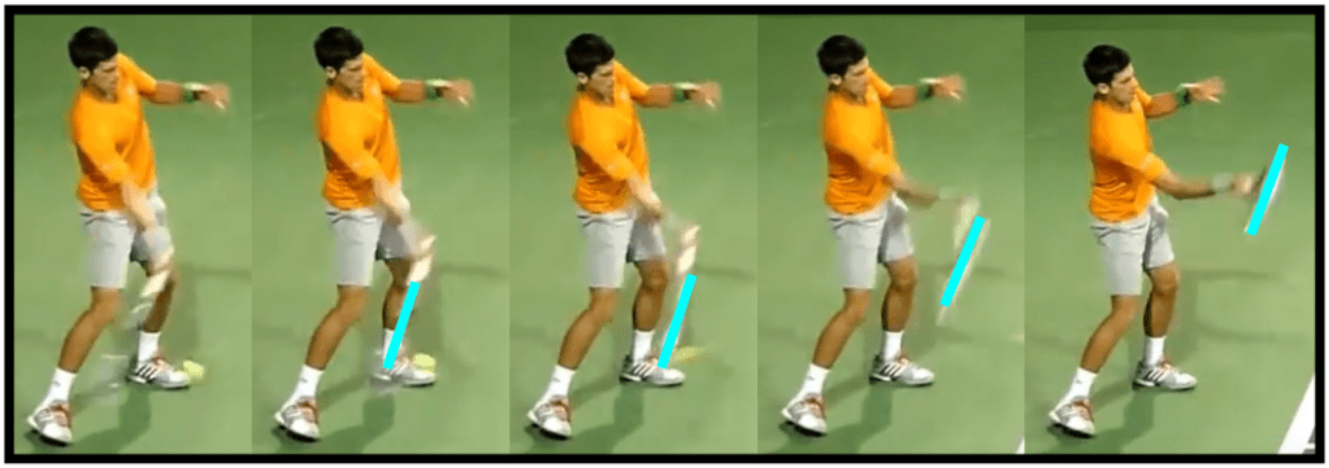
Red: *Approximate launch trajectory (dashed) and rotation (solid) after impact.*

Since our goal is to facilitate this contact in as fault tolerant a manner as possible, "rolling over" instead of maintaining our string angle is especially damaging; if we do that, the contact zone during which a quality shot will be produced is extremely narrow.

Maintain A *Consistent* String Angle

The string angle should *barely change* through the hitting zone. The longer you maintain your string angle through the hitting zone, the more fault tolerant your swing will be.

This is simply explained by collision physics. If a ball strikes an open racket face, it will deflect up more, and if it strikes a closed racket face, it will deflect down more. There is a certain range of string angles such that the ball will fly over the net and come down before landing out. In order for your forehand to work, you need to contact the ball while the strings are at an angle in this range.



The angle between Novak Djokovic's string bed and the ground remains remarkably consistent throughout his entire hitting zone, leading to a very large timing window for striking the ball and producing a quality result.

The *longer* you can maintain a correct angle through the hitting zone, the *less precise* you have to be with your swing timing. Look at Novak Djokovic's forehand above. If he were to contact the ball in *any* of frame 2, 3, 4, or even 5, his shot would go in. His string angle is roughly the same for his *entire* hitting zone. This is just one of the many things he does that allows him to *practically never miss*.

What String Angle to Choose?

The lower the ball is, the more open your strings should be, and the higher the ball is, the more closed your strings should be. Again, this is simple collision physics – you want low balls to fly higher off your racket and high balls to fly lower off your racket.

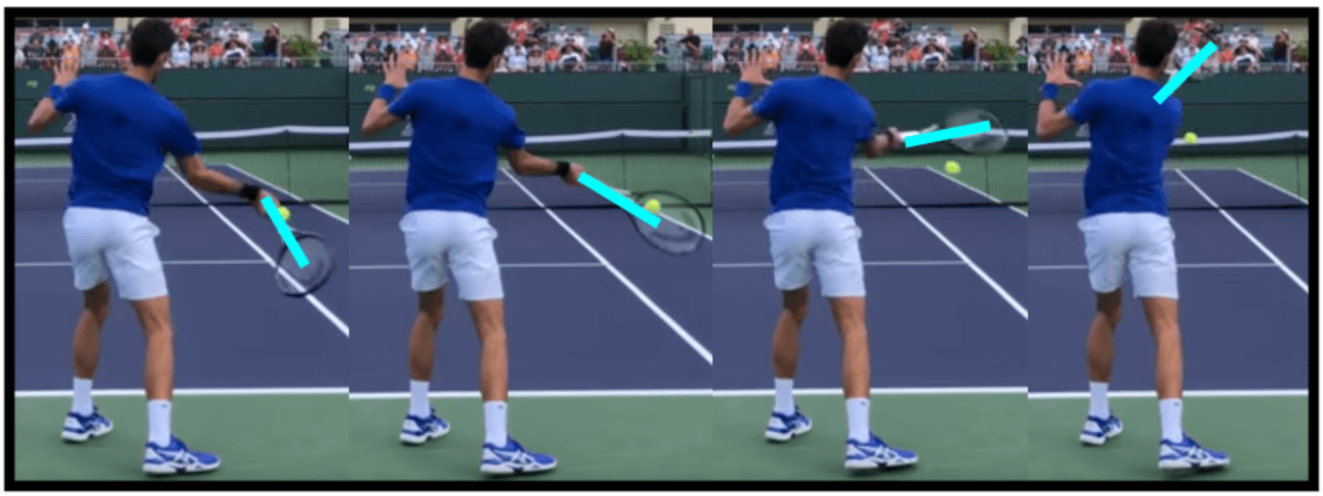
The change in string angle between shots is very minor. On low balls, for example, topspin shots are played with a neutral or even slightly closed string angle, not an open string angle. There is a *ton* of frictional force pulling the ball up through contact; it's not necessary to actually *deflect* the ball up at all, even on a low ball. Only topspin lobs require an open string angle.

Rafael Nadal playing a knee height forehand with an approximately neutral string angle.

I will reiterate – the most important aspect of string angle on any shot is that it is *unchanging through contact.* Whatever shot you've elected to hit, however you've decided to orient your string angle through contact, ensure that the angle barely changes through the hitting zone, and the shot will be fault tolerant.

Don't Players Rotate Their Rackets?

Yes, but in a different plane, a plane which does **not** change their string angle through contact. Elite players rotate their rackets (passively) *about their forearm* through the hitting zone, not about the handle.



Novak Djokovic's racket rotates counter-clockwise through the hitting zone, but the rotation doesn't affect the string angle, which remains slightly closed throughout the swing.

Look at the back view of Novak Djokovic's forehand above. The racket *is* rotating, but it's rotating in such a way as not to alter the angle of his strings through contact. From frame 1 all the way to frame 4, his string angle is slightly closed. That string angle remains consistent, even though the racket whips around his forearm throughout the stroke.

Relaxation and String Angle

At first, it's tough to relax your wrist and maintain your string angle. You have to pull the racket with your core movers at a pretty specific vector to avoid your forearm going crazy and rolling all over the place.

On the other hand, it's very *easy* to maintain a consistent string angle with a tight wrist.

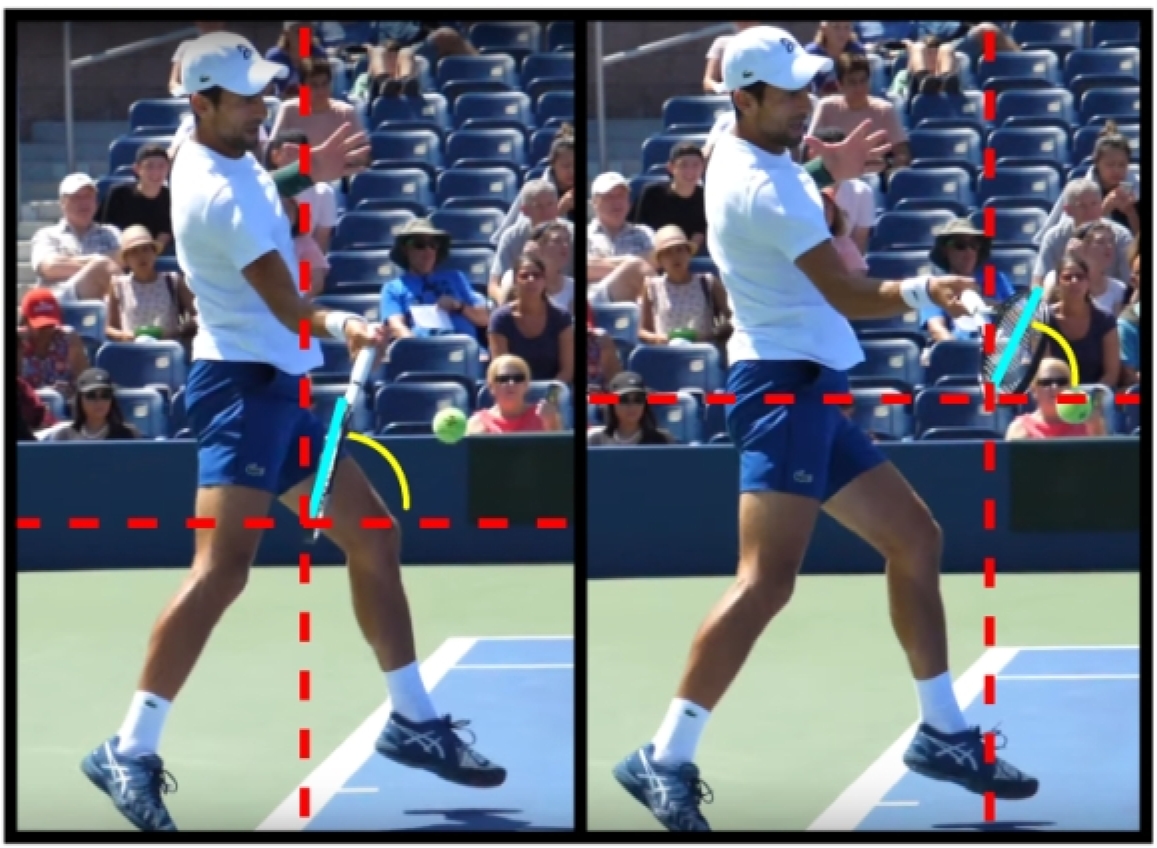
This is why you naturally tighten up when you know you need to make a shot; your brain understands that a tight wrist isn't ideal, but at least you'll be able to maintain a string angle that will keep the ball in play.

But relaxation *is the holy grail* of forehand topspin and power, so it's a worthy goal to practice your string angle maintenance with a relaxed wrist.

Okay, The Angle Actually Changes a Little

So you've finally found a comfortable, whippy, natural feeling swing where the string angle isn't changing very much through the hitting zone. You notice, though, that despite your best efforts, the angle *is still closing a little* as the racket comes around.

That's fine. On a correct forehand, the string angle typically does roll over *a very small amount* through the hitting zone. This is a *very* muted effect and it *certainly* isn't volitional.



Novak Djokovic's string angle closes by only about 10º from the time right before contact, to the time right after contact (from approximately 75º to 65º from horizontal).

Let me clarify the three differences that distinguish this small, acceptable angle change from the typical one that destroys the forehand's fault tolerance.

1. It's Non-Volitional

This angle change happens completely naturally as a result of other forces during the swing. The player is not at all *trying* to change their string angle through contact – they aren't *trying* to "roll their racket over the ball."

That means that this angle change never results in any unnecessary tension in the wrist, never hampers racket head speed, and naturally adjusts itself appropriately based on the exact situation of the stroke.

2. Correct Angle is a *Range*

There does not exist *one* specific string angle at which the stroke must be contacted. Rather, there exists a *range* of correct angles, contact at any of which will lead to a decent shot.

A slightly closed angle will cause the ball to fly lower over the net and land shorter, while a slightly open angle will cause the ball to fly higher over the net and land deeper. Neither one necessarily means you'll miss.

The small angle change that occurs naturally isn't big enough that the angle of your strings *leaves* this acceptable angle range – it's still the case that if you're a little early, or you're a little late, your stroke will *succeed*. As a result, your stroke is still fault tolerant, and thereby still trustworthy.

3. Contact Height Affects Desired String Angle

On a typical shot, the ball is moving *down* through the hitting zone and the racket is moving *up* through the hitting zone[5]. In this situation, the very slight angle change through contact can actually *help* the ball stay in when the swing is mistimed.

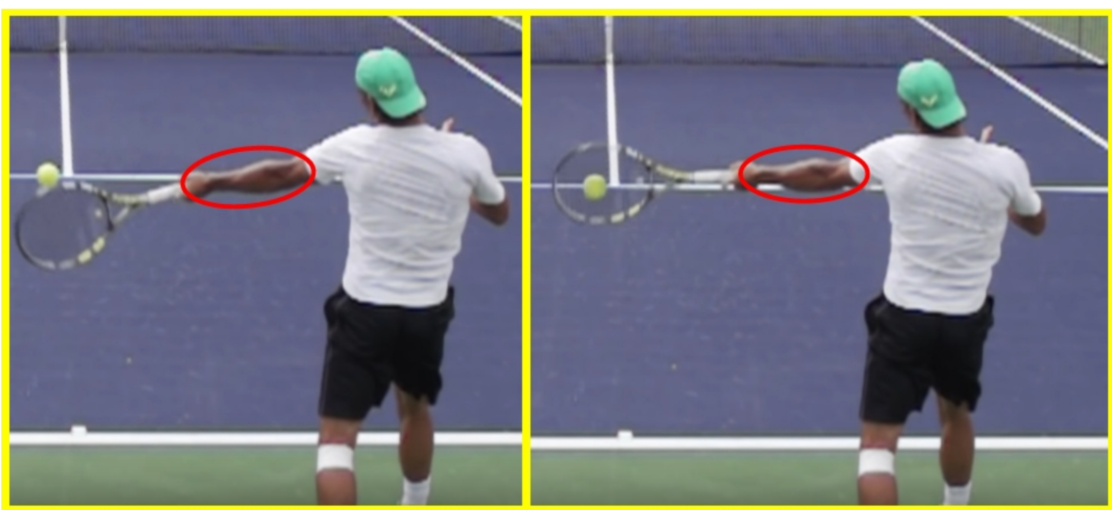
If you contact the ball early in the swing by accident, that also means you contact the ball higher, and in that case you actually *want* a slightly more closed string angle. And, similarly, if you contact the ball late by accident, you'll contact it lower and want a slightly more *open* string angle. In both cases, a slight, non-volitional rolling over often actually helps.

5 This is different if you're hitting on-the-rise, which is one of the reasons that on-the-rise shots are more difficult.

Putting "Roll Over It" to Death Once and For All

Topspin isn't generated by "turning the racket over." It is generated by an upward frictional force on the back of the ball. The racket may in fact turn over during the forehand, but this is part of the stroke's *follow-through*, *not* part of the forward swing. Additionally, it's also true that the racket *does* in fact rotate through the hitting zone, but this rotation happens in a plane that *does not* effect the angle of the strings, and therefore it's not a rotation which detracts from the stroke's fault tolerance.

The rotation whereby the racket "rolls over through contact" constitutes racket rotation *about the handle.* This is the kind of rotation that destroys fault tolerance. The rotation that *should* occur, on the other hand, is rotation about the axis of the *forearm*[6]. When the racket flicks around *about the forearm* through the hitting zone, it gains extra vertical speed through contact *without* having its string angle change dramatically. When the racket is rotating in this manner, you can still hit a little early, or a little late, and have your shot go in.



Left *– Before contact; the string angle is such that, even if contact is made in this position, the shot will likely go in, albeit off-target left.*

Right *– At contact; note the entire racket-hand-forearm system rotates together, leading to a minimally changed string angle through the contact zone.*

Lastly, this rotation is *passive,* not active. It doesn't happen because the player is using any of

6 The rotation isn't *exactly* about the axis of the forearm, but it's very close. As we've discussed, the racket does turn over *a little,* and the forearm-racket angle will often *passively* open up by about 10-30º through the contact zone. Again, all of this happens as result of the relaxed hand, and shouldn't be performed consciously.

the small muscles in their wrist or hand to actively force the racket around. Instead, it's simply a function of *other* forces generated earlier in the swing. The wrist merely relaxes and acts as a hinge to *transfer* these other forces into racket rotation – it does not *create* any of that force itself.

Luckily, Bad Habits Can be Fixed

Our girl from the beginning of this section did get somewhat of a happy ending. After two months of working together, her strokes were *vastly* improved, and, even more importantly, she ended each session pain free. Our average mini-tennis rally length increased from less than three shots at the beginning, to roughly twenty by the end. She was quite fond of mini-tennis once she'd mastered her new, simpler, fault tolerant contact.

Her feel for the ball in unique situations became much better, and, now that she was actually practicing the *right* stroke, her consistency improved as we drilled. Given where she could have ended up if we'd never connected, though the ride was rocky at times, I'd say things worked out pretty well in the end.

A Brief Comment on Grip

In order to facilitate fault tolerant contact, most players use some version of the Semi-Western grip, because, for most, it's a grip by which it's very natural to create the slightly closed racket face that's required. You might notice that we've just dedicated 20+ pages to discussing forehand contact and are only just now mentioning grip at all. That's because, when it comes to coaching the forehand, grip is mostly a red herring.

Contrary to what you've been told, the grip is *not* an important part of the forehand. If it were, you wouldn't have such a wide variety of grips on the professional tour. Rather, it's the orientation of the racket face at contact which is important. A player can create this orientation with an Eastern, Semi-Western, or Full Western grip.

It doesn't matter.

Most students will be better off using whatever grip feels most natural to them, than they will be attempting to consciously force themselves into one specific "superior" grip. Any time spent forcing a different grip would likely be better spent practicing the stroke itself while using the natural grip.

When it comes to coaching the forehand, grip is mostly a red herring.

If I *were* forced to choose, I would say that the Eastern and Semi-Western grips are slightly superior to the Full Western grip. The Eastern grip specifically is a very underrated grip, because it's impossible to hit a quality forehand with an Eastern grip *without* a complete understanding of how the forehand actually functions. This naturally leads beginning tennis players away from it, because, at the beginning, we all have significant flaws in our swings. With the Semi-Western and especially with the Full Western grip, you can get away with certain bad habits for a lot longer.

That said, if Fabio Fognini's beautiful Full Western flat forehand proves anything, it's that grip really is *truly subjective.* Each athlete will have their own grip with which their forehand functions best, and they should use it.

You Can Now Hit a Fault Tolerant Forehand

Wait, what? Are we done? We haven't even gotten to the unit turn or hip rotation. We haven't discussed driving with the legs or whipping the chest around?

That's true, because all of those things act to *enhance and improve* the forehand stroke. They make it better, but, fundamentally, they aren't *necessary.* And each of those improvements mentioned in that first paragraph must actually be abandoned in certain situations. If you're sprinting at full speed for a ball, you can't use hip rotation, because your legs are busy running. If a ball gets blasted at you, you don't have time for a big chest rotation. If you try, you'll almost assuredly catch the ball late and miss wildly.

The preceding discussion of fault tolerant contact constitutes all the information that's *universally* applicable to every forehand, at every level. Everything that comes hereafter concerns how to make the stroke *better.* The following sections explain movement patterns that you can implement to drastically increase your racket head speed while *still creating this same fault tolerant topspin contact*. The following techniques are *only* useful if they can provide this speed increase *without* sacrificing that fault tolerance.

The above discussion of fault tolerant contact constitutes all the information that's universally applicable to every forehand, at every level.

Luckily, in the modern, rotational forehand stroke, that's exactly what we have – movements that allow us to explosively whip the racket around our body and create fault tolerant topspin contact extremely consistently *and* at a breakneck speed. To learn how to recruit the entirety of your musculoskeletal system to accelerate the racket through fault tolerant forehand contact, read on.

Interlude

The Rotational Kinetic Chain Provides Unparalleled Speed

Why Rotate?

Rotation is the engine of the modern forehand. By violently twisting the body around, a player can utilize the entirety of their musculoskeletal system, specifically the strong muscles of the legs, hips, and abs, to accelerate the racket. During the modern forehand, we turn *away* from the ball during preparation so that we can explosively rotate *back towards* it during the forward swing. The arm whips around the body as this happens, and ultimately the kinetic energy from our trunk rotation is transferred into the racket.

Engaging this rotational chain vastly increases racket head speed through contact, thereby also vastly increasing how hard we can hit. Further, it's very easy to transfer some of this increased racket head velocity into *vertical* racket head velocity, which creates topspin.

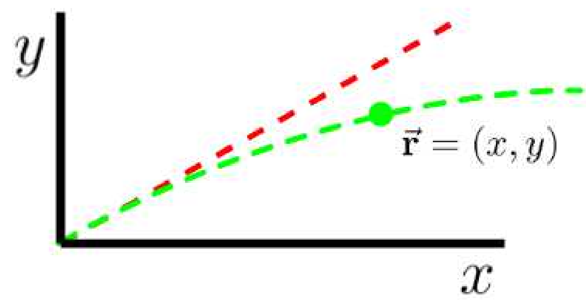
By relaxing the hand as we rotate, we allow the racket to flick around the wrist through the hitting zone – this motion, aptly named the "windshield wiper" motion, occurs passively when our hand is relaxed during the swing. The result of relaxing and allowing this racket rotation to occur is that the racket is moving extremely quickly *vertically* through contact, dramatically increasing topspin potential.

The kinetic energy from our trunk rotation is transferred into the racket during the swing.

By rotating into our swing, we create so much extra racket head speed that we can both hit harder *and* generate more topspin, and by using the largest muscles in our body to power this rotation, we can implement it without impairing the fault tolerant contact that we discussed in Part 1.

The Brain Behind the Scenes

The human brain is quite a remarkable piece of biological machinery. It's capable of storing about 2.5 petabytes of data, and it processes roughly 10^{15} logical operations per second, 100 times more than the iPhone 12. Much of this computing power is dedicated to balancing and moving its unbelievably complex body through space. Before we continue and discuss the rest of the rotational kinetic chain, we first need to pause and briefly address the different manners by which the brain "understands" athletic movement.



$$\overrightarrow{\mathbf{F}_{\mathbf{drag}}} = -\frac{1}{2}\rho C_D A|\vec{\mathbf{v}}|^2\hat{\mathbf{v}}$$

$$m\vec{\mathbf{a}} = -mg\hat{\mathbf{y}} + \overrightarrow{\mathbf{F}_{\mathbf{drag}}}$$

$$\frac{d^2\vec{\mathbf{r}}}{dt^2} = -g\hat{\mathbf{y}} - \frac{1}{2m}\rho C_D A\left|\frac{d\vec{\mathbf{r}}}{dt}\right|\frac{d\vec{\mathbf{r}}}{dt}$$

$$\vec{\mathbf{r}}(t) = ?$$

Each time you throw, your brain is solving this differential equation many times, using different values for initial velocity. Once it finds a trajectory that matches your goals, it signals the arm (and the rest of the throw chain) to produce the amount of force necessary to launch the ball at that initial velocity. To you, it just feels like "aiming."

When you do something like, say, throw a baseball, hundreds of muscles are all coordinating together to produce the result. When you're trying to aim your throw – say, at a catcher's glove – your brain uses the visual information from your eyes to perform a real time calculation of the ball's projected parabolic flight path, and then sends specific signals to those hundreds of muscles to create that flight path upon release. Further, let's say you miss your target, and now you need to adjust. Your brain re-performs a *new* parabolic trajectory calculation based on the differential between the expected and the observed result, and then sends *new* signals to those hundreds of muscles to alter the throw.

But what does it feel like to you? It just feels like throwing. Throwing, and aiming, and maybe correcting your aim by "aiming more to the left," if you missed off to the right. All of this unimaginably impressive computational work happens independent of your perceptions. Your *brain* is doing this, but the part of you that you consider you – that part doesn't actually participate in these calculations. That part just feels like you're "throwing."

Understanding vs Application

In our remaining discussion of the forehand, we'll be discussing primarily how we can use rotation to enhance our fault tolerant contact. Before this discussion, it is important to draw a clear distinction between the two different types of forehand competence. They are:

1. Being able to *produce* a quality forehand
2. *Understanding* what movement patterns produce a quality forehand

There's clearly a lot of overlap here; most people who understand the movements that produce a quality shot can, in fact, produce that shot. Likewise most people who can produce a quality shot understand the movement patterns that got them there, but the overlap is not 100%.

Further, even if both competencies are possessed at a baseline level, one of the two can still be more complete than the other. A very natural, athletically gifted player may be able to produce a flawless stroke with only a rough understanding of the movement patterns they're using, while less athletically natural, intellectually gifted player may possess complete knowledge of what their body is *supposed* to be doing, but can't quite get it to actually *do* that.

During the stroke production itself, almost all muscle level analysis should be ignored.

For what we've discussed to this point, these competencies are roughly aligned. Actually envisioning *the contact you're trying to create* is often a useful endeavor. For other aspects of the swing, too, there's significant overlap between the two kinds of competence. For the most part, utilizing hip rotation is about as simple as *trying to use hip rotation.* For other movement patterns, though – things like engaging the abdominal rotators and stabilizing the racket with the pectorals – these things manifest significantly differently between the two competencies.

As we move on with our discussion of how the rotational kinetic chain can be implemented to enhance forehand contact, some of the things we'll talk about won't be useful when actually *producing* shots on court. The exact manner by which the rotational kinetic chain functions is for your education and for your *observation* of swings. It isn't supposed to be something you actually *think* about to *produce* those swings. In the same way it wouldn't be possible to teach a baseball player to throw by instructing them how to precisely time a contraction of each of hundreds of muscles, it's not possible to teach a tennis player to hit a forehand by having them consciously contract the links in their kinetic chain in some precise sequence.

We discuss the technical aspects of the movement because a theoretical understanding of the what's happening can help us observe and adjust anything that might be wrong with it *after the fact.* During the stroke production itself, almost all muscle level analysis should be ignored. Instead, each athlete needs to discover what activating certain muscles properly *feels like* to them. As coaches, we can try to guide them along the way, but, ultimately, in order to master a complex movement pattern like a forehand, each athlete needs to map their own personal proprioceptive observations onto physical results – they need to figure out what it feels like to do it right, and then do that over and over.

As a last comment on conceptual understanding, although we do go into biomechanical detail here, we still don't go into any *deep deep* detail when it comes to specific muscle activation – the most specific we go in this book is referring to large muscle *groups,* like the "hips" or the "abdominals" – any discussion of what a *specific* muscle is doing during the swing would be surely be purely academic.

Essential and Non-Essential Movement Patterns

The rotational kinetic chain is complicated, and every athlete will implement it into their stroke in a slightly different way. Every athlete's body is different. Every athlete's tendencies are different. That does *not* mean that there isn't a *right* and a *wrong* way to do things. The subtlety here is that "right" and "wrong" only exist *after* a certain goal has been chosen. Right and wrong are *relative* to the shot, or in general, to the movement pattern that's being attempted.

Here's a simple example from sprinting – if your goal is to accelerate from a stopped position, a hinge at the hips is necessary; the lower you keep your body, the faster you can accelerate. If your goal is to maintain a top speed you've already achieved, then an upright posture is better. If you were to abbreviate both of these situations as merely "trying to run fast," you'd be utterly lost as to whether an upright posture is "right" or "wrong."

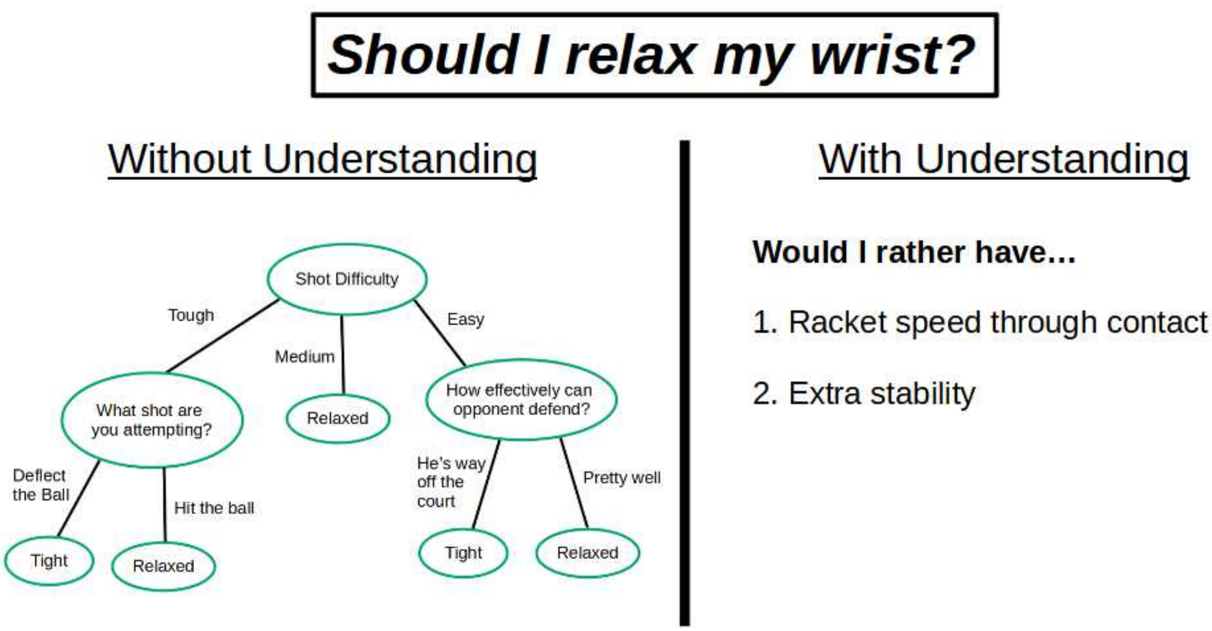
Tennis is the same way. Movements and techniques are either essential or non-essential depending on the stroke being attempted. For example, let's take the hip twist portion of the forehand. The hip twist is *fundamental* to the rotational chain. It doesn't matter how a player hits stylistically; if they want to engage their entire body in the racket head speed generation process, they *must* twist their hips away from their target during preparation and then back towards their target as they swing. If they do that, it's *correct,* and if they don't, it's *incorrect.* There's no leeway here, no room for player preference. The hip twist is *core* to the rotational chain.

This doesn't mean, though, that a player can never *elect* to abandon hip rotation altogether, and thereby forgo some of their rotational power. Perhaps they've decided that their situation calls for a simpler, less powerful stroke. A non-hip-engaged chain will always produce less power than an analogous stroke which engages the hips, but, especially under certain circumstances (like striking a forehand while on the dead run), this sacrifice may be appropriate. In one context, generating the most racket head speed, a hip twist is *essential,* but in another context, say, blocking a 130 mph serve back into play, hip rotation is *non-essential.*

Fundamental Understanding is Paramount

Almost no stroke mechanic is *always* good or *always* bad, even something as infamous as wrist tension on the forehand. Imagine that your opponent is 30 feet off the court. They've hit an extremely short, floating ball back to you, and all you need to do is tap the ball over the net to win the point. When the goal is merely to tap the ball over the net, there's no reason for relaxation. We *don't want* excessive kinetic energy transferred into the racket, and we *don't want* the wrist to act as a hinge like it typically does. All we want is an extremely stable, predictable, racket-arm system that will deflect the ball over the net 100 times out of 100. In this very niche situation, wrist tension is totally fine. The optimal strike for this particular forehand is more similar to that of a volley than it is to our typical, rotational, explosive stroke.

These kinds of distinctions are what make coaching from any foundation *other than* a fundamental understanding of cause and effect so difficult. Without a clear understanding of the *why,* technique decisions become an incoherent if-else tree of cases, conditions, and exceptions.



Almost any specific technique can and should be voluntarily abandoned in certain situations, thereby sacrificing whatever advantage it created in order to gain some other advantage. These kind of voluntary sacrifices make a coach's job quite difficult. Did the student intentionally forgo their hip rotation, or did they do it by accident? Sometimes, students themselves don't even have a great sense of what their body is doing, so it falls on the coach to determine which movement patterns are being abandoned intentionally, and which the student really would be better off employing.

The Two Failure Modes to Avoid

In summary, the issue with so many coaches is that they don't know which aspects of a stroke are essential, and which are non-essential, and they don't know how the relative importance of these aspects varies as the specific goal for the shot changes. This leads to two failure modes.

1. The coach unnecessarily and harmfully corrects things that need not be corrected, things like the personal stylistic choices of the player.

2. The coach *fails* to correct things that *really do* need to be corrected, ascribing those things "personal stylistic choices," when really they're just flaws.

Good coaches work *with* a player's natural tendencies, not against them. If the coach understands the *purpose* of each of the mechanics that they're teaching, they'll be able to determine what is *fundamental* to a particular stroke attempt, and what is discretionary. Further, they'll be able to discriminate between any discretionary movements tacked onto an otherwise sounds stroke. Some of them won't matter much at all, while others may actually *impair* the stroke when added. Hand positions, feet positions, grips – they're all *right* in one context and *wrong* in another. A coach should never force their student into a specific style, when a different style, a style to which the student naturally gravitates, could accomplish their fundamental goal just as effectively

This doesn't mean a coach *never* needs to force their student to change *anything.* There are still wrong answers. If a student is attempting an explosive swing, they shouldn't be under-rotating. The wrist should be relaxed for all high velocity shots. These, among other things, are not discretionary techniques – a student's forgoing them doesn't constitute "personal style," but rather technical inefficiency. It falls on the coach to identify these kinds of mistakes, mistakes which *must* be corrected before an elite stroke can be produced, and fix them.

Part 2

Forehand Preparation

Preparation is *Solely* a Means to an End

Preparation *only* exists because our ready position isn't an optimal starting position to generate power or spin on the forehand – if it were, we'd just use that ready position instead going through the rigmarole of turning sideways and putting our hand back. Forehand preparation is *entangled* with the forward swing. It should not be thought of as independent, its own movement pattern separate from the rest of the stroke.

We prepare for our forehand in a particular way to *facilitate* certain movements. We have a goal in mind, a certain kind of rotational movement pattern that we want our body to attempt, and the role of forehand preparation is to get our body into a position such that we can execute that movement pattern.

In general, the simpler the preparation, the better.

That's all preparation is. It's a means to an end. "Turn sideways," "prepare your hand," "left arm across your body" – these aren't *rules.* These are simply movements that better allow us to perform the forward swing like we want to. In some situations, we'll have to abandon these techniques, while in others, we can exaggerate them. The important concept is *not* the preparation techniques themselves, but rather *what* they allow us to do *later* in the swing. That's what we'll explain in this section. Internalize that idea, and you'll be able to dynamically adjust your preparation to fit the situation, without having to think about it.

The Two Goals of Forehand Preparation

In order for a player to explosively rotate towards a target, they need to first turn *away* from that target. For the core movers of the body – the legs, hips, and abdomen – turning away is all that's happening during forehand preparation. The player is turning their trunk away from the target so that they can explosively rotate back towards it during the forward swing.

In addition to turning away from the target, the player must prepare their racket. The goal is to position the racket such that, when the trunk explosively rotates back towards the target, the racket will be able to absorb that energy and whip around the body unimpeded – if the racket is prepared successfully, all of the force from the prime movers' rotation will be transferred into racket head speed during the swing.

Those are our **two goals** during forehand preparation:

1. Turn away from the target, so that we can turn back towards it.

2. Start the hand in a location such that it can whip around the body freely.

In general, the simpler the preparation, the better. The above two goals are *all* that preparation has to accomplish. In the following sections, we discuss various common and efficient techniques by which these two goals are accomplished.

That's the *why*; now let's move on to the *how.*

The Unit Turn

The “turn away” part of forehand preparation is typically named “the unit turn.” This is a good name, because the entire body turns together during this phase of the stroke.

The hips turn to the side, away from the target, so that they can explosively twist back towards it during the forward swing. The shoulders, arms, hands and racket all come along for the ride – the entire body turns away as a *unit*.

Robin Soderling was famous for taking a massive shoulder turn – turning his shoulders almost 90º farther than his hips – and crushing forehands because of it.

Most players will turn their shoulders farther than their hips, leading to greater abdominal engagement during the forward swing. You can think of it as further winding the kinetic coil, thereby allowing for even more explosive torque when that coil finally unwinds. The farther a player turns their shoulders relative to their hips, the harder they can hit, but also the harder the stroke is to control.

Each player needs to experiment and decide what level of shoulder turn is appropriate for their particular game. The more turn, the more coordination and timing it takes to handle the rotational energy you’ll generate, but also the faster you can swing.

Dynamically Adjusting Your Turn

The “unit turn” is not a singular, static entity. Rather, it’s merely a name for a *class* of body movements roughly described as turning the trunk away from a target. The entire point of the unit turn is to turn *away* from the target so that we can explosively turn *back* towards the target when it’s time to swing. This means that if your goal *isn’t* to rotate as you swing – perhaps you’re jammed and just need to block the ball back – then an extremely abbreviated unit turn, or no unit turn at all, is appropriate.

Additionally, this means that you only turn the parts of your body you intend to use for the swing. Let's say you're on the dead run – you can't use your hips for power, because they're busy sprinting. On a forehand like that, *only* your upper body should turn away from the target during preparation, because only your upper body is available to explosively turn back towards it during the swing. A swing performed this way will generate almost all of its power from abdominal rotation, rather than getting a significant amount of it from hip rotation like usual.

On this running forehand winner, Fernando Verdasco's hips are completely occupied by sprinting to his left. Due to this, his hips do not participate in the swing by rotating; he turns only his chest away from the target during preparation (left) and then turns only his chest back towards the target during the forward swing (right).

The Arm, Hand, and Racket

As the hips and chest turn to the side, the arms, hands, and racket move "back" in an absolute sense – they got closer to the back fence – but *relative* to the rest of the upper body, they remain mostly still. The apparent motion of the arms, hands, and racket is merely a *result* of the turning of the hips and chest, *not* a result of consciously moving those body parts themselves.

During a proper unit turn, the arms, hands, and racket move *very little* of their own accord, and all of that movement is *up* and *out* movement; hardly any of it is *backward* movement. Their final position relative to the upper torso should be very similar to what it was in the ready position: the hands will still be in front of the chest, holding the racket loosely. The only difference is that the hands may be higher and farther from the body.

Rafael Nadal does three things during his unit turn – he turns his hips and shoulders away from his target, he raises his hands up to shoulder height, and he separates his hands from his trunk, creating space between his elbows and his chest.

Extra, unnecessary motion of the hands and arms during the unit turn is the cause of *a lot* of forehand inconsistency worldwide. The unit turn is fundamentally a turn of the *hips and chest*; it's totally fine to move the hands away from your body and to move them up a little, but, with respect to moving *back,* the upper body extremities are just along for the ride.

A Few Notes on Arm Movement

Before we move on to the racket take back, let's discuss two small caveats with arm "movement" during the unit turn.

First, the height of the hands during the turn can vary from player to player. It's okay to turn with your hands anywhere from chest to head height. Moving the hands vertically to the height from which it feels comfortable to start the takeback is totally acceptable.

Second, if the player likes to keep their hands closer to their body in the ready position, it *is* important to get them *away* from the body at some point before the forward swing, and the unit turn is a good time for this. One of the primary goals of forehand preparation is to prepare the hand to flick around the body unimpeded; to do that, it needs to be *away* from the body.

By the end of the turn, both elbows should be at least partially elevated and away from the body. This position will allow the player to remain balanced as they whip their torso around during the forward swing, and the space between the hitting elbow and the trunk will allow the hitting arm to whip around the trunk unimpeded during the subsequent forward rotation.


At the time that Roger Federer's left hand leaves his racket, his left arm is fully extended across his body, which has forced his entire chest to turn to his right.

The Non-Hitting Arm as a Cue

Many players have a harmful instinct of preparing their racket by *moving their arm back*, rather than by turning their entire body away from their target. To combat this, focus on turning *with your non-hitting arm.* If you ensure that your non-hitting arm is stretched *across* your body as you turn, that'll *force* you to turn your *entire* body sideways – you won't feel that non-hitting arm stretch if you instead just pull your hitting-arm back.

Additionally, it can help to keep both hands on your racket until you're fully turned. Only once you've finished your turn and are ready to extend your elbow do you allow your non-hitting hand to come off the racket. Again, this will *force* you to turn your *entire* upper body sideways, which is what we want.

Preparing the Hand

After the unit turn, the trunk has been turned away from the target and the elbows are separated from the chest. At this point, the next step is to prepare the hand for the forward swing. Unless the ball is exceptionally low, this typically entails lowering the hand below the height of the ball, such that it can easily flick *up* through the ball through contact.

There are three good ways to lower the hand, and one bad way. Lowering the hand by sitting down lower with your legs, by tilting your torso, or by extending your elbow will all preserve the swing's efficiency. What *won't* work is lowering the hand by bringing the arm in closer to the body. That arm positioning makes it far more difficult for the racket to whip around the body freely during the forward swing, and you'll lose both power and topspin by adjusting that way.

"Backswing" Confusion

The way swings look on TV is misleading (and *especially* the way backswings look). That misleading visual is the cause of *a lot* of forehand backswing misconceptions. In fact, even the term "backswing" itself is misleading, because there's no real "swing" happening during the backswing. The movement is more accurately described as something like "placing the racket behind the ball and preparing for the forward explosion."

It doesn't *look* like that on TV, though. It *looks* like the players are using their arm to take the racket back in a big circle, and then they're using that big circle to generate racket head speed. It's hard to tell exactly which muscles they're firing when. It's hard to tell exactly what's doing the work of accelerating the racket, and what's merely a passive movement happening as a result of other actions.

Even the term "backswing" itself is misleading, because there's no real "swing" happening during the backswing.

In the following sections, we'll clear up what a proper "backswing" should look like.

Wrist Lag is NOT Backswing

Once you lower your hand to a satisfactory level, your hand preparation is *done;* the next time your hand moves, it will do so as a passive result of being pulled by bigger muscles during the forward swing. Instead of stopping here, though, many players attempt to *force* their hand into the "wrist lag" position using conscious effort. They incorrectly believe that placing the hand into the wrist lag position is part of forehand preparation.

It is not.

On our blog, faulttoleranttennis.com, we have a section called the **Coaching Cues Hall of Shame.** It consists of harmful coaching cues that well-meaning coaches will ignorantly tell their students, cues which typically end up hurting those students' games more than helping them. "Point your butt cap at the net" is one of such cues. It creates harmful wrist tension and destroys racket head speed, and thereby topspin. The butt cap *does* point at the net early in the swing, but that position is a *dynamic position*; it should not be entered volitionally.

TV Can Be Deceiving

As we've already discussed, many bad cues originate from watching the pros; to the untrained eye, it *looks* like Rafael Nadal is turning his racket over through contact, even though he isn't. As we watch on TV, there are two fundamental information barriers that routinely lead to bad coaching cues.

1. **It's difficult to tell the exact timing of the movements that we're seeing[7].**
2. **We can't tell what the player did *intentionally*, via conscious effort, and what happened *passively* as an unconscious result of those intentional actions.**

The butt cap does point at the net early in the swing, but that position is a dynamic position; it should not be entered volitionally.

7 As an example, which we'll explore further in **The Forward Swing**, Rafa's strings *do* turn over, but they turn over as part of the *follow-through*, not as part of the swing itself.

The Butt Cap Myth

The “point your butt cap at the net” fallacy originates from the second barrier. It’s cued because, at a certain point in the swing, every professional player’s butt cap *is* in fact pointed towards the net.

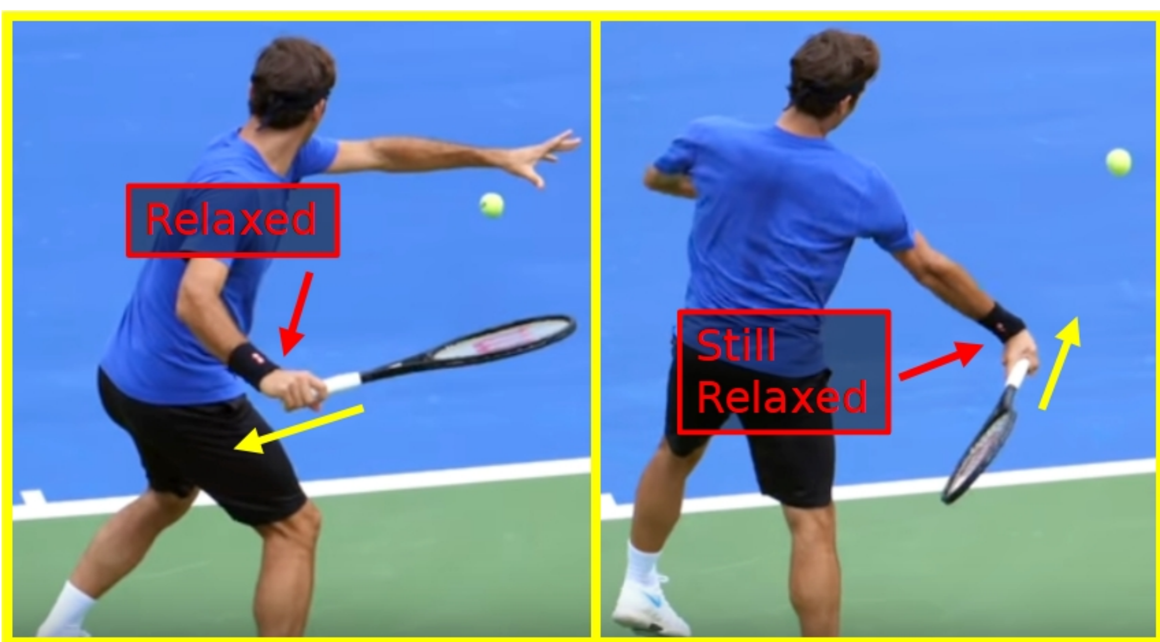
The problem? This butt cap pointing is *passive*, not active.

When our wrist is properly relaxed during the take back, our butt cap will point *towards our body*, not at the net. In order to force the butt cap to point at the net instead, we would need to tighten and extend the wrist and pre-rotate the forearm. Doing this both ruins the swing and can get you hurt.

The Butt Cap Reality

Adding tension to force the butt cap towards the net ruins our wrist’s performance as a hinge, and thereby causes our racket head speed to crater.

When, instead, we relax during the swing, the racket will *naturally* lag backwards as a result of inertia; it wants to stay at rest when your bigger muscles pull it forward. This natural, *non-volitional* lag is what results in the famous "wrist lag position" that’s so common on the tour, not any sort of intentional action.



Roger Federer, a great example of proper wrist relaxation and wrist lag on the forehand, has a relaxed wrist at the end of the backswing (left), and a relaxed wrist during the forward swing as well (right).

The wrist lag position is a position that's entered *without* conscious effort, and it only occurs *during the forward swing.* It does *not* occur during preparation. Attempting to consciously force your hand into the wrist lag position prevents the wrist from effectively *transferring* the force from your body's rotation into the racket, and *transferring* force is the wrist's job during the forehand. The small, relatively weak muscles of the wrist and hand do not *create* any racket head speed on their own.

So *ignore* where your butt cap is pointing during your stroke.

It's not a useful piece of information. When you take your racket back, relax your hand. When you throw it **out**, **up** and **through** the ball, maintain that relaxation. Use just enough tension to maintain a consistent string angle through contact, and to prevent the racket from flying out of your hand. Other than that, keep the wrist loose, throw it in the right direction, and it will perform properly during the swing.

Hand Preparation and Elbow Extension

Unlike wrist lag, elbow extension during hand preparation is one of the few movement patterns that can be done either volitionally or non-volitionally. The hitting elbow will always extend (at least partially) at some point, but when it does is up to you.

Some players, like Roger Federer, elect to mostly extend their hitting elbow consciously during the backswing, and then allow it to fully extend as a result of momentum during the forward swing. Other players, like Rafael Nadal, do not extend their elbow during the backswing and instead allow all the extension to happen as a result of momentum during the forward swing.

Roger Federer (left) extends his as part of his preparation, whereas Rafael Nadal (right) elects to keep his elbow bent before initiating the forward swing.

No matter how a player elects to handle their elbow extension, again, notice that there's no real "swing" involved in this part of the backswing. Either the player waits and allows the momentum from their rotation to extend their elbow, or they extend it before they swing and essentially *place* the racket back.

Also notice that the racket's position in the hand has barely changed from the ready position, even after the elbow is extended. The wrist, forearm, and hand don't *do* anything during forehand preparation – they aren't actively manipulating the racket's position at all. Instead, these small muscles and tendons in the arm merely serve to hold the racket while the body turns, and while the elbow (for some players) extends to push the racket back into its starting spot.

Elbow Extension Recommendations

If a particular amount of elbow extension feels natural to you, use that. For most players, though, they can mess around with how much they extend their elbow before initiating rotation, and they all feel kind of similar. If you're in that camp, I'd recommend using something close to what Roger Federer uses – Nadal's fully cocked elbow in the backswing does create some extra whip through the hitting zone, but it's harder to control.

Federer, on the other hand, uses a simpler stroke, and he leverages that simplicity by taking the ball extremely early when he wants to be aggressive. This style – the style by which you use a more abbreviated swing and create offense with timing and angles, rather than pure racket head speed – tends to fair better for most players.

Elbow Extension and Returning

Often, there's no need to extend your elbow at all on a forehand serve return (or when returning a similar fast, flat shot). Many players who typically play the forehand by allowing their elbow to extend through the hitting zone, often play the forehand *return* by keeping their elbow bent. This greatly abbreviates the swing, making it easier to time. When the elbow stays bent, the arm and pectoral muscles do very little to accelerate the racket during the swing, and instead the racket-arm-chest position stays the same through the entire rotation.

Novak Djokovic returning a body serve with a significantly bent elbow on the forehand.

This technique is especially useful for returning a body serve. On that serve, the returner usually doesn't have time to get out of the way of the ball. As a result, they are forced to make contact far closer to their body than is necessary for the typical rotational whip swing, during which the elbow extends by momentum.

Avoid *Fully* Extending Before Swinging

Almost every player keeps at least some degree of bend in their elbow before rotating forward, and you should too. Tennis is a very explosive game, and the forehand is a very explosive movement. In order to protect our bodies during these explosive movements, we want to allow each of our limbs and joints maximal *freedom of movement.*

Locked limbs are vulnerable limbs.

If the elbow is *fully* straightened during the backswing, then any additional force which acts to further straighten it during the forward swing will act to *hyper-extend* it. Additionally, if it's fully straight during the backswing, and then you realize you need a bent elbow to return something like a body serve, you'll need to quickly and awkwardly use your bicep, forearm, and hand to pull the racket into that bent elbow position, which is far more awkward than simply using the bicep to *stabilize* the already bent elbow to prevent it from extending as you rotate.

In general, locked limbs are vulnerable limbs. It's the same reason we never play tennis with straight knees, and we always maintain a slight hinge at our hips as we move. Small angles ensure that our *muscles* are the entities absorbing the various forces generated while we play, rather than our tendons, ligaments, joints, etc. Maintain a small amount of elbow bend while you prepare the hand, and your arm will have plenty of freedom to adjust itself to whatever forces happen to be generated during that particular stroke.

Using a Fully Bent Elbow

Some players never extend their elbow – they elect to maintain close to 100% of their elbow bend throughout the entirety of the forward swing. The players who keep this bend for their entire stroke sacrifice a small amount of racket head speed for added consistency – by not allowing momentum to extend the elbow through the stroke, the stroke is simpler and the core movers do more of the work. The players who allow their elbow to extend as they swing create a little more whip through contact, at the cost of a stroke which is slightly more difficult to control.

Jack Sock blasting an inside-out return winner using his explosive bent arm forehand.

Both work, and elite forehands of both kinds exist. Often, bent elbow users actually hit *harder* than their elbow extended counterparts, because the simpler motion allows them to inject more *effortful* power into the stroke while maintaining control. It really is a matter of personal preference.

Similar to our discussion of grip, if I were forced to recommend one over the other, I would certainly recommend allowing the elbow to straighten through the hitting zone. The straightening creates a longer arm, and thereby a longer lever, for transferring the rotational force from the body into the racket. This longer lever allows for more *effortless* power generation, and thus, in the long term, will be more energy efficient overall.

But, again, if Jack Sock's Full Western, bent elbow, rocket of a forehand tells us anything, it's that the above sentiment is very subjective.

Adjusting to Contact

The key to effective stroke production is efficient biomechanical alignment. For every shot, there are certain body alignments which produce racket head speed *most* efficiently and *most* consistently. The more often we can align our body properly before swinging, the higher quality, and the more fault tolerant, our shots will be. Our goal during forehand preparation is to get our body into these effective biomechanical alignments.

> ***On the forehand, our biomechanical advantage is our ability to whip the racket around the torso unimpinged.***

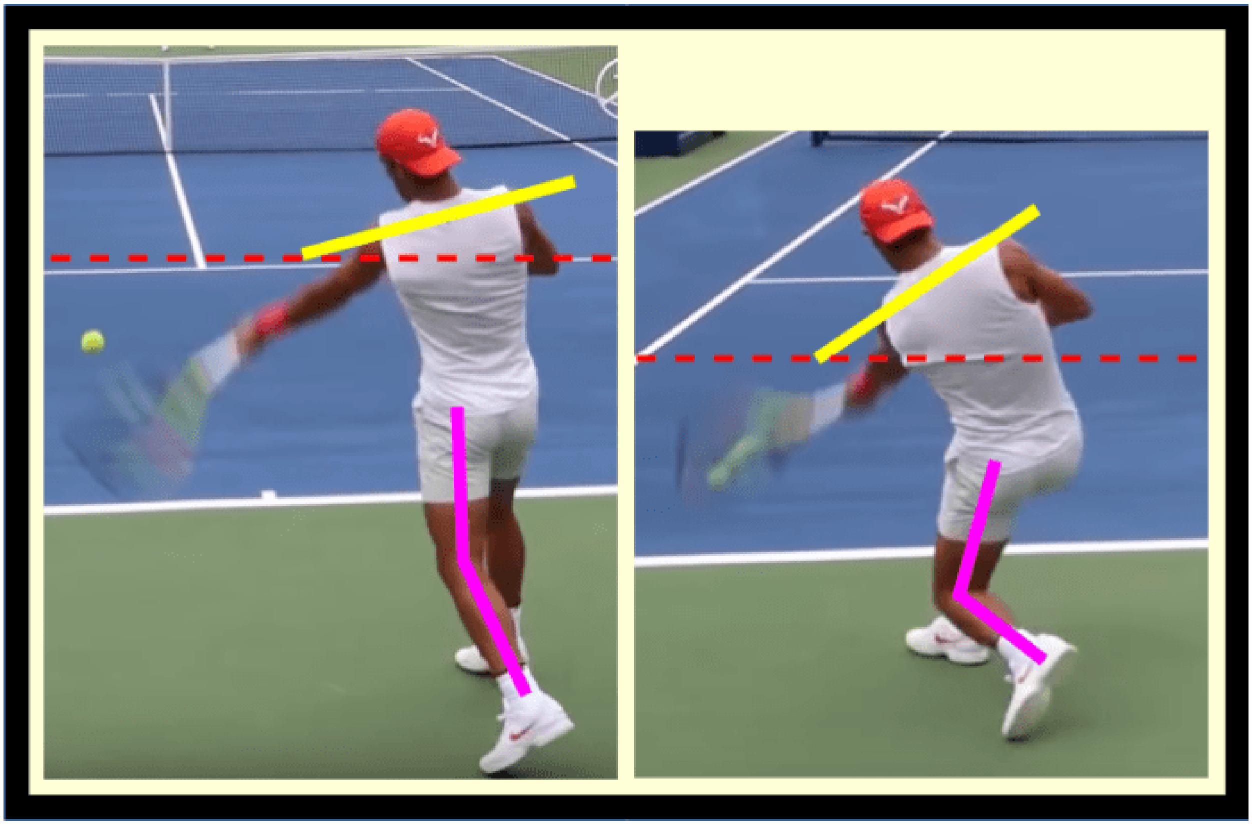
The tricky part is routinely getting ourselves into these positions while playing many different balls, at many different heights, from many different positions on the court. We need to contact the ball in a myriad of different locations, and yet, in doing so, we need to use a remarkably *consistent* relative upper body position.

On the forehand, our biomechanical advantage is our ability to whip the racket around the torso unimpinged. In order to allow this, we need to maintain adequate space between our torso and arm. If the angle between the torso and the arm gets too narrow, the arm won't naturally flick out and around as we rotate our trunk. On a shot like a body serve return, sacrificing this natural flick is necessary – there's not enough time to harness our rotational whip – but on a low forehand, there's no reason to sacrifice this advantage.

Therefore, we need to adjust our contact height *without* significantly changing our arm angle. This is accomplished by adjusting with the larger muscle groups. We always adjust with the largest muscle group we can. This means we have two primary methods of adjustment:

1. **Adjust with the legs, either by sitting down or by widening them.**
2. **Adjust with the torso, by tilting it.**

The vast majority of our forehands will be played with an identical torso-arm angle, regardless of contact height, with adjustments made via the legs and torso. Legs and torso adjustments are the only two effective adjustments, as they are the adjustments that let us alter the contact height without impeding our torso-arm-racket alignment – the alignment which facilitates our rotational whip.



Rafa Nadal adjusting his body for a regular height (left) versus a low (right) contact point. On the low ball, his torso is far more tilted, and his legs are far more bent, allowing him to use the same relative upper body position that he uses on the higher ball.

Suboptimal Adjustments

Adjustments of the arm and wrist are a last resort.

When sufficient adjustment can't be made by using only the legs and torso – perhaps we're on the dead run or the ball is coming so fast that we don't have time to move – then we adjust with our arm (not the hand or wrist).

We can bend it, elongate it, or move it closer or farther from the body, as called for by the situation, but we *do not* violate the fundamental theorem of tennis. Adjusting with the arm is not ideal, but since we still maintain an angle between the forearm and the racket, and since we still contact the ball out in front of us, the shot will still go in. It won't be great, but the rally will continue.

Only as a *last last* resort do we adjust by straightening out our wrist and violating the fundamental theorem of tennis. A common case of this is when the ball is so far away that, even on the dead run, we are *just barely* are able to make contact – we simply need every inch of legs, arm, and racket fully straight in order to reach the ball. If you find yourself in this situation, try to catch the ball in front of you and pray it goes in. The shot is not going to be fault tolerant.

The Fundamental Theorem as a Cure For Improper Adjustment

Focusing on the fundamental theorem of tennis helps many players break bad adjustment habits. All too often, I see players using their wrist to adjust to low balls. They elongate their racket-forearm angle (violating the theorem) in order to get the racket head low enough to hit a low ball, and in doing so lose all stability and control. To avoid this, ensure you're always throwing a relaxed 90° racket-forearm system through the ball. For most players, thinking about this causes them to naturally adjust with the larger parts of the body, the legs and torso, rather than with the wrist.

> ***All too often, I see players using their wrist to adjust to low balls.***

Arm adjustments do not violate the fundamental theorem of tennis, but they are not optimal. You *can* move your arm closer to or farther from your body without interfering with the forearm-racket system, and thus these shots will still be relatively fault tolerant and frequently go in, but they'll be lacking in power because they don't harness the rotational kinetic chain nearly as well as shots for which adjustment was made with the legs or torso.

Inhale and Wait

Inhaling and waiting may seem trivial compared to the rest of our preparation discussion, but it isn't. In fact, the fact that this advice *seems* trivial is likely the primary reason that so many tennis players have never *actually been taught it* in the first place. That's a shame, because inhaling and waiting is the last critical key to preparing the body for an effective, explosive, fault tolerant forward explosion.

If possible, inhale through the nose, rather than through the mouth. Nasal breathing oxygenates the body better than mouth breathing – you actually get more oxygen per ounce of air inhaled through the nose than you do through the mouth.

Of course, if your nose is congested, or if you're so out of breath that the nasal pathway simply isn't wide enough to get enough air, just inhale through the mouth and don't worry about it. That said, during practice, during conditioning, and during any non-maximally strenuous rally, begin forming the habit of nasal breathing, and your conditioning will thank you.

Nitric oxide, a chemical produced by the sinuses during nasal breathing, is essential for many health markers; everything from being an anti-inflammatory agent to reducing blood clotting.

The Inhalation-Exhalation Pair

We inhale during preparation so that we can exhale during the stroke. Sound familiar? It should; each aspect of preparation forms a diode with its corresponding aspect of the forward swing. Inhaling is no exception.

Inhaling during preparation helps prepare both the body and the mind for the upcoming stroke. The inhaled air itself has the obvious purpose of providing oxygen for the explosive movement to come, but it also provides the less obvious function of bracing the core, increasing stability before the swing.

As we explode during the swing, we exhale, releasing the air and relaxing the previously braced muscles after they've contracted to drive the body's rotation. This relaxation allows our arm and hand to freely whip around the body, unimpeded by tense muscles, and it prevents any injury that would be caused by abruptly trying to prematurely *stop* the kinetic chain we've unleashed.

Each aspect of preparation forms a diode with its corresponding aspect of the forward swing.

Success in the Absence of "Rhythm"

Waiting is also a critical piece of forehand preparation that is often overlooked. Not every shot will occur in perfect rhythm. In fact, the better your opponent, the *less* rhythm you'll have to work with on your forehands. Forehand preparation is not one, fluid, continuous motion that occurs the same way every time. *Far* from it. Instead, as we keep mentioning, it is a *means to an end.*

Proper preparation constitutes getting the body in position, turning away from the ball, preparing the hand, and then *waiting* until the ball is in the proper place to initiate the forward swing. This waiting doesn't have to be completely static – the "preparing the hand" phase can be done more quickly or more slowly in order to time the incoming ball – but, fundamentally, the last step of preparation is a short *waiting* period between body preparation and swing initiation.

On balls that come extremely quickly, this waiting period may be zero. On balls that come extremely slowly, this waiting period may be long. Your cue for when to initiate the forward swing is *not* that a fixed amount of time has passed since you started your forehand preparation. Rather, it is a *visual* cue. You are visually tracking the ball during your preparation, and once the ball gets to a certain location, you initiate your swing.

Visual Tracking

When the ball is on your opponent's side of the court, your primary visual focus should be on your opponent's body. As they set up for the ball, their distance from it typically gives you a lot of information about where the ball is likely to go. Further, once they begin their hip rotation, the degree of that rotation also provides information. Lastly, of course, your opponent's racket face through contact fully telegraphs the ball's future trajectory, though this information is only provided milliseconds before you could observe that trajectory itself.

The longer you play tennis, and the more varied and unique types of opponents you play against, the better you'll get at reading bodies and rackets. It just takes time for your brain to assimilate all the information. If you've seen 10,000 forehands come off a racket, you'll be better at reading that shot than someone who's seen 1,000 forehands will be, and likewise, if you've seen 100,000 forehands come off a racket, you'll be even better still.

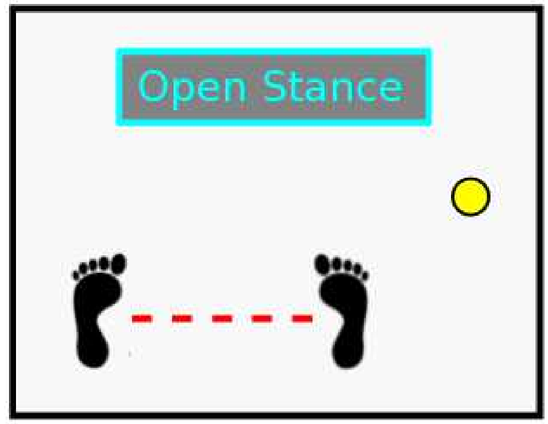
Once the ball leaves your opponent's racket, the ball itself demands close to 100% of your visual attention. Occasionally, you may want to take a quick glance at your opponent to observe from where they're defending, but, in general, nearly *all* of your visual focus belongs on the ball from the time when it is struck by your opponent, to the time *when it leaves your own strings.* Tracking in this way greatly improves your stroke's fault tolerance. The more visual information you provide your brain concerning the incoming ball's flight path, the easier it is for your brain to map out an appropriate swing.

Further, by watching the ball *all the way into contact,* you allow your brain to utilize visual information to make adjustments throughout the *entire* swing process, all the way up to contact. You'd be amazed at just how many quick, small adjustments you can make, pretty much automatically, just by watching the ball all the way into your strings. If, as an experiment, you *force* yourself to *not* watch the ball through contact, you'll be amazed at just how often you mishit the ball or frame it. Small, mostly subconscious adjustments made as a result of proper visual tracking are very under-rated. It's physically impossible to *initiate* the swing perfectly every time, but that doesn't mean the swing needs to fail. These small, in-swing adjustments based on vision play a huge roll in maintaining the stroke's fault tolerance.

A Brief Comment On Stance

Stance is similar to grip – it doesn't matter that much. With an open, semi-open, or neutral stance, a player can correctly turn away from the ball, prepare the hand, and then explosively rotate back towards the ball during the forward swing.

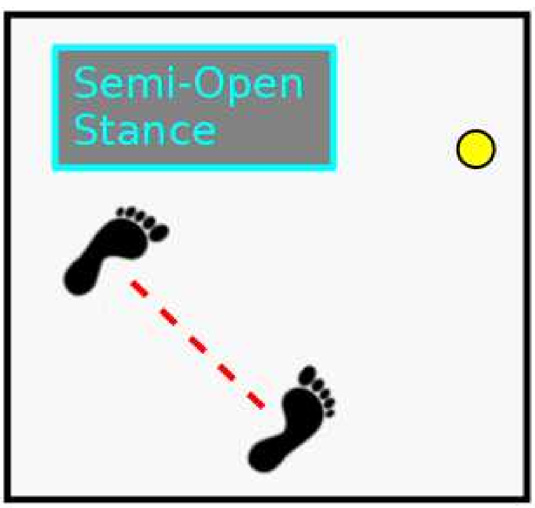
Many players will use different stances on different forehands throughout the match. It's mostly a matter of personal preference; each player should determine what feels best for their particular body. That said, there are still some pitfalls to avoid when it comes to forehand stance.



In the open stance, the feet are parallel with the net, and contact is made about an arm's length away from where the hitting foot starts.

Get the Hips Back Around

The fault tolerance of forehand contact is maximized when the hips are *square* to the target. That means that if you initiate your swing from something like a neutral stance, in which your hips are pointed sideways during preparation, you need to ensure you get them rotated all the way back around by contact.



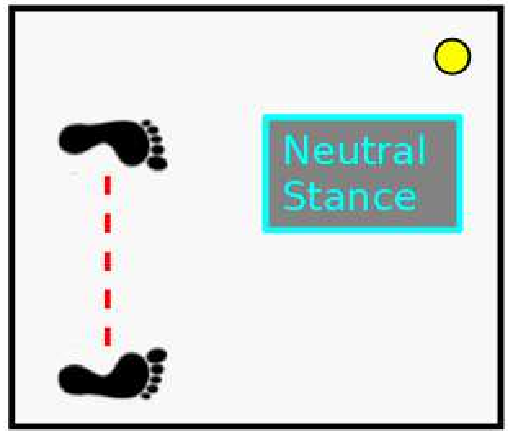
In a semi-open stance, the feet are set on a diagonal to the target, and body rotation is needed to reach the optimal contact point.

This is totally fine, of course – that hip rotation will generate lots of racket head speed, but always understand that the *reason* to use the neutral stance is *so that* you can perform that explosive hip rotation.

If you have a habit of *not* rotating your hips as you swing, then if you prepare using a neutral stance, your hips will end up *still sideways* at contact, which greatly hampers the stroke's fault tolerance. The best solution for this, of course, is to train yourself to lead your swing with hip rotation, but a good stop gap would be to temporarily switch to a fully open stance. With a fully open stance, your hips are facing the target during your entire preparation and swing; you won't have to rotate them in order for them to point at the target by contact.

Using Many Stances

Preparation exists to *prepare* for the forward swing. The forward swing you'll want to attempt will vary based on the situation. You'll want to use varying degrees of hip rotation on varying shots, and thus you should begin with varying stances.

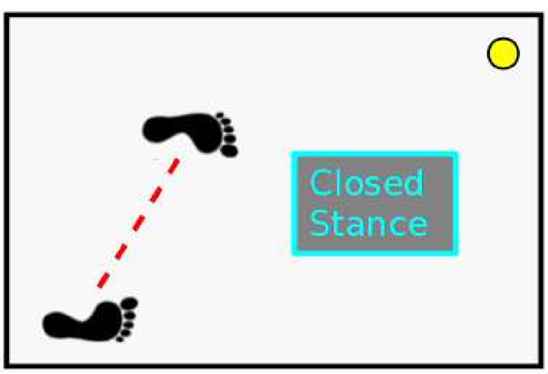


In a neutral stance, the feet are set perpendicular to the net. The optimal contact point is far from the back foot's initial placement, and therefore a lot of body rotation is required to reach it.

The important point here is that the *feet* are not what's critical – what's critical is where the *hips* are aligned. The hips have freedom to turn while the feet remain planted, but only so far. You can utilize hip rotation out of any stance, and you can fail to utilize hip rotation out of any stance. Stance, like grip, ultimately is a matter of personal comfort. Just ensure that whatever stance you choose, your hips are *square to your target* by the time you reach contact.

A Warning About the Closed Stance

Earlier, I mentioned that the open, semi-open, and neutral stances are all fine when it comes to forehand stance preparation. What I didn't include was the closed stance. That's because it's quite difficult to get the hips all the way around from a closed stance. It's doable – if you drive hard off of your hitting leg's glute and allow that hitting leg to follow-through around your body, you *can* get your hips around, but this isn't a very natural feeling motion and, for most players, should be avoided.



In a closed stance, the back foot is behind the front foot, relative to the trunk. So much rotation is needed to reach optimal contact that this stance is not recommended for the modern forehand.

Most players who play the forehand with a closed stance don't do it because they want extra space to whip their hips around; instead, they just leave their hips under-rotated during contact and thereby destroy their stroke's fault tolerance. Instead of trying to coach these students to use their hips more, it's probably better to start by having them play the forehand from a neutral stance instead.

One area in which the closed stance does excel is balance – it's a stance from which it's very natural to sit low and remain stable. This makes it a viable option on low forehands, but, again, ensure that the hips *get back around to the target* by contact. Your back foot's follow-through provides a good indicator of how well you're rotating your hips out of a closed stance – if it follows through around your body after the stroke, resulting in you ending in an open stance, you're probably getting sufficient rotation on your swing.

That's It

Forehand preparation is really quite simple. There may be a lot of *text* in the preceding sections, but most of that text is telling you what *not* to do, or explaining the various misconceptions that exist about things like "backswing." Forehand preparation fundamentally constitutes two goals:

1. We turn the prime movers away from the target.

2. We prepare the hand.

It isn't trivially simple – especially for uncommon contact heights, we do have to remember to adjust our hand using our legs and torso, rather than by moving our arm or wrist around – but overall it really is quite simple.

Nearly all of the forehand-killing backswing mistakes aren't caused by players *failing* to take preparation steps that they *should* be taking. Rather, they're almost always caused by players doing *extra* things during preparation that they *shouldn't* be doing.

Any extraneous, unnecessary movement added to forehand preparation will make the following forward swing less consistent, and many of the extraneous movements recreational players tend to add are actually worse than that. Things like adding tension to point your butt cap at the net, or using your arm to pull the racket way back behind the back, *actively hinder* the forward swing.

And if we've learned anything in this section, it's that preparation's *only* function is to *assist* the forward swing, to prepare the body such that said swing can be executed as efficiently as possible. So keep it simple. Turn your body, and prepare your hand. Inhale, relax, and wait, and when you finally swing, the resulting strokes will be far more consistent.

Part 3

The Forward Swing

Exhale and Relax

The more things you have rattling around inside your head, the more techniques you're trying to implement or movements you're working to master, the easier it is to *forget* the crucial first step of the forward swing – exhale and relax. But this step *is* truly essential. In order for the rotational kinetic chain to work as intended, a strong, braced muscle contraction is *only* needed at the very beginning, when the hips, trunk, and abs initiate the movement. Through this contraction, we exhale, so that by the time the racket is executing its final whip through contact, our body is relaxed, and no muscle tension is impeding that whip at all.

The Counter-Intuitive Role of the Hand

One of the most counter intuitive aspects of the forehand swing is just how relaxed the hand and wrist should be during the stroke. Correct tension constitutes holding the racket just tightly enough to prevent it from flying out of your hand. The optimal way to generate racket head speed is to generate energy using the largest, strongest muscles in the body – the large muscles in the legs and trunk – and then *transfer* that energy into the racket. This is a far more efficient strategy than generating energy using the smaller muscles in the arm[8], and it's *especially* more efficient than using the *tiny* muscles in the wrist.

We touched on this briefly in **Preparation**, but I'll reiterate here: instead of tensing the wrist, and trying to extract some extra small modicum of force from those tiny muscles, we completely relax it to use it as a *hinge*. The wrist's role is to *transfer* energy from the rest of the kinetic chain into the racket. The looser it is, the less energy is lost in this transference. This is why a tight wrist leads to such a slow swing. A tight wrist is an inefficient hinge when it comes to *transferring* force. The racket cannot freely whip around it. With a tight wrist, it doesn't matter how hard we drive with our legs and trunk to start the motion, most of that energy is going to be *lost* before it ever gets transferred into the ball.

8 On the dead run, when the legs and trunk both must be fully extended (and thus completely unloaded) in order to get you to the ball in the first place, you'll have to use your arm to generate force. This is why shots on the dead run are struck so weakly, usually as forehand slices; the prime movers that typically accelerate the racket are unavailable.

Lead With The Hips

You've turned your body away from the target, you've prepared your hand, you've inhaled, waited, and finally it's time to swing. What now? Hint – the answer is *not* to pull with your arm.

The key that unlocks forehand racket head speed is body rotation, and body rotation starts with a rotation of the hips. There are two different kinds of hip rotation, and most players use both on their stroke. First, if the forehand is played out of a neutral or semi-open stance, the player will drive forward off of their back leg, rotating the entire body back towards the target. Sometimes, the player will allow this leg to follow-through around the body, thereby ending the swing in an open stance. Other times, they'll allow the back leg to kick back behind the front leg to counter balance the rotation. Either way, the forward swing starts by initiating rotation by driving off of the back leg.

> ***Finally it's time to swing. What now? Hint – the answer is not to pull with your arm.***

The second kind of hip rotation occurs when the hips have been wound farther than the feet[9]. Contraction of the glute muscles will forcibly straighten out the hips with respect to the feet, unwinding this angle. Most players play the forehand out of some version of the semi-open stance, and then further wind their hips past their feet from there. The result is that they initiate their swing by both driving off the back leg and contracting their glutes, driving their entire body around and forcibly straightening their hips relative to their feet.

When on the court and attempting to produce the stroke, I recommend internalizing it as "twisting the hips back towards the net." That will produce the above movement patterns without having to actually consider all the technical details. Further, if you've wound your hips farther than your feet, you'll naturally unwind them, and if you haven't, you'll naturally only implement the first kind of rotation. Either way, as long as you initiate your twist back towards the target by utilizing the strong muscles in the hips, your stroke will be explosive.

9 The concept at play when the hips are turned away farther than the feet is the same concept at play when the chest is turned farther than the hips. The chest-hip separation angle stores extra energy in the abdomen, and when the abs unwind that angle, that energy is released. The hip-feet separation stores extra energy in the glutes in a similar manner.

How the Feet Liberate the Hips

It sounds so simple – lead the swing by rotating the hips. If it is, then why is it so often done wrong? As it turns out, rotating the hips on a forehand isn't as trivial as it initially seems, primarily because it's really easy for the feet to get in the way.

A full hip rotation is actually *impossible* if both of your feet are firmly planted on the ground. Trying to rotate your hips while your feet stay planted will put a large amount of torsion force on your knees and ankles; don't do it.

Therefore, your feet will need to come off the ground, at least partially, in order to hit a proper forehand. In this section, we'll discuss four common footwork patterns used on the forehand. Each of these patterns is simply a manifestation of letting the feet move freely.

> ***A full hip rotation is actually impossible if both of your feet are firmly planted on the ground.***

The feet naturally want to get out of the way when you rotate your hips; your brain knows not to send that torsion force into your ankles. As long as the player allows their feet to move freely, and doesn't keep them locked to the ground, these patterns will occur naturally. That said, it's still worthwhile to learn what these patterns are, what they look like, and when to use them. Unconscious competence can only get you so far. In the end, every athlete should know *why* they're doing what they're doing, even if they do it naturally, so that they can replicate it on command.

Now, let's get into the four common patterns by which the feet move to allow hip rotation.

1. The Non-hitting Foot Comes Up

This is probably the most common footwork pattern used during the hip twist, especially on neutral rally balls. The non-hitting foot comes up off the ground, while the hitting foot remains on the ground. As the player twists their body around, they rotate on the ball of the hitting foot, and the non-hitting foot doesn't restrict this rotation at all, since it's in the air.

Both Novak Djokovic (left) and Roger Federer (right) using the same left-foot-up footwork pattern to allow necessary hip rotation on an inside-in forehand.

This is a great footwork pattern for rally balls and neutral balls. Typically, a player can't move forward into the court using this footwork pattern, so other patterns are used to twist on shorter, slower balls.

2. The Player Pivots On Both Feet

The double foot pivot is most useful when the player's weight is moving forward, likely because the player is looking to be aggressive. If a player wants to follow their shot through into the net, then picking the front foot off the ground and rotating about the back foot isn't a great option; that won't generate any sort of forward momentum through the shot.

Instead, the better option is to pick both heels off the ground and simply allow the feet to twist along with the hips as they rotate. This footwork pattern works very well on aggressive shots from the baseline and on approach shots inside the court, since, on both of those kinds of shots, the player wants to transfer their weight to their front (non-hitting) foot and take the ball early.

Djokovic striking a hard, cross-court forehand by pivoting on both feet together, keeping his weight on his front foot.

This stance is stable and low impact. It doesn't require jumping off the ground, and it works particularly well when returning slow, low balls like slices. On every full forehand swing, a hip twist is required, but due to the nature of a low ball, it would be very difficult to get either of the feet off the ground while striking it. Thus, for low balls, picking both heels up and pivoting on the balls of both feet works best.

3. The Hitting Foot Comes Up and Kicks Back

Federer utilizing his classic "hop step" to strike a high, slow forehand aggressively and continue forward to net.

Another aggressive footwork pattern that's commonly used when a player wishes to come forward involves a conscious drive off the hitting leg. The player initiates their twist by driving so hard off the hitting leg that it comes up off the ground, and as the player subsequently twists their hips, this off-the-ground hitting foot naturally kicks back behind them.

It's extremely important that the player's non-hitting foot is not restricted during this process, as, again, if it's firmly planted on the ground, an ankle snapping amount of torsion force is going into that ankle (or maybe the knee). A common, natural way to handle the large amount of force placed on the front (non-hitting) foot during this pattern is to allow it to come up off the ground as the force pulls it.

This particular footwork pattern, often known as the "hop step," is a favorite of Roger Federer, specifically when hitting approach shots. Don't be misled by the name, though. There is no *conscious* "hop" going on here; the player doesn't *try* to hop off the front foot at any point, but rather just *allows* the forces that want to pull the foot off the ground during the follow-through to do so.

4. Both Feet Come Off the Ground

There are two different applications for the both-feet-off-the-ground forehand that are pretty disparate in nature. The first is if the player is lightly on the run – not on the dead run, but still required to hit without setting their feet.

In this case, it's usually optimal for the player to perform a small jump as they rotate their hips back towards the target. This allows their feet their full range of freedom during the twist. Having the feet in contact with the ground *while* moving sideways *and* twisting the hips is a recipe for injury, and it's best avoided by taking a small jump when striking a forehand on the move.

Nick Kyrgios is famous for the jumping forehand – a shot during which the lower body and trunk generate such a massive amount of force that both feet end up off the ground, while the arm and wrist stay relaxed to transfer that force into the racket.

The second case is when the player wants to hit a shot really, really hard, and both legs are driving so hard off the ground they're going to fly off of it. This shot is most commonly used on high, slow, short balls, and it's very effective when executed correctly.

The important thing to remember about the jumping forehand is that *all* the tension is in the legs and trunk, and *none* of the tension is in the arm or hand. The exterior links in the kinetic chain must stay completely relaxed in order to efficiently transfer the force from the prime movers into the racket.

The Abdominal Coil – Unwinding the Upper Body

After the hips initiate the rotational explosion, the abdominals continue it by unwinding the upper body. This doesn't happen immediately, though. While the shoulders and chest *will* rotate forward passively during hip rotation (they are, of course, connected to the hips), this does not require the abdominals to do anything more than *stabilize* the core. During the hip rotation phase of the kinetic chain, the *relative* angle of the shoulders and hips is maintained. A few milliseconds *after* hip rotation initiates the swing, the abdominals fully engage: they act to *actively* twist the shoulders and chest back toward the target, thereby *unwinding* this angle.

Recall that during the unit turn, the shoulders are typically turned farther than the hips, creating a shoulder-hip angle separation – the hips might turn away by 45° while the shoulders turn away 65°. Abdominal rotation during the forward swing is the counterpart to this aspect of forehand preparation. The reason we turn the shoulders farther than the hips during the unit turn is precisely so that, during the forward swing, the abdominals can unwind that extra turn. As the abdominals unwind this turn, they further torque the upper body around and contribute their energy to the forward swing.

The quintessential example of a shoulder-hip angle separation, Robin Soderling winds his shoulders almost 90º farther than his hips.

The most helpful proprioceptive cue for activating this rotation is *pulling the non-hitting elbow out and around the body.* This ensures that the entire upper body rotates towards the ball as a unit, rather than just the hitting-arm going alone. During this phase of the swing, both elbows and the chest rotate back towards the net *together.* They turned away as a unit, and they turn back as a unit.

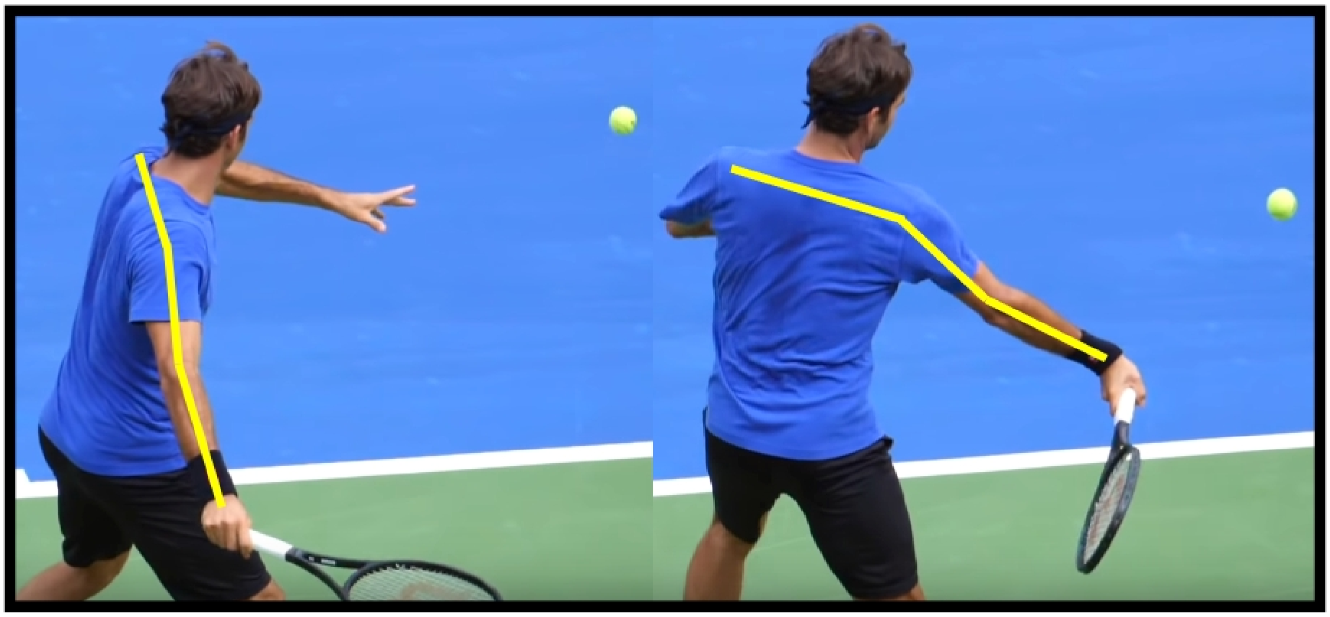
Rafael Nadal prepares with his non-hitting elbow well on the hitting side of his body, then pulls it out to the right (he's lefty) as he initiates his forward swing, thereby whipping his hips and chest back towards the net and violently accelerating the racket.

During the rotational explosion, each muscle group *feeds* its energy into the next link in the chain, in sequence. The hips come first, followed by the abdominals. That said, *thinking* about this timing *during stroke production* is almost definitely useless. The timing sequence, by which the hips fire, then a few milliseconds later the abs fire, then a few milliseconds later the chest fires – that sequence is typically not performed by thinking about it. Instead, internalizing it as something like "pulling the non-hitting arm around the body" tends to be far more helpful. It's best to take advantage of our body's *built-in* movement functions as we learn how to coordinate this chain.

Unit Turn, Unit Swing

Many players fail to sufficiently rotate their hips and chest back towards the net as they swing. This common problem is usually solved once the player learns to pull their non-hitting elbow around their body *with* their hitting elbow. Once that's drilled in, the problem of under-rotation typically goes away. However, sometimes that faulty *under*-rotation is then replaced with *over*-rotation (or, sometimes, premature rotation). This, of course, is equally counter-productive.

The final piece to this rotational puzzle is understanding that the entire upper body rotates *forward* together during the swing, just like it rotated *back* together during the unit turn. When I recommend you "pull the non-hitting arm around," that's a *solution* for the problem of *under-rotation.* However, if a player is already rotating properly, and that player is coached to "pull the non-hitting arm around," they may start rotating too early, or too much.



During abdominal rotation, during which the angle between the hips and the shoulders unwinds, the relative position of Roger Federer's left arm, his chest, and his right arm is remarkably constant; the entire upper body is rotating together.

Ultimately, your goal during abdominal rotation on the forward swing is to learn what it feels like to rotate your entire upper body – your chest and both arms – forward together as a single unit, just like you rotated them *backward* as a single unit during the unit turn. This rotation will be explosive, stable, and effective. You won't have arms flailing this way and that, and the rotation will be driven by the strong, stable abdominal muscles in your core. It's a rotation that can be performed extremely explosively and extremely quickly *without* injury, unlike, say, pulling the racket forward using the hand and arm.

Out, Up, and Through

Every player has been coached to "brush up the back of the ball" in order to generate topspin. While this advice is technically correct, it's very misleading as to *how* this famous "brushing" actually happens[10]. Most players understand that they need to swing **up** and **through** the ball, but what many haven't been taught is that it's also essential to swing **out** to the ball. There are three critical vectors to the correct swing path, not two.

Out, up, and through.

Of these three directions, the **out** part is the most overlooked. Despite it's lack of popularity though, it is *essential* to generating a heavy, fast forehand, and it's actually far *more* important than the **up** direction. A swing containing the **out** vector imparts strong forces on the racket that cause it to *naturally* whip around the wrist through the hitting zone. This racket rotation is the less understood, but the more important of the two fundamental movement patterns that are used to generate topspin in tennis.

The Two Topspin Movement Patterns

The two movement patterns that generate topspin in tennis are very different in their execution. They are:

1. **Upward translation of the racket-arm system**
2. **Racket rotation about the forearm**

The better known of the two is upward translation of the racket-arm system through contact. As the player hits, if the racket-arm system is moving **up**, then, of course, the racket itself is also moving up. Therefore, at impact, friction between the strings and the back of the ball will generate torque on it and create topspin ball rotation. This is the "swing low to high" idea which we've all heard countless times.

However, the *other* movement pattern that generates topspin is far *more* important, especially at the highest level – *racket rotation about the forearm*. Through the hitting zone, the racket passively whips around the wrist, about the axis of the forearm. This rotational whip

10 Trevor Noah humorously recounting this exact experience: https://www.youtube.com/watch?v=37lqCgW1h9o&t=55s

constitutes the traditionally named “windshield wiper” motion, and it *also* creates an upward racket motion through contact and thereby a resulting upward frictional force on the back of the ball.

Whether or not the entire arm-racket system is translating up, when the racket is rotating about the forearm during contact, the racket *itself* is still explosively moving upward through the strike (even if the arm is not). *This* upward racket motion, the motion generated by racket rotation, is responsible for a lot more of the total upward frictional force imparted at contact than any upward translational motion is.

Topspin on the Tour – Primarily Racket Rotation

You’ll often see tour level players hit heavy topspin shots while *barely even using* an upward translational component in their swing – their follow-through ends up *below* their non-hitting shoulder. How is that possible? Heavy topspin without the racket-arm system moving drastically **up** through contact? How did they generate all that spin without swinging “low to high?” It works because they’ve generated almost 100% of their spin using the *racket rotation* contributor to spin, rather than by translating the racket up.

Dominic Thiem striking a waist height, cross-court forehand with heavy spin, despite finishing his swing below his non-hitting shoulder. His spin is generated almost entirely by racket rotation through the hitting zone.

In the previous image, Dominic Thiem is utilizing almost entirely racket rotation to generate topspin, but, often, heavy spin players like he and Rafael Nadal will use *both* topspin movement patterns together: they both swing the entire racket-arm system **up**, *and* they relax, swing **out**, and allow the racket to whip about their forearm. The result is that they finish these shots above their head, and by utilizing *both* techniques, they generate some of the highest RPM forehands on the tour.

Rafael Nadal utilizing both racket rotation and upward translation to generate maximum spin on a knee height forehand. He sits low with his legs and tilts his torso in order to maintain his rotational whip, despite the knee height contact point.

Elite players also utilize the *same biomechanics* on every forehand, regardless of contact point. Racket rotation is easy to lose, especially on low balls, if adjustment to low contact is performed incorrectly. Instead of adjusting to a lower contact height using the arm – pulling it closer to the body to get the racket lower – elite players adjust by sitting down lower and tilting their torso. This way, the arm can flick **out** and around the body in the *same motion relative to the trunk* at every contact height, even a low contact. By adjusting this way, elite players do not lose the anatomical advantage of racket rotation when contact needs to be made lower than usual.

The Role of the Out Vector

So what does **out** itself have to do with any of this?

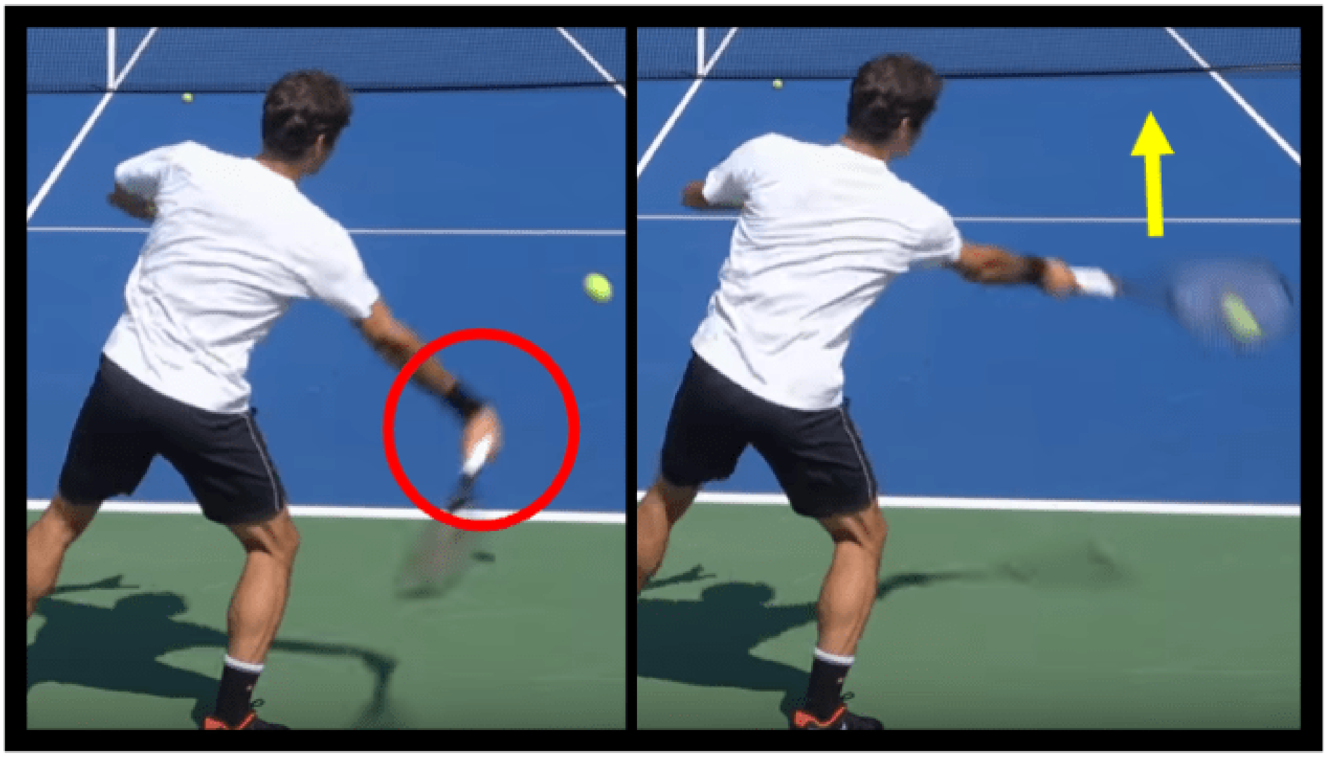
The **out** vector is the result of the volitional action by which we generate this racket rotation, the most important contributor to topspin. By consciously swinging **out** to the ball and contacting it laterally *away* from the body, we generate the forces that cause the racket to naturally whip around the wrist.

As we begin our hip rotation, our racket is thrown outward by the force the hips create. Eventually, this outward racket velocity has to be stopped, lest the racket fly out of our hand. The stopping is performed subconsciously, by holding the racket just tightly enough to prevent it from flying away. Friction between the handle and our hand forcibly stops the racket's outward motion, abruptly pulling it back **in** towards our body. This abrupt pull changes the racket's flight path and causes it to rotate about the point of contact that's pulling it.

Depending on the exact orientation of the arm, racket, and wrist, the resulting rotation can happen in different planes. The goal is to orient the racket-arm-wrist system such that the racket face rotates through contact roughly about the axis of the forearm, so that the string angle isn't dramatically altered by the rotation. This typically means holding the racket with a relaxed, neutral wrist, and swinging diagonally up to the ball..

You'll want to experiment with the **out** vector of your swing while practicing shadow swings. Eventually, you'll be able to feel the moment at which the racket's outward velocity is stopped, the moment when that energy is then transferred into racket rotation around the wrist. Mess with slightly different vectors and swing paths, and observe, in each case, how this rotational whip happens. Also note how difficult it is to create this whipping action close to your body, and how much more naturally it occurs when imaginary contact is made both out in front and laterally away from the body. The more familiar you get with the cause and effect behind this whipping action, the easier it'll be to apply it when striking an actual ball.

By consciously swinging out to the ball and contacting it laterally away from the body, we generate the forces that cause the racket to naturally whip around the wrist.



Roger Federer's relaxed wrist lags behind his hand as it is swung out to the ball. Then, as the racket hits the edge of its range of motion and can no longer continue out, it rotates around the forearm and is thereby explosively thrown both up and forward through contact.

This rotational whip is *far more muted* if the racket is flung to a contact point that is laterally closer to the body; in that case, there's no *outward* velocity on the racket during the initial stages of the swing, and thereby no transfer of that outward velocity to rotational whip once the racket can't go any farther out. As long as the racket is still thrown **up** through contact, the racket will still rotate *a little*, but not nearly as much. The fact that the outward velocity is *forcibly stopped* by friction with the hand – that's what creates the *whipping* effect through contact. Without the **out** vector, that doesn't happen.

Conscious Effort vs Automatic Results

As tennis instructors, it's important that we make it *abundantly* clear which movements during a swing are *volitional*, and which ones are *automatic*. Racket rotation about the forearm through the hitting zone is *automatic* – it is NOT created via conscious effort, and the wrist and hand musculature do not play ANY role in *generating* this rotational force (though they do stabilize the racket and the string angle while that rotation is happening).

This is an extremely common misconception – many coaches harmfully instruct their students to "use your wrist," despite the fact that the wrist's job is just to relax and to allow forces generated by *other* parts of the body to act on the racket and whip it around that relaxed wrist. Wrist lag, racket rotation, and essentially the racket's entire swing path after the volitional initiation of the forward swing constitute *dynamic positions.* They are entered automatically, as a result of other intentional actions taken before them.

Confusion Surrounding "Wrist Action"

When tennis commentators talk about "great wrist action," what they really mean is that the player does a great job of *relaxing* their wrist and allowing their wrist to act as a hinge that effectively *transfers* force into the ball. They do not mean that the wrist is in any way *generating* force itself.

And when they talk about a player having a "strong wrist," that's simply a reference to the fact that the harder a player swings, the stronger their wrist must be to *stabilize* the racket during that swing. Again, it is NOT implying that the player's "strong wrist" is *creating* that racket head speed by using its strength.

The action that actually *generates* the racket head speed is the trunk rotation which flings the racket **out** and **through** the ball. Therefore, the action that a tennis player should *think* about while playing is just that – swing **out** to the ball and strike it *away* from the body. That's it.

When Nick Kyrgios jumps off the court and slaps a 100 mph forehand winner, the commentators often compliment his "live wrist." What they are really complimenting is how relaxed it is, and how efficiently it transfers the power from his core into the ball.

Do that with a relaxed wrist, and as your racket gets to the end of its range of motion, it will *naturally and automatically* flick around. Any *conscious attempt* at using the wrist or the arm to create the windshield wiper motion will typically just lead to harmful wrist tension, ineffective shots, and injury.

So How Do I Actually Apply This?

The physics at play during the windshield wiper motion are pretty complex, so ignore the physics themselves when you're on the court (but understand them off the court). Instead, just remember the cause and effect:

Cause – swinging **out** to the ball with a relaxed wrist.

Effect – the racket will whip around the wrist and rotate through the contact zone.

We use this knowledge to debug our own forehand. If, during a particular practice session, you feel like you aren't getting sufficient topspin on your shot, a failure to properly swing **out** to the ball may be the culprit.

We've all experienced the irritating phenomenon that, on certain days, our stroke feels great, but then, other days, all of its fault tolerance has evaporated, and it's a struggle to even get the ball into the court. Perhaps you've accidentally started violating the fundamental theorem of tennis. Perhaps you're fatigued (or just being lazy), and you aren't playing low and wide enough. Or perhaps you've subconsciously altered your contact point, pulling it in towards the body, thereby preventing that dynamic, rotational racket flick which generates so much spin.

Remembering to swing ***out***, **up**, and **through** the ball, instead of just **up** and **through** it, is one more tool in your toolbox for fighting that inconsistency, and for consistently maintaining that heavy, fast, fault tolerant forehand that makes tennis so enjoyable to play.

Dominate the Slicer with Heavy Low Forehands

At Fault Tolerant Tennis, we teach fundamentals. Our aim is *not* to provide assorted tips and tricks that apply only in certain cases, but rather to educate our readers about the **essential biomechanical movement patterns** that underpin each shot. We teach *principles* here – principles that lead to fault tolerant, effective strokes in *every* situation, whether the shot is being played high, low, wide, or close.

That said, the low forehand specifically *still* deserves its own section, because it is such an important, frequent, and *mistake prone* shot.

Dan Evans, the #1 ranked British player in the world, is famous for almost exclusively slicing his one-handed backhand.

The Common Pitfall Against Slicers

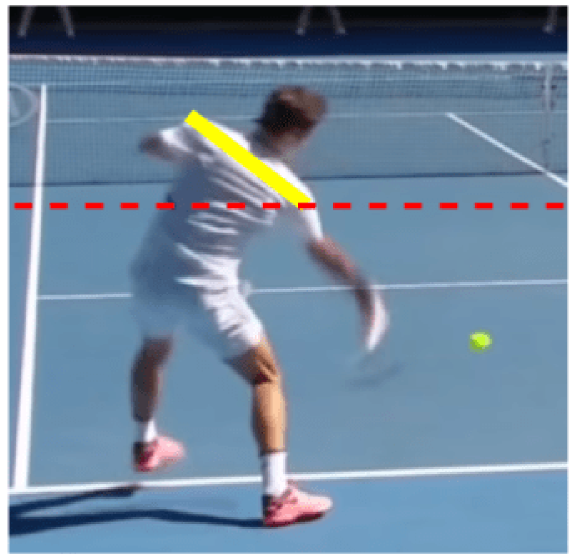
In a match against someone who only hits slices, you'll be forced to play nearly all of your forehands from knee height or lower. Without an effective, fault tolerant low forehand, these matches are *extremely frustrating*.

The mechanics that make the forehand fault tolerant, the mechanics we teach in this book, are very natural on waist height and shoulder height forehands. That's why, in a normal match against a non-slicer, your forehand feels fine. Almost all of those forehands are played from waist height or from shoulder height, and, as a result, you naturally apply fault tolerant mechanics.

Everything we've discussed – from preparation, to body rotation, to swing path – once you've practiced a little, using those mechanics will be your *first instinct* on a typical height forehand. On a low forehand, these very same mechanics will initially feel very unnatural; if you're not consciously focused on using on them, you'll probably accidentally *abandon* them. That's when you get a bunch of mishit low forehands, lost points, and a smashed racket.

Utilizing Waist Height Mechanics

To transpose our fault tolerant mechanics to a lower contact point, we need to adjust our preparation appropriately. We've already discussed adjustment, but it bears repeating – you adjust to a low forehand by sitting lower with your legs and by tilting your torso, *not* by pulling your hand closer to your body. Adjusting with your bigger muscles allows you to maintain the same alignment of your racket, arm, and hand *relative* to your trunk that you use on a waist height forehand, thereby allowing you to rotate your hips and trunk in the same manner as on a higher shot.



Roger Federer adjusts to a low forehand by bending his legs and tilting his torso, allowing him to rotate his body the same way he does on a waist height forehand.

When preparing for the low contact point, many players' first instinct will be to pull their hand in, rather than tilt their torso. This kind of hand adjustment *is* sometimes necessary on *exceptionally* low balls, but just be aware that the closer you pull your hand in, the more you *inhibit* the transfer of energy from the trunk's rotation into the racket.

Rotational Acceleration

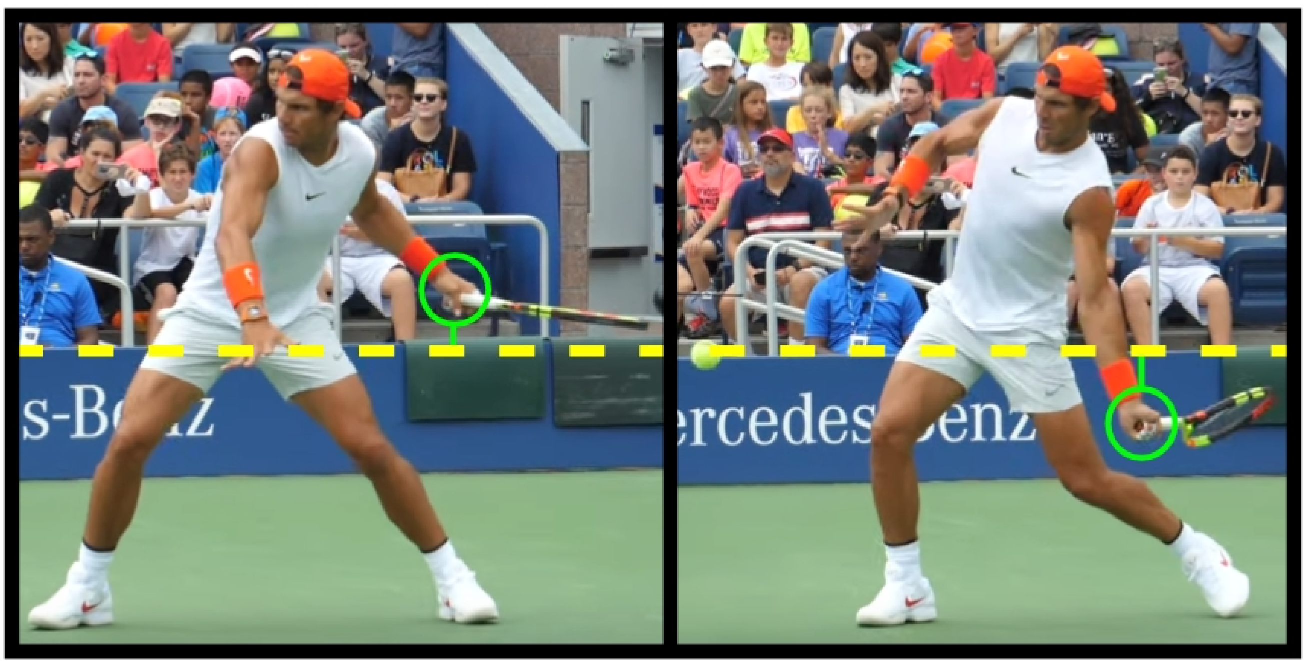
At a given angular velocity, translational velocity is always greatest *farthest* from the center of rotation (the bigger the circle, the faster the points on the circle are moving). You can prove this to yourself at home – pinch your elbow into your side, bringing your racket in close to your body, and then twist forward like you would on a swing. Now hold the racket farther away from you, and do the same. Can you hear all that extra wind through the racket when it's far?

The more the racket is allowed to flick *away* from the body, the more the trunk's rotation will accelerate it.

Differences Preparing the Hand

On any forehand, it's nice when we can prepare our hand below the anticipated contact point, because then it's really easy to throw it up through the ball. However, this isn't actually *strictly* necessary.

On the low forehand specifically, the **out** vector of the swing actually helps drive your hand *down*, thereby making it possible for it to move *up* through the contact point, even if it didn't *start* below the contact point in the first place. This is a physical consequence of your tilted torso – as your elbow extends and your racket flicks **out** (away from your torso), that **out** angle is now diagonally down, rather than straight out.



Rafael Nadal prepares for a knee height forehand with his hand a small distance above the eventual contact point. As he rotates with a tilted chest, his elbow extends, naturally pushing the hand down, thereby allowing it to still move slightly upward through contact.

Since that elbow extension will drive our hand down, even if we prepare our hand at the level of our contact, or even slightly above it, the stroke can still succeed. The **up** vector of this kind of swing will be muted, since the hand will barely be below the ball at any point, but that doesn't mean we can't generate topspin.

Windshield Wiper Spin is Enough

Due to the racket's rotational whip (about the axis of the forearm) through the hitting zone, the racket head *itself* will still be moving **up** extremely quickly through contact, even though the *hand* isn't moving up through contact much at all. Thus, the strike will still generate significant topspin.

Just the rotational whip of the racket alone, the passive "windshield wiper" motion generated from rotating the trunk and throwing the racket **out** – just that vertical racket head speed will often provide enough topspin to keep the ball in play, even *absent* a strong **up** vector in the swing path.

But How Low is Too Low?

Sometimes, though, the ball is so low that there's simply not enough room for a topspin swing path; if you tried, either your racket would hit the court, or the ball would hit the bottom of your frame and fly out. It'll take practice to train your instinct for when this is the case, but make no mistake, there *does* exist a height below which a successful topspin forehand is impossible.

David Ferrer playing a forehand that is so low that a successful topspin stroke is impossible; he blocks the ball back over the net instead.

When you find yourself there, you'll have to slice or push the ball back. All of our non-topspin specific fundamentals – from the fundamental theorem of tennis, to adjusting using your prime movers first, to tracking the ball all the way into the strings – they apply just as much on a forehand slice as they do on a topspin stroke. Stay focused, stay relaxed, and catch the ball out in front of you.

Maintaining The Swing Path

As you swing at a low forehand, your mind is yelling at you that, "this ball is low, so you've gotta *really* pull it up." While this thought is *technically* accurate, in practice, it tends to be a recipe for disaster. Even when the contact point is low, our conscious focus needs to be on the fundamentals which create *fault tolerance* in our stroke, not on "getting the ball up." The over focus on **up** tends to lead to three common mistakes:

1. **Forgetting to extend *through* the ball**
2. **Forgetting to swing *out* to the ball**
3. **Opening or changing the string angle through contact**

1. Forgetting the *Through* Vector

Over focus on the **up** vector will often cause you to accidentally forgo the **through** part of the swing. As a result, you'll spin the ball down into the net – the shot will have plenty of topspin, but your purely **up** swing path won't supply the required *forward* momentum to drive the ball back over the net.

Luckily, this mistake falls into the class of issues that's fixed simply by thinking about it. Instead of allowing your mind to yell, "Get it up! Get it up!" at you, focus on your forward extension. Tell yourself, "**out, up** and ***through***," and ensure that the "**through**" part of that cue is strong in your consciousness as you swing.

Rafael Nadal's impressive extension away from the body, during the follow-through after a knee height forehand, indicates a strong through vector was utilized during the forward swing

2. Forgetting the *Out* Vector

The mind's natural over-emphasis on **up** will *also* often make you forget the **out** vector, and forgetting the **out** vector also severely damages the stroke's fault tolerance. As we've discussed, the **out** part of the swing path is actually *just as responsible* for topspin generation as the **up** vector.

By allowing the racket to flick **out** to the ball, rather than just straight towards it, we impart forces on it which cause it to whip around the axis of the forearm through the hitting zone, creating the "windshield wiper" motion. That motion results in a ton of vertical racket head speed through contact, and thereby topspin, but without the **out** vector, it doesn't happen.

It's extremely easy to forget about both the **through** vector and the **out** vector when swinging at a low forehand, because your mind is so strongly focused on **up,** but lose those other vectors, and you lose the stroke's fault tolerance with them.

3. Subconsciously Turning the Strings Up

The mind's natural over emphasis on "getting the ball up" will also tempt you to alter your string angle, angling the strings towards the sky. Unless you swing really, really slowly, this string angle will cause the ball to sail out.

I know it's counter-intuitive, but *even on a low forehand*, the strings, through the hitting zone, should be either parallel to the net, or even still slightly closed towards the court. At contact, there exists a *ton* of frictional force between the racket and the back of the ball while the racket is flicking **up** and **through** it; that friction *will* drive the ball up. You've just gotta trust it.

Through contact on a knee height forehand, Rafael Nadal's strings are parallel to the net.

If the string angle is altered such that the strings are pointing up at contact, then a rotational swing utilizing a relaxed lagging and snapping wrist simply won't work – the ball will fly off the racket at an angle such that it will always go long.

No "Rolling the Racket"

If you try to fix your open string angle problem by rolling your racket over the ball through contact, that will completely *destroy* the stroke's fault tolerance; unless you time the hit *perfectly*, you're going to miss.

This kind of racket rotation (as opposed to rotation about the axis of the forearm) is the quintessential forehand fault tolerance killer – hit the ball just slightly too early, and it's going long, but hit it slightly too late, and it'll hit the net.

Maintain your consistent string angle through contact, just like on any other forehand. Trust the friction, and trust your natural topspin, the result of your relaxed wrist and your **out, up, and through** swing path.

The Low Forehand is Not Fundamentally Different

To the uninitiated, the low forehand seems like a completely unique shot: a different stroke to practice, separate and distinct from the waist height forward. This is not the case. What *is* the case, though, is that the biomechanical techniques that make the forehand so efficient and fault tolerant are very easy to *accidentally abandon* on the low forehand, because on the low forehand they're initially quite counter intuitive.

And that's why the typical player's game falls apart against slicers.

In a normal match, they're using a fault tolerant stroke – they often slightly mistime or mishit the ball, but they probably don't even notice, and it doesn't matter, because the resulting shots are still effective. The difference against Mr. Slice is that they've accidentally *abandoned* that fault tolerant stroke, and now, unless they strike those low forehands *perfectly*, they miss them.

The biomechanical techniques that make the forehand so efficient and fault tolerant are very easy to accidentally abandon on the low forehand.

To beat the slicer, you have to keep the points neutral, even when forced to play low ball after low ball. To beat a particularly good slicer, you'll also have to hit passing shots on some of those low balls. With fault tolerant mechanics, this is easily doable. Though it might not come naturally at first, the very same mechanics that we utilize on waist height forehands *can* be used on the low forehand as well, and they *will* give you that same fault tolerance advantage – not every strike will have to be perfect; they'll still go in.

Practice implementing these mechanics on low forehands during your training, and eventually you'll be able to hit 10, 20, or 50 low forehands in a row without missing. It'll take some conditioning – the leg musculature required to get very low, very often is certainly non-trivial, but hey, that just takes more training. After that, it's just a regular match. Once you stop missing low rally balls, you force the slicer to try to find offensive opportunities the same way you yourself are trying to. The slicer can no longer win just by letting you miss out of neutral situations.

Master the low forehand, and matches against slicers will be a lot more fun, certainly more fun than mishitting every ball and not knowing why.

A Brief Comment on Follow-Through

The follow-through receives far too much attention throughout the tennis coaching world. As far as contact itself is concerned, the follow-through physically, literally doesn't matter, because it transpires after the ball is already gone. We *only* discuss it and evaluate it at all because it can give us useful *information* as to what happened *during the hitting phase*. But, again, the follow-through *itself* doesn't affect the swing. It's merely a diagnostic tool at our disposal.

The Forced Follow-Through Mistake

Some players actually end up taking the whole rotation thing *too* far. I don't blame them, of course. After all, how many times have we used the word "rotation" so far in this book? But while the primary energy generator for the swing *is* certainly a rotational kinetic chain, *contact* is translational, not rotational. It is the back-to-front force generated by the racket that drives the ball up and over the net. That force is tangent to the racket's circular path around the body, and it needs to be pointing at our target at the moment of contact.

> ***The follow-through itself doesn't affect the swing. It's merely a diagnostic tool at our disposal.***

The racket should never be actively *pulled* across the body in any way. This issue often arises because a student is unknowingly using *conscious effort* to try to mimic the *entire* swing as they see it, either in others or on TV. This includes using conscious effort to mimic the around-the-body follow-through. This results in an interesting problem – instead of the player's goal through contact being to *flick through the ball*, their goal through contact is to *get the racket into its follow-through position*. The misaligned goal results in the player's applying the *wrong* force vectors through the swing[11].

11 Ironically enough, you can often observe the flaws caused by a forced follow-through in the offending player's follow-through itself. Added jerkiness, lack of relaxation, and a general disjointed nature of the racket's movement after contact are all clues that the player may have too much of their mental focus (any of it) dedicated to the *after* contact part of the swing.

The rotational kinetic chain is merely a *tool* we employ to increase the speed of *translational* contact. That rotation *is* the most efficient method by which a player can explosively fling their racket up and through a ball, but the *end goal* is to get the racket to flick up and through the ball, *not* body rotation *for the sake of rotation itself.* The racket *only* ends up finishing on the non-hitting side of the body because the forces generated during the swing force it there after contact. It happens *after* contact, as a *passive* result of the forces we used to fling the racket up and through the ball.

Jennifer Brady's Beautiful Follow-Through

At Fault Tolerant Tennis, we love Jen Brady's forehand. She hits her forehand like a man, and that's a really good thing. Not just in the sense that Jen Brady's forehand is awesome, and the forehands on the ATP Tour are also awesome. It's more than that. Jen Brady employs on her forehand *specific, efficient, explosive biomechanics* that are mostly unique to the male tour.

One righty, one lefty. One male, one female. One bent-elbow, one straight elbow. Despite their nominal differences, Jennifer Brady (left) and Rafael Nadal (right) employ very similar forehands.

In an article on faulttoleranttennis.com, we go through and explain these efficient biomechanics as we analyze her beautiful, effective stroke, and in doing so also dispel the myth that "female athletes aren't strong enough" to use these mechanics. If you'd like to read the entire thing, I've linked it in a footnote[12].

12 https://faulttoleranttennis.com/jen-brady-the-girl-who-rips-3500rpm-forehands/

Here, we'll just discuss her follow-through. Jen's racket naturally turns over as part of her follow-through because of the way she swings through contact. This turn over is not volitional, and, as we've discussed at length already, a player should NEVER consciously *try* to turn their racket over as they swing. This turn happens *naturally* as a result of the relaxed wrist and body rotation during the forward explosion.

Like all great forehand strikers, Jen throws her hand **out** to the ball and then **up** and **through** it. We've already discussed how the outward racket path creates racket rotation when the racket hits the end of its range of motion, but we haven't yet addressed what happens to that rotation *after* the ball is gone. Unless the rotation is *actively stopped with forearm tension,* it continues into the follow-through. The result is that the racket continues rotating across the athlete's body, and the forearm naturally pronates to accommodate that.

Jen Brady showcasing great extension and racket turn on her follow-through, passive indications of a correctly executed swing.

It is, of course, extremely destructive to use any sort of tension to prevent the racket from doing what it naturally wants to do after the ball is gone. After contact, the strike is complete – just relax and don't get injured.

By observing Jen's follow-through, we can deduce that she is swinging **out** to the ball, she is generating significant racket whip around her hand, and she is relaxing and allowing the whole system to naturally decelerate after contact.

Her follow-through also displays great *forward extension* of the arm *away* from the body. Her elbow finishes *well* in front of her chest, indicating that she's successfully transferred the *rotational* force from her trunk into back-to-front *translational* force through the ball. This tells us, definitively, that she is not making the forced follow-through mistake that we just discussed.

Forward Swing Progression

Unwinding the kinetic chain is an extremely complex *physical* process, but that doesn't necessarily mean it has to be an extremely complex *mental* process. Here's what I recommend as a progression for simplifying the complex process of understanding and executing the forward swing.

1. Internalize the forward swing as "twisting towards the net."

2. Lead the twist with your hitting leg. Drive off the ground; contract your glutes.

3. Rotate back together, rotate forward together.

4. Ensure that the racket extends *forward through contact*.

Let's discuss both the biomechanical implications of each, as well as the simpler ways by which these biomechanics tend to be internalized while actually out on the court.

1. Twist towards the net

Luckily, human beings have a pretty good intuitive understanding of the "twist" movement pattern. In coaching, we always want to hijack these pre-coded movement patterns whenever possible. Twist, throw, jump, lunge, sit – these are actions we all take every day; they are far better coaching cues than commands concerning specific joint angles and limb positions are.

At its core, the forehand is a twist action. The player twists *away* from the target during preparation, and then twists *back towards* the target during the forward swing. For some players, all you have to do is tell them to "twist," and the entire swing looks perfect. You'd be amazed how many 5-year-olds, after only being told to "turn sideways" and then "twist back towards the net," execute an almost flawless looking modern forehand, with a proper contact point and a good looking follow-through, without needing to consciously think about any of it.

2. Lead with the hitting leg

When just the idea of "twisting" alone isn't enough, we need something more specific to focus on. This is where “lead with the hitting leg” comes in. Some people don’t naturally possess (at least at first) an athletic understanding of what exactly the “twist” action entails, and so guiding their proprioceptive awareness to the right place is necessary.

Often, “lead by driving off the back leg” is all that’s necessary for a player to sync up their twist action.

That place is the hitting leg. When a player consciously initiates their swing by exploding off of their hitting leg, they are far less likely to leave their hips under-rotated, and that’s what’s important when it comes to forehand rotation. The hips *need* to get back around. If they don’t, the player sacrifices both significant racket head speed and significant fault tolerance.

Often, “lead by driving off the back leg” is all that’s necessary for a player to sync up their twist action. After implementing that idea, they will often fire the rest of their rotational kinetic chain properly, without needing to be cued on the upstream body parts. It feels quite natural, after driving off the back leg to initiate rotation, to then continue by using the abdominals to straighten out the trunk, and then the chest to extend through contact.

3. Rotate back together; rotate forward together

Moving farther out the chain, the next issue we see is a failure to rotate the upper body as a unit. This failure typically causes one of two issues during the forward swing. In some cases, the non-hitting elbow is pulled around the body too early in the swing, causing the chest to open up too early, thereby desynchronizing the abdominal rotation from the rest of the kinetic chain. Other times, it doesn’t come around at all, and the chest remains under-rotated by contact, wasting some of the abdominal rotation and impairing the contact’s fault tolerance.

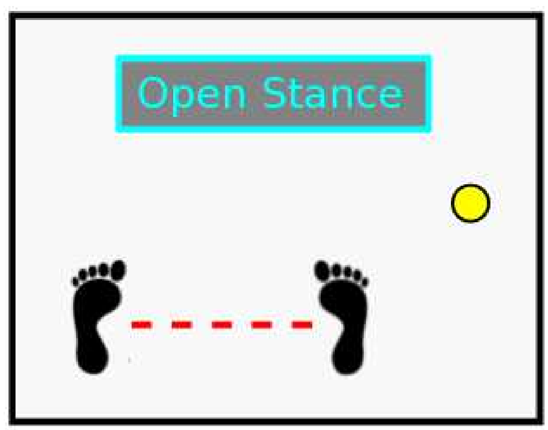
The remedy to both of these issues is the mantra “back together, forward together.” To solve under-rotation, this idea necessitates pulling the non-hitting arm around the body as you swing. You’re always going to swing with your hitting arm; when you ensure your non-hitting arm comes with it, that forces your entire chest to rotate as a unit.

When early rotation is the issue, the solution is to *not* bring the non-hitting arm around *before* the hitting arm. The hitting arm, the non-hitting arm, and the chest all rotate forward *together.* Often, a player who develops early rotation is a player that *used* to under-rotate, and now they're so focused on bringing their non-hitting arm around that they do it too early. The important distinction here is that a proper forehand is *not* about "bringing the non-hitting elbow around;" that's just a solution for the common problem of under-rotation. Rather, a proper swing consists of rotating the entire upper body forward *as a unit.*

Too Much to Think About

This is also the point in this progression at which you should expect significantly more practice to be required for mastery. "Twist" is easy; most players can be told to "twist" and then implement it without much thought at all. "Twist by driving off your back leg" is still pretty easy. "Twist" itself is so automatic that the additional "back leg" portion really only constitutes one *new* instruction to add to your thinking. It's *one* new thing, and humans beings are very good at working on *one* new thing.

"Twist, and lead the twist with your back leg, and ensure your upper body rotates *together,* rather than one elbow going too early" is now *two* new things. Unfortunately, the way human brains work, *two* things is not linearly harder than *one* new thing, but rather exponentially harder. You'll often find that, on one stroke, your back leg drive will be great, but you'll leave your shoulders under-rotated, and then on the next stroke, your upper body turn will be awesome, but you forgot to drive off your back leg. It's just going to take practice to get both parts working together, and the forehand isn't going to be super fault tolerant while either the hips or the shoulders remain under or over-rotated.



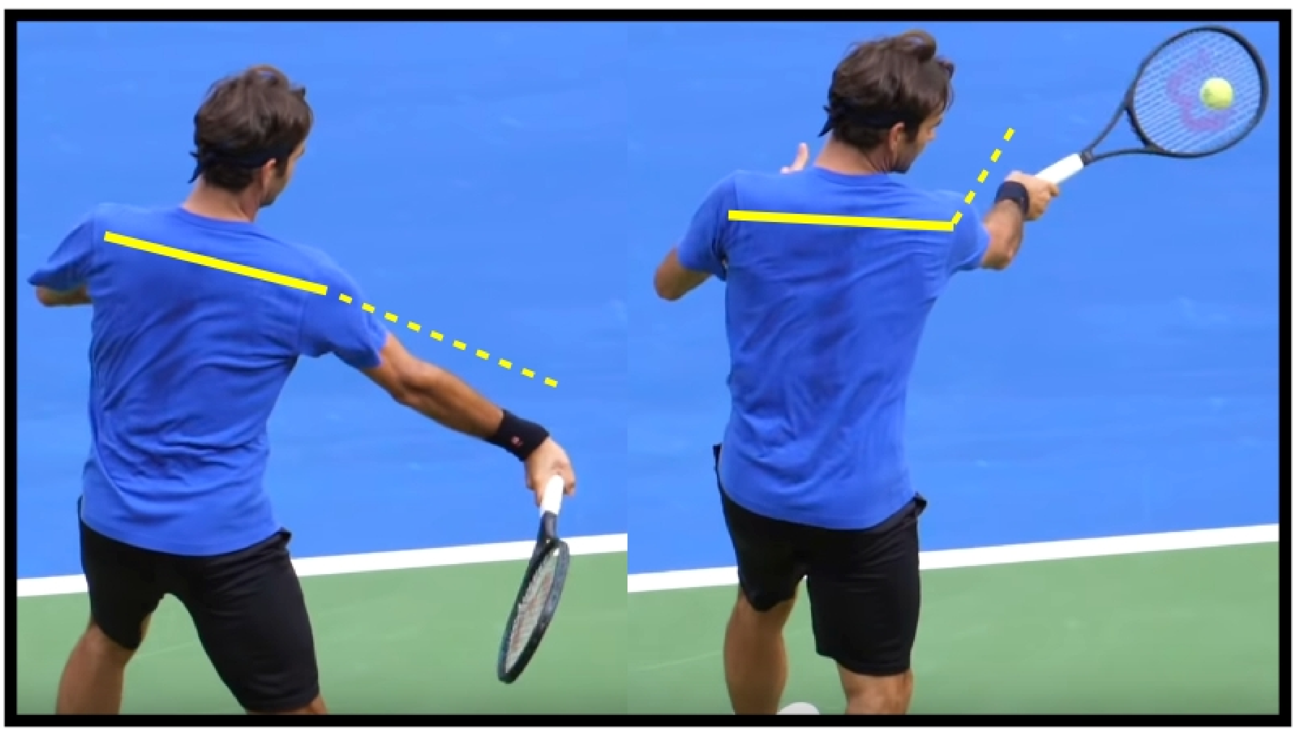
In a fully open stance, the hips remain square to the target during preparation, making it a great stance to use for a player who just wants to focus on coordinating their upper body.

One useful way to streamline the process is to temporarily hit the forehand exclusively out of an open stance. That way, you can *just* focus on rotating your upper body, since your lower body can stay as is and be fine. You'll lose the extra power you get from hip rotation, but you *will* have a fault tolerant, match usable forehand where you only have to focus on *one thing* to improve it – twisting the chest and both arms together. Once you master that, you can add back in your hip rotation and, again, you'll only have *one thing* to work on.

4. Extend through contact

One helpful cue to synchronize the end of the kinetic chain, especially in cases where a player is making the forced follow-through mistake, is "extend through contact." The player should envision the racket flicking *forward* through the ball. In their mind, the swing is finished at contact. The follow-through isn't part of it.

Visualizing the *translational* (back-to-front) contact you want to make with the ball as you strike it will help cue you to *not* prematurely pull your racket around your body, and instead to utilize your kinetic chain to get your racket flicking **through** the ball as fast as it can.



During hip rotation and abdominal rotation (left), Roger Federer's arm and chest form a roughly 180º angle. As contact approaches, the hitting side pectoral muscle drives the racket forward relative to the chest, decreasing this angle by about 70-80º.

There's also a small issue pertaining to muscle activation during the chain that we haven't addressed yet, which the "extend through contact" cue typically resolves. If you rotate on your forehand with a completely relaxed hitting side chest muscle, your racket, hand, *and* arm will lag behind the body way too much. It'll stretch your arm back into a compromised position, and it'll feel awkward and powerless. In order to get the entire chest-arm-racket system to rotate together as a unit, the hitting side pectoral muscle needs to stabilize the arm, not letting it lag backwards. This brings to light an interesting contrast between the arm and the wrist – the chest should keep the arm stable, not allowing it to lag backwards as the upper body rotates. The wrist, on the other hand, should relax, and it *should* allow the hand and racket to lag back as the arm pulls them forward.

Not only does the chest need to stabilize the arm as it comes around, but at a certain point, the chest itself should drive the racket forward, thereby decreasing the relative chest-arm angle. This, like most of the body alignments we've talked about, is much better internalized as an English phrase, utilizing more intuitive language – "extend through contact" – than it is as "use your hitting side pectoral muscle to decrease the angle between your upper arm and your chest."

The Forehand POWER Checklist

So you've read this far. You've implemented everything you've learned about fault tolerant contact, rotational preparation, and the forward swing. You're finally striking the ball *the same way every time*. The natural next question is – okay, now how do I hit it *harder*? This checklist provides six ways to do so.

Before We Start…

First, let's distinguish between the two distinctly different *kinds* of "forehand power."

The first is *effortless* power.

Effortless power is the power we gain simply by using proper biomechanics. It's gained while barely sacrificing any consistency or fault tolerance, and utilizing it only slightly raises the shot's degree of difficulty. All players should strive to practice the mechanics that yield effortless power.

The second is *effortful* power.

Effortful power is volitional. You've decided you're in a situation where hitting really, really hard is appropriate, and so you're going to *try* to hit really, really hard. The most obvious example of an appropriate situation for effortful power is a short, high, floating forehand. I want to be clear that effortful power is a *personal choice*; even on a floater, not every player elects to crush it.

Some players hardly ever choose to hit really, really hard and play almost every shot with lots of margin and spin, while other players (Nick Kyrgios and Jack Sock, for example) treat almost every high, slow, forehand as if it should break the sound barrier. Almost every item on the list that follows creates *effortless* power when applied, but can also be further exaggerated to create additional *effortful* power. The decision as to whether or not to execute said exaggeration falls to you.

Power is subordinate to contact point.

In addition to distinguishing between the two types of power, I want to stress that *everything* that follows is *subordinate* to a proper contact point. As you implement the tips below, it's very easy to start messing up *where* you're contacting the ball, relative to your body. In order for the modern swing to work as intended, the ball must be contacted *away* from the body – both laterally away from the body and out in front of it. That's the only contact point that allows the racket to harness the trunk's rotation and then naturally flick through the ball, striking it in a topspin way.

While striking a blistering jumping forehand winner, Nick Kyrgios contacts the ball well away from his torso.

If, in the course of implementing these checklist items, you start hitting the ball late, or start catching it too close to you, then any power gained from implementing the change will be offset by the less efficient motion.

Also, you'll notice that nothing we talk about here has anything to do with the *arm*. Power does *not* come from swinging harder with the arm; only tension, misses, and injury do. You can use your abs, you can drive forward with your legs, you can rotate with your hips – there are many efficient ways to generate power, but pulling harder with your arm isn't one of them.

1. Drive off of the back leg

This is a proprioceptive cue – to increase power, you should actually *think* about driving hard off the back leg and *feel* the hitting-leg butt muscle contracting. This drive will both push your whole body forward and cause your hips to rotate back towards the target. Both of those consequences – the hip rotation and the forward momentum – are essential for effortless power.

Here's an experiment you can do to teach yourself the movement: stand up, plant your feet, and then twist your hips sideways like you would on a forehand. Now tense your hitting leg butt muscle and watch it straighten out your hips. This is the primary muscle contraction that starts the rotational kinetic chain.

On any forehand that's not played out of a defensive position, you want to drive off your back leg. Almost all forehands use some degree of leg drive, but the amount of leg drive increases as a player wishes to be more and more offensive. On rally balls, you may not want to spend any of your mental focus on driving hard off the back leg. On a particularly easy ball, on the other hand, you can elect to inject *effortful* power, by actively focusing on maximizing the strength of that hitting leg contraction: loading it lower or twisting it farther away, and then exploding off of it as hard as you can.

This is a very quick, explosive contraction. It's not about driving off the hip *for a longer period of time,* but rather driving *harder* during the quick, explosive turn. As long as you're doing that, the harder you drive, the harder you'll hit. Just be sure all your extra power doesn't come at the expense of your contact point – that racket still needs to flick through the ball *in front* of your body and *away from* your body in order for the power from your leg drive to actually translate.

2. Relax

Force production necessitates tension; driving really, really hard off the back leg requires a really, really tight contraction of the muscles in that leg, specifically the quads, glutes, hamstrings, and calf.

The strange part about tennis, though, is that while *some* of our body parts forcefully contract to generate force, *other* body parts need to stay *completely relaxed* to *transfer* that force into the racket. The wrist and the hand are two of the body parts that stay relaxed. The more relaxed they are, the faster the racket goes. Hold the racket just tightly enough to prevent it from flying out of your hand, and use just enough wrist tension to maintain the string angle through contact.

I recommend mostly relaxing the arm, too. You *can* try to pull with your chest at the tail end of the kinetic chain, but this adds tension and can be difficult to time. Most players, just by exploding forward with their back leg and rotating their trunk towards the target, can generate plenty of force, with no need to use the arm as anything other than a rope to transfer it.

3. Breath in, then exhale through contact

Always breathe in, relax, and then exhale as you swing.

The forceful contraction necessary to accelerate the racket happens at the *beginning* of the swing. Exhaling through this contraction is fine – your core won't lose any stability until all the air is gone.

As the racket comes through the hitting zone, exhale the rest of your air. Any contraction necessary to generate force is *already complete*, and therefore you now want your body completely relaxed so as not to impede the subsequent racket whip at all through contact.

4. Turn your shoulders

Let's start with the effortless power version of the shoulder turn. A large amount of forehand power is generated when the abdominals rotate the player's chest/shoulders *forward* towards the target as they swing. This requires that their chest/shoulders be turned *away* from the target during preparation, such that they can rotate *back towards* the target during the stroke.

Ideally, you always want to turn your shoulders *farther* than you turn your hips. This creates something called "hip-shoulder angle separation," and this separation encourages the hips to unlock first, and then the chest/shoulders to follow, creating a rotational whip through contact.

Roger Federer (left) turns his shoulders farther, relative to the hips, than Juan Martin Del Potro (right) does. Both still lead the swing with their hip rotation, and both hit exceptionally hard.

The angle separation isn't necessary for effortless power, though; it just helps add yet another link to the rotational kinetic chain. As long as the hips unlock first, even if the shoulders and hips turn away by the same amount, the hip rotation contributor of the rotational kinetic chain will still accelerate the racket as expected.

4.5. Turn your shoulders a lot

All players *initiate* the swing with their hips, whipping them around *before* the shoulders, but after that, each player tends to have a different preference for how much *abdominal* torque to further incorporate into their kinetic chain. Each player has a different preference for their personal balance between power and control, and each needs to experiment with their own preparation.

Robin Soderling was famous for taking a massive shoulder turn – turning his shoulders almost 90º farther than his hips – and crushing forehands because of it.

That said, a huge shoulder turn *is* very helpful in generating power. Next time you have a high, slow, short forehand, try turning your shoulders away more than usual. Just your shoulders, not your hips. You should feel the muscles in your core stretch as you do this.

It might take some practice to get the timing down, but unlock all that loaded abdominal power correctly, and you'll absolutely crush the ball. Just remember to relax, extend through contact, and watch the ball all the way in, otherwise you're more likely to hit the back fence than strike a winner.

5. Pull your non-hitting elbow away

Turning your shoulders a lot during preparation is great, but what if you're not even utilizing the turn you're getting? Both hip rotation *and* chest rotation create power, after all. It would be a shame to forgo one of them by under-rotating. The easiest way to ensure that your chest whips back around towards the target is to consciously yank your non-hitting arm away from your body, thereby initiating that chest rotation.

Rafael Nadal prepares with his non-hitting elbow well on the hitting side of his body, then pulls it out to the right (he's lefty) as he initiates his forward swing, thereby whipping his hips and chest back towards the net and violently accelerating the racket.

6. Don't over-rotate the hips

Optimal power is achieved when 100% of the force generated by hip rotation is transferred into the racket. In general, this is actually pretty difficult to do (although doing it with less than perfect efficiency will still result in blistering forehands).

Attempt to have your hip rotation stop right as they're pointed at the target, or slightly past it. Rotational power is great, but a rotational *whip* is better. A whip isn't just thrown hard at the target – it's momentum is violently *halted* mid-flight in order to transfer *all* of its kinetic energy into its tip.

Your rotational whip doesn't have to be perfect – many tour players often hit winners with their hips ending over-rotated a bit – but for that *maximum* biomechanical power, stop the rotation at precisely the correct time, and *all* of the trunk's rotational energy will transform into racket head speed through the ball.

Andy Murray (left) hitting a 124 mph winner inside-out, and Fernando Verdasco (right) hitting a 103 mph winner down-the-line. Each player's hips remain pointed only slightly farther than their target, deep into their follow-through, indicating an efficient transfer of hip rotation into racket acceleration.

Again, over-rotating the hips when hitting hard is *strongly preferable* to leaving them under-rotated. That said, especially the easier and higher the ball is, really try and whip them around to exactly their precise destination, and no farther. If you want to slap 124 mph forehand winners like Andy Murray, you'll have to over-rotate less.

Are You Still Relaxed?

We started this section with relaxation, and we'll end it with relaxation, too. Relaxation is the key that unlocks power, topspin, *and* consistency on the modern forehand. Everything we've just discussed about how to properly harness the power in your rotational kinetic – *none* of it matters if all of that energy is wasted before it ever gets into the racket.

A relaxed wrist is a wrist that functions effectively as a hinge to facilitate the transfer of energy from trunk rotation into racket head speed. Unfortunately, this relaxation is often the first thing to go when you start focusing on a new technique or movement. No matter what else you're thinking about, no matter what technical improvement you're currently working on, breathe in, relax, then exhale as you swing.

It's also important to clearly state that the above biomechanics *will allow* you to relax while still retaining control of your shot. Sometimes, players tense up because they feel like it's the only way they can control their racket head. Typically, this indicates that they aren't properly utilizing the biggest muscles in the body to manipulate the racket, and are instead doing things like pulling with their arm and trying to generate force with their wrist. Manipulate the racket with the strong muscles of the legs, hips, and abs, and only minimal wrist tension will be necessary to control it.

No matter what technical improvement you're currently working on, breathe in, relax, then exhale as you swing.

As you cultivate a knack for treating your entire body as one system, from feet all the way out to racket, it'll start to feel very intuitive how you should go about manipulating the racket through space. You'll learn how to use those bigger muscles to raise, lower, and translate the racket head. Gain that sense, that proprioceptive understanding of what it feels like to manipulate the racket without using your hand, and you won't be nearly as tempted to tighten up as you swing. You'll be able to let the forehands fly.

Part 4

Training the Forehand

What if I Want to Hit Really, *Really* Hard?

Before our forehand power checklist in **The Forward Swing**, we distinguished between *effortless* power and *effortful* power. Obviously, in order to hit really, *really* hard, you have to use both, but the manner by which you employ them isn't immediately obvious.

For every player, there exists a physical upper limit to how much *effortful* power they can inject into their stroke before their margin for error drops off a cliff. If they were to *try* to hit any harder than that, their miss percentage would skyrocket. The speed at which they can hit while staying *below* this effortful power threshold constitutes their upper limit for speed on match usable forehands.

You *cannot* successfully inject any more *effortful* power into the shot after this point, no matter how hard you try. Even if you're focusing on all the right things – driving off your back leg, winding your abs up during preparation, whipping your torso around with your off-elbow, and staying relaxed – at a certain point, *trying* to hit harder will cause your success rate to crater.

The Power Limit

The cratering success rate when trying to use too much *effortful* power can manifest in a few different ways. First off, as a general rule of physics, the faster your racket is moving, the harder it is for you to time your stroke. With a faster swing, a contact zone of the exact same length now constitutes *less time* during which correct contact can be made.

> ***Even if you're focusing on all the right things, at a certain point, trying to hit harder will cause your success rate to crater.***

This is one reason that baseline, mechanical level fault tolerance is so important. The more fault tolerant your stroke is, the harder you can swing, even though doing so often introduces small timing faults. Since the shot is struck in a fault tolerant manner, these small timing faults don't break it. Even when you hit hard, you can still maintain a sufficiently high success rate, because your shot *doesn't have to be timed perfectly* to succeed.

Second, as we discussed back in **Forehand Contact,** any small mechanical flaws are *magnified* by racket speed. Even non-timing related execution mistakes, like a slightly incorrect string angle or a slightly sub-optimal contact point, will affect the stroke's result more at greater swing speeds – a ball that would have landed on the line at 80% effort might fly a few inches out at 100% effort.

Lastly, and probably most importantly, it's not actually easy to maintain fault tolerant mechanics while trying to inject a massive amount of *effortful* power. Most of the time, after a certain point, conscious effort to "try and hit harder" manifests as tightness in your hand, tightness in your arm, and an overall jerkiness and discoordination in the rotational kinetic chain. The result of this, of course, is that despite your mental perception of putting in *more* effort to swing *faster*, you'll actually end up hitting the ball *slower* than if you'd put in *less* effort. The rotational kinetic chain *is* your racket speed engine, and when the mechanics that create that chain are compromised by extra tightness and discoordination, that engine stalls.

Your Mind Will Lie to You

Unless you're very experienced, this limit on your useful effortful power is likely lower than you think. Your mind will lie to you. It will ignorantly (and arrogantly) inform you that, if you just dial up the effort another few notches, you can crank out an extra 5 or 10 mph. In all likelihood, you can't.

You have to experiment *during practice* to determine what level of effort corresponds to the hardest *consistent* shot you can hit – shots below an 80% success rate are mostly useless in tennis, except out of defensive positions. Once you find this level of effort, you need to train *what it feels like* into your memory, and you need to cultivate the discipline to stay under it during your matches. That's your top speed. For now, at least.

If we can't inject more *effortful* power into our shot without ruining our success rate, then if we still want to hit harder, there's only one variable left to tweak – *effortless* power. In order to hit really, *really* hard, you need to gain 15 pounds of muscle[13], and then swing the exact same way you swung before. That's it.

Once you've mastered everything in this book, and you've found your personal level of controllable *effortful* power, you've done all you can with respect to power at your current physical strength level. You can no longer hit harder by *trying* to hit harder. At this point, you'll be able to hit hard, for sure, but the athleticism required to *really* crack it constitutes more than just proper technique.

TV Strikes Again

Strength is a very under-rated part of tennis athleticism, partially for good reason. Technique *really is* more important, categorically. For the large majority of the player-base (especially those at the recreational level), any time spent on strength training would be largely wasted with respect to their tennis game, because they haven't *learned* how to effectively *recruit* their muscles yet. Without first mastering *how* to recruit the body's strength to generate racket head speed – by employing the rotational kinetic chain – a player won't be capable of effectively *harnessing* any new strength they might build into a measurable advantage on the court.

Roger Federer posing for a Gillette commercial. This is what the upper back of a world class athlete looks like.

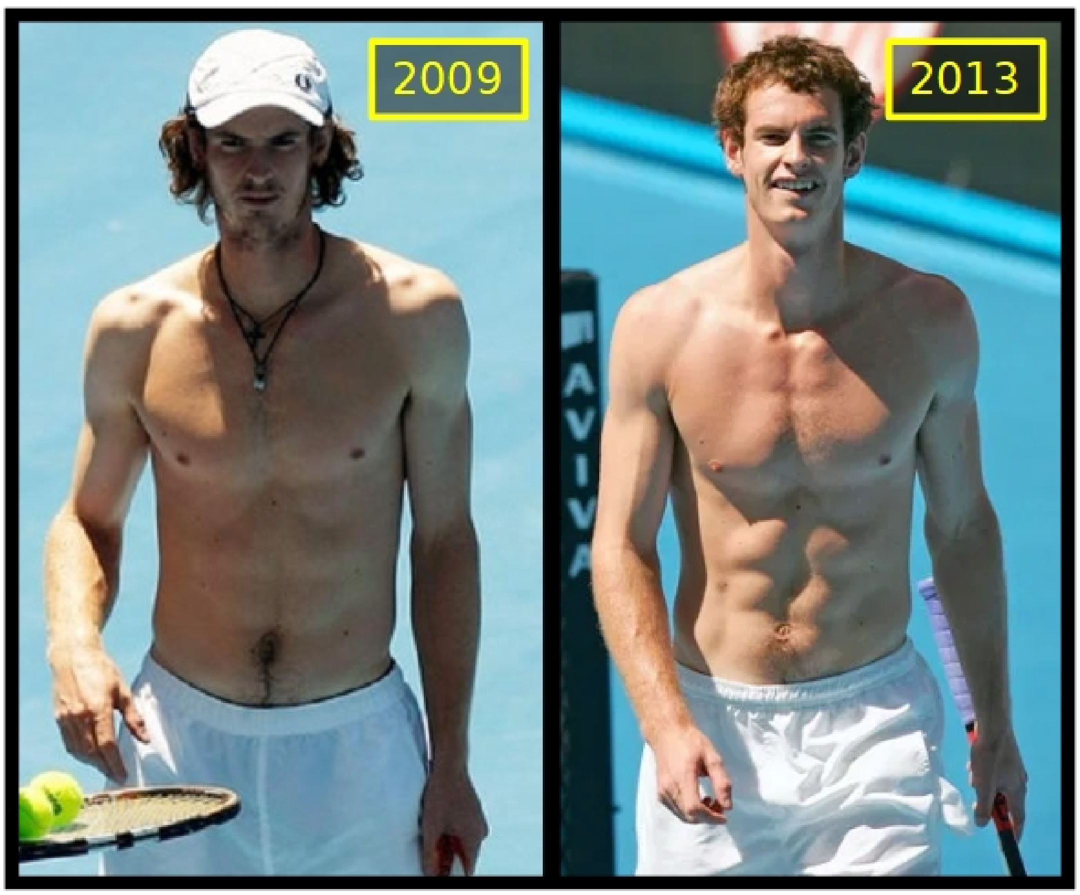
In the same vein, a strong player with poor mechanics will lose to a weaker player with excellent mechanics ten times out of ten. That's just the way tennis strokes work. If the choice is one or the other, strength *or* technique, technique will win every time. However, at the highest levels of the game, of course,

13 This "15 lbs (6.8kg)" number is just an approximation. It's a useful starting point for, say, a fit 5'10" (1.78m) male playing at 150lbs (68kg). They would be a much better player playing at 165lbs (74.8kg), provided they could maintain their lean body mass/fat ratio while gaining the weight.

this selection is a *false* dichotomy. In order for any player to really unlock their potential, they need to develop *both* strength *and* technique.

Back in **Forehand Preparation**, we discussed how misleading visuals on TV often lead to common misconceptions in tennis. Under-emphasis on strength is another of such misconceptions. They say that when it comes to modeling, "the camera adds 10 pounds." Well, when it comes to tennis players, the camera *subtracts* 10 pounds. Through your TV, it's actually very difficult to see *just how strong* the players on the tour really are.

Take Roger Federer for example. When you think of him, what are the first *athletic* traits that come to mind? Coordinated. Graceful. Fit. Quick. But not *strong.* It's just a trick of the cameras, though – just like the model you're seeing on TV *isn't* fit, but rather is unhealthily skinny, the athlete you're seeing on TV isn't *merely* fit, but is actually *also* unbelievably strong.



Andy Murray's upper body in 2009 versus in 2013, when he won is first Wimbledon.

Andy Murray further proved the importance of strength when he transformed his body prior to winning his first Wimbledon title in 2013. Check out the before and after, above, comparing his upper body musculature in 2009 to that in 2013. He's lean in both, he's *fit* in both, but he's significantly *stronger* in 2013. Now, sure, this is just one data point – a single tennis athlete who used strength to transform himself from great, to all-time great – but it's still quite significant.

Don't let the TV deceive you. Once you've mastered your fault tolerant technique, the *tennis specific* part of your power generation is essentially complete. The next step has nothing to do with tennis specifically. The next step is to get stronger.

Drop Feed for Forehand Development

There exists an *extremely* common issue impeding forehand development across the world today: forehand practice doesn't translate well into matches. Drop feed solves this issue. Just drop the tennis ball on the hitting side of the athlete's body, roughly at their contact point, and have them hit in either direction as effectively as they can.

Drop feed's defining attribute is that the dropped balls have *zero* pace; any time a player hits a dropped feed over the net, they've generated 100% of the forward ball speed on their own. As we'll see below, this property is what makes drop feed such an effective tool for fault tolerant forehand development.

The Common Forehand Development Issue

Here's the typical case – to start, every player's forehand is not fault tolerant, meaning that their stroke cannot tolerate a small fault and still succeed. If they swing a little early, or a little too late, they miss. If they prepare a little too close, or a little too far, they miss. If their string angle is off by 4 or 5 degrees, they miss. Unless they strike the ball *perfectly,* the stroke fails.

And the faster they swing, the more amplified this effect.

Therefore, the work around for a stroke that lacks fault tolerance is to swing *slower*. The slower a player swings, the less every fault I just mentioned affects the ball's trajectory. By swinging slower, the player has effectively restored enough of the swing's fault tolerance to make it usable in a match. It's at this point in a player's tennis development that they typically follow one of two paths.

The slower a player swings, the less every fault affects the ball's trajectory.

1. The Idealist

Some players, for whatever reason, reject the slower swing. Perhaps they've always wanted to hit really hard, so a slow swing doesn't interest them. Maybe their favorite player is Jennifer Brady, and they're committed to learning how to rip 3500 rpm forehands like she does.

Whatever the reason, to them, slowing down the swing (and thereby fundamentally altering it) to "get the ball in" is *unacceptable*. Due to this, they actively seek out coaching and instruction to fix the explosive version of the stroke.

It's often a very messy road, varying significantly depending on the quality of instruction they find, but, ultimately, they tweak and practice the real thing until it's usable in an actual match. This kind of self-evaluating, self-starter typically ends up fine (and may discover drop feed training on their own).

2. The Match Player

Then there's the very common case of the player who doesn't *realize* how non fault tolerant their explosive stroke really is, and how much they're altering it to make it match usable. To compensate for their poor mechanics and remain competitive in matches, this player *does* slow down their swing, allowing them to keep the ball in play. The unfortunate result of this choice is that, during nearly all of their hours playing tennis, they're *not actually practicing* a modern, explosive, rotational shot which can develop into something elite.

Instead, they're simply redirecting their opponent's pace and blocking the ball back over the net. They might "turn sideways," and they might "swing low to high," but those things aren't being done to generate explosive racket head speed – they're just being done because there's a vague idea that you're *supposed* to do them.

> ***During nearly all of their hours playing tennis, they're not actually practicing a modern, explosive, rotational shot.***

The result is that the common match player develops a strange hybrid forehand between an actual swing and a defensive block shot, but the only remotely fault tolerant shot they have is that defensive block shot. And the defensive block shot doesn't actually *require* any of the rotational mechanics that they've tried to implement. They can't actually utilize the rotational kinetic chain to accelerate the racket in a fault tolerant manner, but yet they still employ various randomly selected techniques from that movement pattern because that's what they've been told to do. They've been coached to "turn sideways," they've been coached to "swing low to high," etc, but those modern forehand mechanics aren't being implemented in a way that makes any athletic sense. Rather, they're just grafted onto a slow, purely translational, defensive block shot, the goal of which is just to keep the ball in play during matches.

That's also why this kind of player struggles so much on any high or short ball. They don't *possess* a fault tolerant stroke to use in an offensive situation. If they find themselves in one, they only have two options: block the ball back in like usual, or miss.

Drop feed to the rescue

So why is the forehand drop feed drill the solution?

This drill *forces* you to swing fast. Since there's zero pace on the ball to start, if you swing slowly, the ball won't go anywhere. Players with non fault tolerant strokes slow down their swing to restore consistency, but it only works because their opponent's ball has some velocity on it that they can redirect, some pace that they can use to block that ball back over the net. In this drill, there's no pace to block. It'll *force* the player to swing, and it'll *force* them to confront the flaws that are destroying the fault tolerance of that swing.

But Balls in Real Matches Have Pace

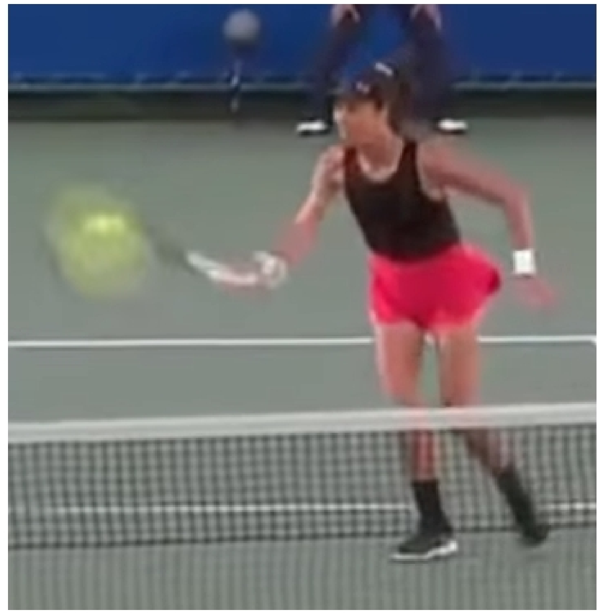
Most do, but some don't. There's a video online[14] of a challenger level player *losing a set* to a player with an injured shoulder, who could only serve underhand. It's a clip that exemplifies everything wrong with modern tennis coaching.

It's also a great example of why our primary goal is *not* to drill and drill and drill the exact kind of ball we think we're going to see in a match. That *kind of* works, and if you play against someone who gives you that exact ball, you'll do fine, but tennis isn't always that simple.

What happens when you play a slicer? A junk baller? Dare I say, a *pusher*?

14 https://www.youtube.com/watch?v=u1BW4Nayfm4

That's why, instead, we train *fault tolerant mechanics*, which apply no matter how the ball is hit at us. We aren't training "the waist height forehand" or "the running forehand" (at least not as its own concept). We are developing a *proprioceptive understanding* of the *principles* by which we move our body in space to produce our desired contact with the ball. Someone who understands those principles will easily adapt and crush an underhand serve.



Hseih Su-Wei, a player known for her craftiness, touch shots, and feel, hitting a forehand slice winner past Elise Mertens in the Osaka 2019 round of 16.

Shouldn't Training Mimic Match Conditions?

Sometimes, but not usually.

Training isn't about practicing exactly what we do in matches. There's a time for that, but most players spend far too much of their time on that part of the equation. For optimal performance, we want to spend a large amount of our time training things that are *harder* than what we do in matches. That way, the shots we need in matches become completely routine.

We train on drop fed, low balls, with zero pace, because they are *extremely punishing* of bad mechanics. By doing so, we take full advantage of our mind's reward/punishment system. In training, every time we use poor mechanics, we *want* to miss – we *want* our brain to feel that pinch of negative emotion, so that it associates that movement pattern with "wrong."

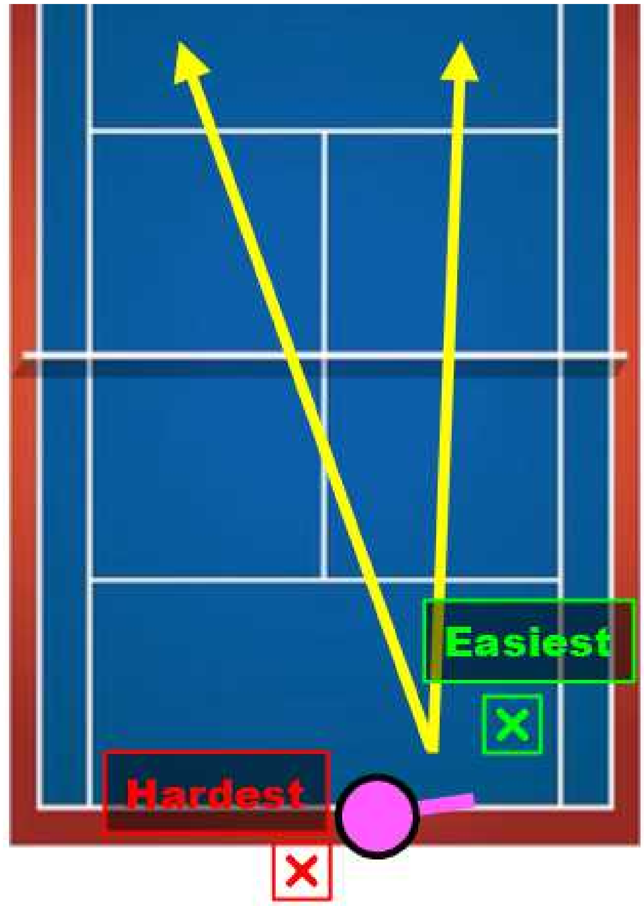
In match conditions, it's the opposite. Even when we use poor mechanics, we *want* the ball to go in. We *don't want* our technical issues, like an inconsistent string angle, exposed. That's why we do things like slow down our swing. The issue is that we make those adjustments *without noticing* the degree to which we're altering the swing on a fundamental level, or remembering that what we're doing really isn't ideal. Players who adjust in matches like this often notice that they play fine during regular baseline rallies, but then they frequently miss their approach shots and their offensive shots. What they may not realize is that the very same technical flaws that cause them to miss these “easy” shots are present in their baseline forehands as well, they’re just papered over by the fact that it’s actually the *baseline* shots are *relatively* easier, compared to slow balls with no pace.

You can get the ball back in play quite consistently by slowing down your swing, even with technical flaws present in the stroke, flaws that would ruin its fault tolerance if you tried to swing out. In a match, your brain will send you *reward* signals each time you do this. Your goal was to hit the ball back over the net, and you successfully hit the ball back over the net. That’s why it’s very difficult to fundamentally improve a stroke in match conditions – often, when you strike the ball mechanically poorly, it still goes in, and so your brain still tells you that you did a good job.

Eventually, Live-Ball is Superior

At a certain point, primarily training live-ball will be superior to training drop feed. The better the player is, the more live-ball training is useful, but even a very good player shouldn’t *only* be training live ball. Drop feed, and especially toss feed with awkward feeds, are great ways to perform a quick little refresh on fault tolerant mechanics. Even just 30-40 drop fed balls before a hitting session is a great way to see if, on that particular day, there are any lingering old bad habits or technical glitches present in your swing that will be more difficult to diagnose once you move to live-ball.

Making it Harder



The easiest feeds are dropped to the athlete's hitting side, and in front of them (green target), while the hardest are tossed so as to land both behind their body and on their (initially) non-hitting side (red target).

Once an athlete has made, say, 20 drop fed forehands in a row, it's good to change it up. Lightly toss the balls in different locations around them. The hardest feeds are the ones tossed both at their body, and behind them. The easiest are tossed in front of them, and to their hitting side.

What We're Practicing

Generating 100% of your own pace on a ball is hard enough – moving around the court and then doing it is even harder. Tossing the ball to awkward locations, and forcing the athlete to move to them before hitting, teaches the athlete two critically important skills.

1. How to find the perfect contact point

Since the athlete will be forced to move differently for each ball, they won't be able to execute their movement by habit. Instead, they'll need a mental idea of what the optimal relative position between their body and the ball should be when they swing, and then they'll move to get into that position.

This is exactly what we want. Again, just like with the stroke itself, we are *not* training the act of moving to the shot as any sort of specific series of steps, that the feet must perform. If we did, that series would only apply only in certain situations. Instead, we want the athlete to have a *goal* in mind – get my body into a certain position, relative to the ball – and then we want them to improve at executing that goal, utilizing movement.

2. How to generate pace out of imperfect positions

Of course, in a match, your opponent is doing everything in their power to *prevent* you from getting into that perfect position. Therefore, it's also essential to have a stroke which succeeds when out of position.

Tennys Sandgren striking a brilliant cross court forehand pass, while on the dead run and falling backwards.

The toss feed drill with moving feeds helps train this too. The better a player is at moving, the more awkward the feeds they should train with. Keep dialing up the difficulty – right up to the edge of what you feel is appropriate. Sometimes, the feeds should be so tough that the trainee is *forced* to play the ball out of an imperfect position.

All the time, on the tour, we see players generate crazy power and spin out of what seem to be very compromised positions. This is one of the many skills most believe is simply natural talent.

It is not. It can be trained, and toss feed is a great way to train it.

Drop Feed Will Reveal Your Flaws

Drop feed is the gold standard for ironing out mechanical glitches. There are other essential skills that it doesn't teach, skills like tracking the ball over the net, and timing the swing on an incoming ball, but it's still an incredibly useful tool.

It's *punishing*, and that's what matters. Unconscious incompetence is the most dangerous place for any athlete – not only do you have a technical flaw, but you *don't know* that you have that technical flaw.

Whatever flaws a forehand stroke has, drop feed will expose them; that's its greatest virtue. And once that flaw is exposed, once you know what to work on, you're already more than halfway to fixing it.

Training to Crush the Pusher

In order to beat a pusher, you need **high percentage offensive shots**. Most players at the recreational level do not posses high percentage offensive shots, and yet they *attempt* offensive tennis anyway.

This leads to the frustration death spiral. It *feels* like you're in a winning position – you've got a short, high ball over the middle; all you have to do is hit one hard forehand, and maybe a volley or two, and the point is yours.

But you haven't actually *developed* your flat, hard, forehand, and you haven't actually *practiced* those one or two volleys you'll have to hit. In your current state, this really *isn't* a winning position, even though it feels like one. Until you train your offensive game to the point where you routinely convert these shots at a 90+% clip, you're going to lose from offensive positions *all the time*.

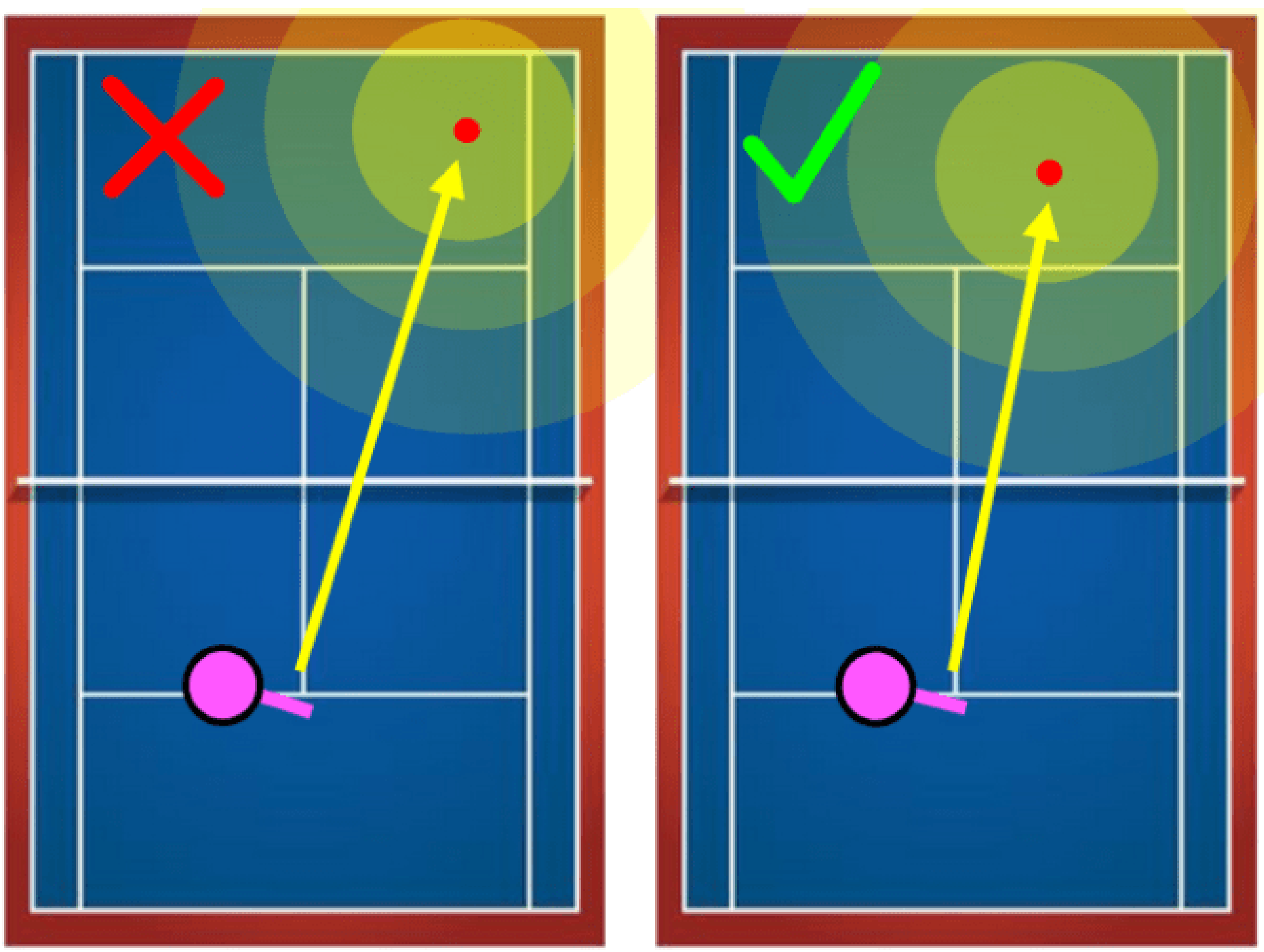
You miss, you feel like you shouldn't have, you get frustrated, miss some more, and then get more frustrated. And the pusher wins 6-2 6-1, even though you were the "better player."

In order to beat a pusher, you need high percentage offensive shots.

Fixing Your Strategy

The first step to beating a pusher is that you yourself must also keep the ball in play. When your opponent pushes, it'll tempt you to attempt shots that you can't actually convert with a high enough frequency to be profitable. We need to stop that.

We've already discussed the *strategy* behind beating a pusher at length on our blog[15], but here's the upshot. To start, select shots that you know you can make with an 80% probability or better. That means that for every 4 times you make the shot, you miss it 1 time. If you force yourself to only attempt shots you know you can make at this 4:1 ratio or better, it'll prevent you from heading down the typical frustration death spiral to a quick defeat.



An intuitive target (left) that is actually far too aggressive for the depicted player's skill level; far more than 20% of the shot's outcome distribution is out. Using a more conservative target (right) will lead to far better outcomes on this approach shot.

15 https://faulttoleranttennis.com/the-pushers-strategy-is-better-than-yours/

While you're trying to find the shots that you can execute at this 4:1 clip, you might have to adjust in real time. If you've just missed 3 of the same shot, move the target in and aim higher over the net. If you've made 15 in a row, you can feel free to be a little more aggressive.

You Have to *Practice* Your Offense

If you've never actually *practiced* your offense, playing against a pusher is *always* going to be tough. The match is going to be tighter than you feel like it *should* be, and you're going to have to play a lot safer to win than you feel like you *should* have to play.

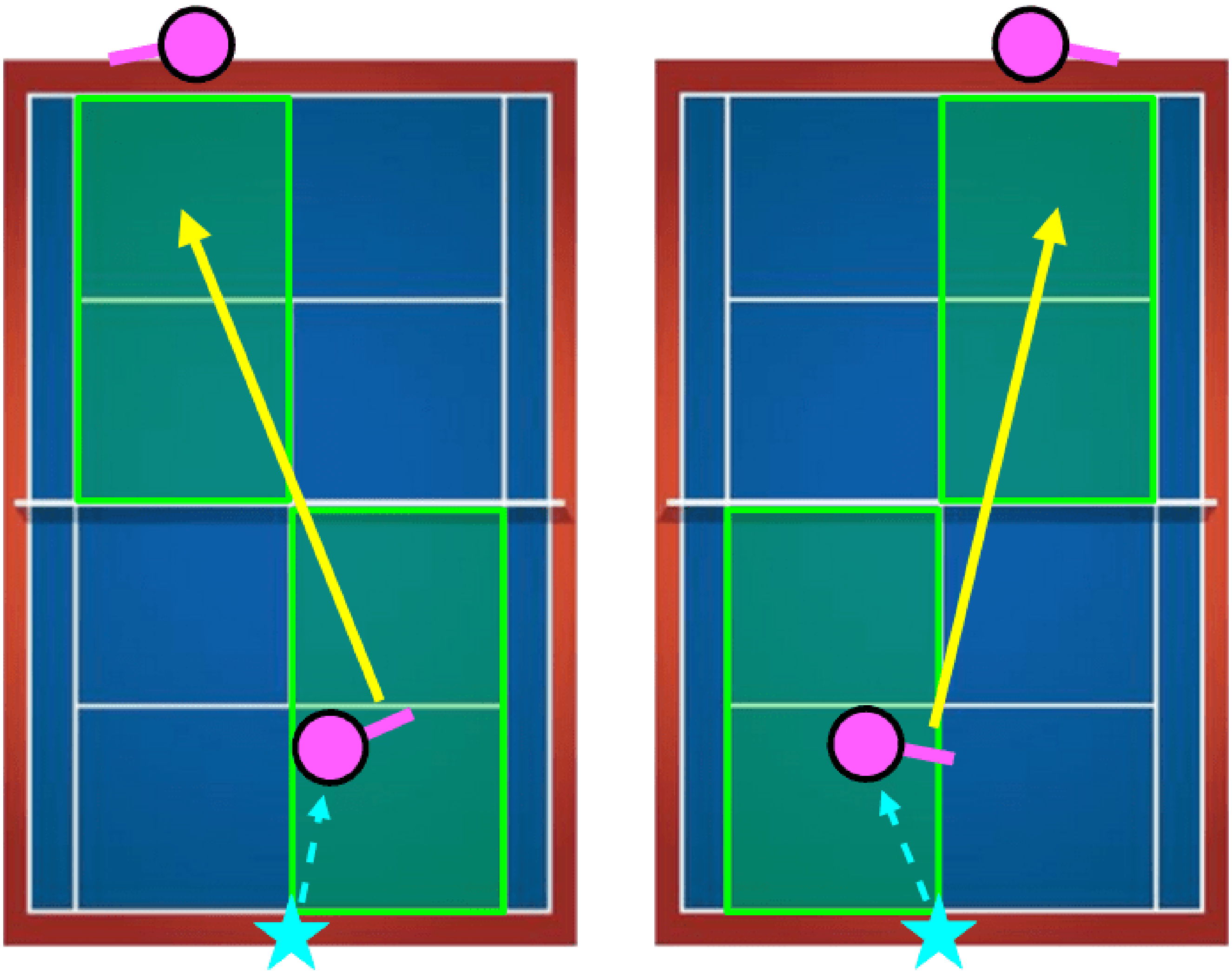
But here's the thing – what you're seeing on court is the ***truth.***

Until you spend the time on the practice court to actually *train* your offensive tennis, you should *expect* that your offensive tennis will be poor when playing a match. So, in that vein, here's the best drill for training it.

Offense-Defense

My favorite drill for practicing offensive tennis is one I call "Offense-Defense." It's useful for the defender, too, but it's less realistic from their perspective. Where the drill really shines is *training your offense.* Here's how it works:

There's one offensive player and one defensive player. Both players start at the center of the baseline. This is important – to maintain the realism of the drill, neither player should move from the center until the ball is fed. The defensive player feeds the offensive player a high, short ball to either side. Then, the two players play out the point cross-court. Any shot that doesn't land cross-court (as indicated by extending the center-service line to the baseline) is out. The doubles alleys are out.



A graphical depiction of Offense-Defense – green is in, everything else is out. Players start in the center, then move once the ball is fed. The drill can be done on the deuce side (left) or on the ad side (right). In this depiction, the offensive player is approaching with a forehand on both sides.

How it Works in Practice

The player who receives the high, short ball will hit a cross-court approach shot and come to net. They are effectively playing offense. The defensive player can try to hit a passing shot (tough because there's so little court), they can try to lob, they can try to keep the ball low – whatever they feel is appropriate in this situation.

Here's why the drill is so helpful – both players are doing everything they can to *win* the point. The rules are set up such that the rallies will be longer than typical offensive vs defensive rallies, leading to more practice than would be possible under true match conditions, and yet the *shots* actually making up these longer rallies are still quite realistic.

Both players are doing everything they can to win the point.

This falls in stark contrast to many other drills, during which one player is *not* playing to win. For example, as a coach, when I feed my student volleys, I'm not *trying* to hit it past them. When I hit it hard at their feet so they can practice digging the ball, I'm not *trying* to get them to miss. While this kind of practice definitely has its place, it doesn't translate to matches *nearly* as well as the practice from a drill like Offense-Defense does. In Offense-Defense, both players are *trying to win*. That's why the skills you train while drilling it carry over so well into matches.

Why not play full court?

We play half court such that an Offense-Defense rally can continue even when both players are genuinely trying to win the point. If the drill were full court, many of the points would just end with an approach shot winner, and each player would hit far fewer balls/hour.

Playing half court beautifully solves this problem. We're essentially playing every rally from *after* the time when the defender has *guessed correctly* where the offensive shot is going. That's perfect, because, in a match, those are the rallies that continue anyway.

The offensive player won't be able to blow the ball by the defender easily in half court, and will typically have to play multiple volleys in a row to win the point. This is also extremely good, because the discipline to *construct* a point using high percentage offensive shots is *essential* for quality net play.

The discipline to construct a point using high percentage offensive shots is essential for quality net play.

Skills That Offense-Defense Trains

Here's a non-exhaustive list of the amazing benefits of Offense-Defense.

On the offensive side

- Hitting cross-court approach shots
- Moving through an aggressive shot to net
- Split stepping at the appropriate time while coming in
- Hitting an approach volley from the service line
- Choosing when to drop volley and when to drive volley
- Constructing a point with high percentage volleys
- Overheads
- Lob retrieval

And on the defensive side

- Digging out a hard ball hit at your feet
- Keeping the ball low
- Reading the net player's racket and body to anticipate the shot
- Defensive touch shots
- Immediate transition from defense to offense on a weak volley
- Lobbing

Adjustments and Suggestions

Depending on your level, playing with the doubles alleys in may be appropriate. The goal of removing the alleys is to force the players to learn to aim their shots, and to force the rallies to continue. That said, the goal isn't for both players to just miss and miss and miss. If you can't consistently hit the half court target without the alley's extra width, add it in to keep the points going.

This drill works great on both the forehand and backhand side. On the backhand side, I recommend alternating feeds between feeds that are higher and closer to the center of the court, on which the offensive player should hit an inside-out forehand, and feeds that are lower and wider, on which the offensive player should approach with the backhand.

Mix it up – throw in the occasional drop shot approach just like you would in a match. Don't treat this as a "drill" in the sense that shots like drop shots, or slices, or trick shots are inappropriate. You'd never hit a drop shot while grinding racket-fed forehands, but it makes perfect sense to hit them in Offense-Defense, especially if your opponent is defending from way back in the court. Offense-Defense is your opportunity to experiment with and improve all aspects of your offensive tennis.

Offense-Defense Actually Translates

I prefer not keeping score during Offense-Defense, so that players don't feel pressured into avoiding the very offensive shots they need to practice. If you can only hit, say, your flat approach forehand at 50% right now, this is the time to get it to 80%, not to abandon it because it's too inconsistent.

As you run up for a short ball in a match, it'll finally feel familiar.

That said, decide what you want to work on – the shots you're going to use that *are not* match ready – and then genuinely compete from there. Do that, and the practice *will* translate to matches in a way that *so much* practice doesn't.

Offensive tennis is typically where a player's ineffective practice gets exposed. After adding Offense-Defense to your repertoire, that will no longer be the case. As you run up for a short ball in a match, it'll finally feel *familiar* – you'll finally feel *confident*, like you've done this 100 times before, because, after drilling with Offense-Defense, you *will have* done it 100 times before. As you split step for your volley, as you move in and strike the volley, and then as you shuffle back, line up the overhead, and blast the final winner past your opponent – each of those shots will have been *practiced.*

The X-Drill for Forehand Conditioning

Playing while fatigued is hard. The more physically grueling a rally is, the more difficult it is to execute your strokes. Fatigue in tennis isn't merely mental, either. Both the creatinine and the glucose in your muscles are constantly being burned in order to contract them, and then subsequently refreshed during periods of rest. You are a *physically less explosive* player on shot 50 of a rally than you were on shot 1.

Playing under fatigue is one of the extremely common situations in which small faults will tend to creep into your mechanics. We preach fault tolerance in this book because it's impossible to play tennis without making small mistakes. If your strokes fail in the presence of those small mistakes, your strokes are fundamentally broken.

> ***If we want matches to feel easy, we train in situations that are more difficult than those we find in matches.***

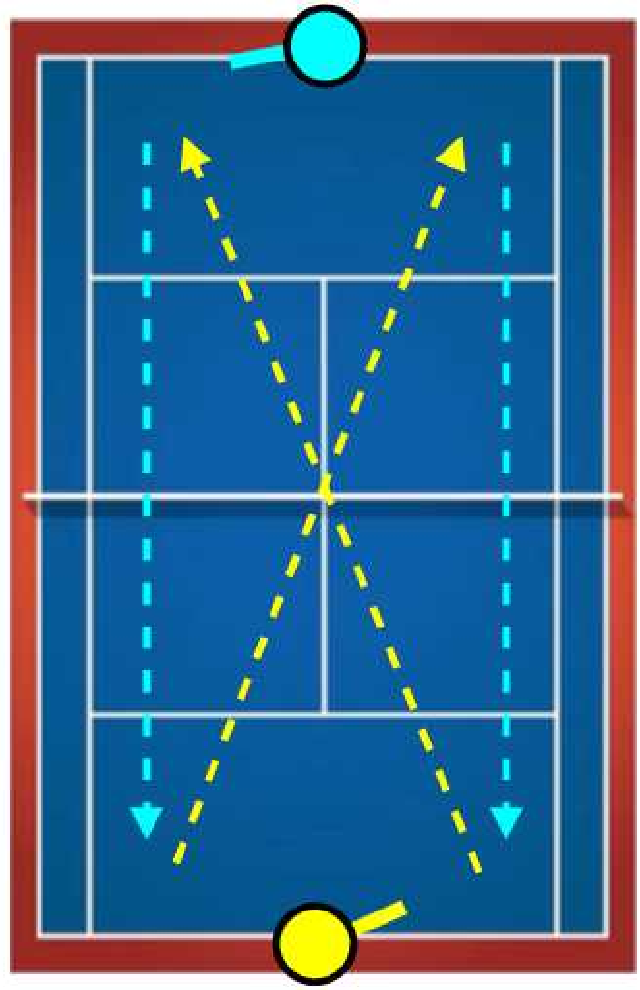
However, just because our stroke *can* handle the faults of fatigue doesn't mean we can't try to train those faults away, and that's the point of X-drill. Our goal is to create a training environment that is *more fatiguing* than that which you'll actually find in a match. That way, when our body adapts to the stress of that training environment, it will be fully prepared for the match environment. Remember, if we want matches to feel regular, we train match play. If we want matches to feel *easy,* we train in situations that are *more difficult* than those we find in matches.

How it Works

The X-drill is pretty simple. Two players rally from the baseline. One player always hits cross-court, and the other player always hits down-the-line. The result is that each player is always retrieving a ball on the *opposite* side of the court from where they just hit.

The drill can be performed either competitively or cooperatively. Competitively, the ball is out if it doesn't land in the appropriate corner after being struck. You can play an entire game out with one player cross-court and one player down-the-line, or you can use a tiebreak system and switch who's hitting which direction every two points.

Cooperatively, both players play their shots high over the net with lots of topspin, and the goal is to keep the rally going as long as possible. Feel free to play balls that are out, but *do not* let the ball bounce twice. You *must* cultivate the habit of sprinting forward for the ball in practice, lest you impair your reactions in an actual match.



A visual depiction of the X-drill; the cyan player is hitting down-the-line, while the yellow player is hitting cross-court.

Why It's Effective

All we've done with the X-drill is take a particular match situation, a long rally, and then dial up the training stress of that situation. All the systems required to succeed in a long rally will be stressed during X-drill, thereby driving positive adaption during recovery. In this regard, it's an extremely efficient drill on two fronts.

First, it lets you *practice* playing while fatigued. You get to learn what that feels like, learn how to control your body when it's in that state. Second, X-drill *trains* all the exact same body systems which are stressed during a long rally. It's just that with X-drill, we stress those systems *more* than they're stressed during a typical match rally, so that we can further drive athletic adaptation.

Conditioning on the Court

I recommend X-drill for conditioning *over* non-tennis-specific training. Something like jogging is certainly great for the heart and lungs, but we need more than that when we play tennis. We need our body to sprint, then rest, then sprint again. We need to perform what essentially amounts to fifty body-weight squats in quick succession, and we need to add in an explosive abdominal twist after each one. All of that high repetition, low-weight conditioning is much better trained on the court, *using the exact movement patterns we need*, than it is in the gym.

There is a place for simple conditioning or simple sprint training. Sometimes, you don't have access to a court. Sometimes, you're dealing with a minor injury. In general, though, the cardiovascular system is very easy to train while *on* the court, so why not do so and train your strokes at the same time? It is getting *stronger* that's hard to do on court, because your body-weight just isn't enough stress on your muscles, but for getting *fitter,* just use X-drill.

Develop the Feel of Federer

Many people assume that "touch" and "feel" are simply natural talents. They are not – like everything else in tennis, touch and feel can be *trained.* While there certainly *are* players who, with practically no coaching, will quickly develop an impressive feel for the game, that doesn't mean that feel is unattainable for the rest of us.

Quite the opposite, in fact. Precisely because players like Roger Federer are such naturals, we mortals are able to learn which biomechanical alignments are useful for developing feel, and which ones aren't by studying those players.

The Mechanics of Feel

The video linked in this footnote[16] (timestamped at 2:22) is the best, most informative practice video I've ever seen when it comes to studying feel. It showcases the most important concept that creates feel in *every* situation – The Fundamental Theorem of Tennis.

Recall the theorem, as stated back in **Forehand Contact**:

1. The racket makes a 90-135° angle with the forearm

2. The racket strikes the ball in front of the body (in a pinch, in front of the forearm)

Roger Federer playing a low backhand slice (left), a regular backhand slice (middle), and a backhand block (right). In each of the three different situations, he contacts the ball out in front of his foot, and he maintains an angle between his racket and his forearm.

16 https://www.youtube.com/watch?v=F5rjKZZihw0&t=142s

No matter what shot Roger is hitting, he *always* catches the ball in front of him, and there is *always* an angle between his racket and his forearm. He adjusts his *entire arm* when he has to adjust – he does *not* adjust by straightening out his wrist.

That's really all there is to feel. Whether you're playing a forehand slice, backhand slice, forehand block, backhand block, drop shot, or even a regular forehand or backhand, gaining a feel for the ball is as simple as gaining an understanding of how to adjust your racket in space *without* straightening out your forearm-racket angle and *without* catching the ball late.

Practicing Feel

Feel is a *skill* that you can practice and master, just like the forehand or backhand.

Mini-tennis is a great drill for it – you and your partner stand on opposite service lines and lightly hit the ball back and forth across the net. Relax your hand, but use just enough tension to maintain your 90-135° angle.

Focus on hitting the ball *in front* and focus on adjusting your *entire arm–racket system* to the ball, rather than just your racket. Play around with taking the ball earlier than feels natural – the correct contact point on a touch shot will often *initially* feel quite far in front of you.

Roger Federer casually playing a short, low forehand (left) and a forehand which has gotten to close to him (right). In each case, he takes the ball well in front of his body and maintains an angle between his racket and his forearm.

None of the power of your shot comes from your forearm. Your forearm merely *stabilizes* the racket during the stroke. You can use a slight body rotation, or you can use your chest to gently pull your entire racket-arm system forward, but do *not* use the muscles of your hand and wrist to swing. That causes misses and injury.

If you prefer a competitive setting, **doubles tennis** is the way to go for feel training. In doubles, you'll see all sorts of low shots, short shots, and shots you need to play quickly, while out of position, all of which necessitate great feel to execute.

Feel Makes Tennis Fun

There's nothing worse than all-out sprinting for a ball, barely getting there, just sliding your racket in underneath it, and then... missing the shot. Feel shots, touch shots, shots out of unique situations – these are the shots that make each tennis point unique, and the shots that make the game so intoxicating to play.

The key to succeeding out of novel situations is the fundamental theorem of tennis. Keep your racket in front, and keep your forearm-racket angle set. Whether you're playing a match, practicing, or even just hitting around, develop those good habits, and I guarantee that, over time, you'll develop the coveted feel that makes tennis so wonderfully enjoyable.

Closing Thoughts

My Hope

It is my hope that, after reading *The Fault Tolerant Forehand,* your understanding of the forehand stroke has been transformed. The first, most basic goal of this book was to provide each of you with a road map for crafting your forehand stroke, from *wherever* it is now, into the elite shot that you've always dreamed of having. More than that, though, I wanted you to gain some insight into the most effective way to approach athletics in general.

This book, fundamentally, is about *principles.* I'm not just trying to get you to *do* things; I'm trying to get you to *understand* the things you're doing. Every piece of technical advice I've provided has been provided *with* an accompanying justification. To be honest, I don't expect you to remember most of the technical advice itself. I don't expect you to remember what unwinding the abdominal coil is, or all the caveats relating to hand preparation and elbow extension, or that sometimes you don't even have to turn your hips. Creating a book of rules to memorize would have been pointless.

Instead, my goal was to instill in any reader a new understanding of what the forehand stroke really is – first, at a very basic level, and then at a more advance one. I wanted you to understand, for each movement you're attempting that makes the stroke more effective, *what* exactly you're doing and *why* exactly it works. That way, the book can be all encompassing. I could never have included every edge case, every specific situation out of which a forehand stroke must be produced. What I've provided instead is a set of principles that *do* apply universally, along with subsequent explanations of *why* those principles apply.

Now, I leave it to you to implement them.

One more thing – if you liked the book, please leave us a review on Amazon![17]

17 And if you didn't like it, don't leave us a review ;)

faulttoleranttennis.com

On our blog, faulttoleranttennis.com, we approach the entire game of tennis from a fault tolerance perspective. You'll find all the same values there that you found in this book:

- We start with fundamentals, then build up from there
- We always explain the *why* before the *how*
- We comment on meta level coaching

If you found *The Fault Tolerant Forehand* useful, here are some articles you may like as well:

- **Jen Brady – The Girl who Rips 3500 rpm Forehands**
  https://faulttoleranttennis.com/jen-brady-the-girl-who-rips-3500rpm-forehands/
- **The Serve is a Throw**
  https://faulttoleranttennis.com/the-serve-is-a-throw/
- **5 Signs of a Great Coach**
  https://faulttoleranttennis.com/5-signs-of-a-great-coach/
- **Dynamic vs Static Positions**
  https://faulttoleranttennis.com/dynamic-vs-static-positions/

Want to get more content like *The Fault Tolerant Forehand*? **Every Friday**, we publish a new article, which we'll send to your inbox totally free of charge. You can sign up for our email list at the bottom of any of our articles.

Questions, comments, concerns? Feel free to email us at faulttoleranttennis@gmail.com.

Image References

Beautiful videos of Roger Federer and Rafael Nadal Practicing – LoveTennis YouTube Channel:

https://www.youtube.com/watch?v=EFY460oquXw

https://www.youtube.com/watch?v=EFY460oquXw

And their associated website: http://www.LoveTennis.com

Federer showing off the Fundamental Theorem of Tennis:

Original video – Channel: dambuhalang ponjob vlog

https://www.youtube.com/watch?v=Dx4F8ftursk

Highlight version that's linked below the "Develop the Feel of Federer" chapter:

https://www.youtube.com/watch?v=F5rjKZZihw0&t=142s

Jennifer Brady/Rafael Nadal Forehand side-by-side:

Jen: https://www.tennisnow.com/News/2020/January/Brady-Shows-Strength-Stunning-Barty-in-Brisbane.aspx

Rafa: https://www.tennisworldusa.org/tennis/news/Rafael_Nadal/63434/rafael-nadal-s-forehand-has-wrong-technique-says-kuznetsova/

Murray and Federer shirtless:

https://arnoldzwicky.org/2013/07/11/more-tennis-hunks/